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1. Introduction

This `architecture models` the engineering-model-go repository as a model-driven toolchain. It captures authored functional intent, export surfaces, and verification-oriented traceability. The `model` is intended to guide implementation work in this repository using stable IDs and support paths.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	in-review	Initial repository self-model with stable top-level domains; implementation details will be expanded iteratively.	15	15	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Security View (`VIEW-SECURITY`)	in-review	Security scope starts from MCP boundary and export-validation supply chain paths.	10	10	7	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Traceability scope is seeded and will be tightened as requirements are added.	8	8	8	high	Counts a unit as covered when direct inferred runtime or code evidence exists.
State Lifecycle View (`VIEW-STATE-LIFECYCLE`)	draft	Captures model validity and artifact freshness lifecycle behavior.	0	0	0	n/a	Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.
Interaction Flow View (`VIEW-INTERACTION-FLOW`)	draft	Core implementation workflow is captured to guide model-driven repository changes.	3	3	0	high	Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal.

4. Terms and Definitions

Term	Definition
`actor` (TERM-ACTOR)	A person or operational role that interacts with functional units.
`ai agent` (ACT-AI-AGENT)	Consumes MCP outputs to plan and execute model-guided edits. Aliases:coding agent

Term	Definition
`allocation trace` (FU-ALLOCATION-TRACE)	Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems. Aliases:allocation
`architecture author` (ACT-ARCHITECTURE-AUTHOR)	Maintains model structure, mappings, and architecture intent. Aliases:model author
`architecture decision` (TERM-DECISION)	Authored architecture decision record with status, context, decision text, and consequences.
`architecture decision records are authored` (FEAT-ARCHITECTURE-DECISION-RECORDS)	Feature indicating the architecture model includes first-class ADR entries. Aliases:adrs authored
`architecture decision records document` (DATA-ARCHITECTURE-DECISION-RECORDS)	Generated document containing authored architecture decision records. Aliases:adr document, decisions document
`architecture model` (DATA-ARCHITECTURE-MODEL)	Authored architecture source describing functional intent and mappings. Aliases:architecture yml
`artifact generated event is raised` (EVT-ARTIFACT-GENERATED)	Event indicating one or more generated outputs were produced successfully. Aliases:artifact generated
`artifact generation` (FG-ARTIFACT-GENERATION)	Generation of AsciiDoc and exchange artifacts from model inputs. Aliases:generation
`AsciiDoc diagram` (TERM-ASCIIDOC-DIAGRAM)	Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.
`AsciiDoc document` (TERM-ASCIIDOC-DOCUMENT)	Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.
`asciidoc generator` (FU-ASCIIDOC-GENERATOR)	Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts. Aliases:engdoc
`AsciiDoc section` (TERM-ASCIIDOC-SECTION)	Structured generated document section or row model used to render authored and inferred architecture content.
`attack vector` (TERM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
`authored mapping` (TERM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
`catalog` (TERM-CATALOG)	Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.
`catalog entry` (TERM-CATALOG-ENTRY)	A stable catalog term with a definition and aliases for linting, linking, and generated references.

Term	Definition
catalog references resolve successfully (COND-CATALOG-REFERENCES-RESOLVE)	Condition requiring all authored references to match known catalog entries. Aliases:catalog resolved
ci pipeline (ACT-CI-PIPELINE)	Runs automated validation and export checks for pull requests. Aliases:ci
ci validation mode is active (MODE-CI-VALIDATION)	Automated pipeline mode focused on deterministic checks and artifacts. Aliases:ci mode
CLI command (TERM-CLI-COMMAND)	Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.
cli orchestration (FU-CLI-ORCHESTRATION)	Command entrypoints that orchestrate loading, validation, generation, and export flows. Aliases:cmd layer
code element (TERM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
codemap inference (FU-CODEMAP-INFERENCE)	Infers runtime, code, and verification evidence from repository sources and test markers. Aliases:inference
compliance engineer (ACT-COMPLIANCE-ENGINEER)	Reviews controls, risks, and traceability outputs for compliance needs. Aliases:assurance
compliance mapping (TERM-COMPLIANCE-MAPPING)	Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.
control (TERM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
control verification (TERM-CONTROL-VERIFICATION)	Authored control verification record with method, status, threat/risk scope, findings, and evidence.
data object (TERM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
deployment target (TERM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
design document (TERM-DESIGN)	Authored design narrative organized by model entities and architecture views.
design view (TERM-DESIGN-VIEW)	View-scoped design narrative attached to an authored functional group or functional unit.
engineering model go (SYS-ENGINEERING-MODEL-GO)	Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts. Aliases:engineering model toolchain, engmodel
event (TERM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Term	Definition
`evidence` (TERM-EVIDENCE)	A file, artifact, or observation used to support control, risk, threat, or verification claims.
`external schema dependencies are available` (COND-EXTERNAL-SCHEMA-AVAILABLE)	Condition requiring external schema artifacts to be present for validation. Aliases:schema available
`flow` (TERM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
`flow step` (TERM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
`functional group` (TERM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
`functional unit` (TERM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
`gemara export is enabled` (FEAT-GEMARA-EXPORT)	Feature enabling OpenSSF Gemara catalog and evaluation log generation. Aliases:gemara generation
`gemara exporter` (FU-GEMARA-EXPORTER)	Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK. Aliases:enggemara
`generated artifacts are fresh` (STATE-ARTIFACTS-FRESH)	State where generated outputs reflect the latest authored model inputs. Aliases:up to date
`generation run is requested` (EVT-GENERATION-RUN-REQUESTED)	Event indicating documentation or export generation execution starts. Aliases:generation requested
`generation workflow` (TERM-GENERATION-WORKFLOW)	Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.
`go toolchain` (REF-GO-TOOLCHAIN)	Go compiler, test runner, and module tooling used by the repository. Aliases:go
`hardware interfaces are authored` (FEAT-HARDWARE-INTERFACES)	Feature enabling hardware items and hardware/software interface modeling. Aliases:hw interfaces
`implementation engineer` (ACT-IMPLEMENTATION-ENGINEER)	Implements code changes guided by model requirements and support paths. Aliases:engineer
`inference hint` (TERM-INFERENCE-HINT)	Authored source and ownership configuration used to discover runtime, code, and verification evidence.
`inferred mapping` (TERM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
`interface` (TERM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Term	Definition
lint run (TERM-LINT-RUN)	Requirement lint configuration that controls parsing and quality checks for requirement text.
LOBSTER activity trace (TERM-LOBSTER-ACTIVITY-TRACE)	Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.
lobster exporter (FU-LOBSTER-EXPORTER)	Generates LOBSTER traceability inputs and reports from model and verification links. Aliases:englobster
lobster toolchain (REF-LOBSTER-TOOLCHAIN)	LOBSTER core and TRLC integration tools used for traceability reporting. Aliases:lobster
lobster trace report (DATA-LOBSTER-TRACE-REPORT)	Traceability report output generated from TRLC and test links. Aliases:lobster report
local development mode is active (MODE-LOCAL-DEVELOPMENT)	Local developer mode for iterative model and code updates. Aliases:local mode
malformed model input (AV-MALFORMED-MODEL-INPUT)	Invalid or ambiguous model content that causes incorrect graph construction. Aliases:invalid model
mcp integration (FG-MCP-INTEGRATION)	MCP server interfaces that expose model-aware tooling for AI agents. Aliases:mcp
mcp server (FU-MCP-SERVER)	Serves model-backed MCP tools with validated inputs and structured responses. Aliases:engmcp
MCP server (TERM-MCP-SERVER)	Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.
mcp server is enabled (FEAT-MCP-SERVER)	Feature enabling MCP runtime APIs for model-aware tooling. Aliases:mcp enabled
MCP stdio transport (TERM-MCP-STDIO-TRANSPORT)	Content-Length framed standard input/output transport used by the MCP command-line server.
MCP tool (TERM-MCP-TOOL)	Named MCP operation with a constrained input schema and structured response payload.
mcp tool call is received (EVT-MCP-TOOL-CALL-RECEIVED)	Event indicating the MCP server received a tool invocation request. Aliases:tool call
MCP tool response (TERM-MCP-TOOL-RESPONSE)	Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.
mcp tool result (DATA-MCP-TOOL-RESULT)	Structured MCP tool response payload carrying model-aware query results. Aliases:tool result

Term	Definition
`model` (TERM-MODEL)	The authored architecture model root that composes functional structure, relationships, inference hints, and views.
`model authoring` (FG-MODEL-AUTHORING)	Authoring and loading of architecture, catalog, requirements, and design inputs. Aliases:authoring
`model bundle` (TERM-BUNDLE)	Loaded architecture, catalog, and companion documents resolved as one working model context.
`model is invalid` (STATE-MODEL-INVALID)	State where validation diagnostics include one or more errors. Aliases:invalid
`model is valid` (STATE-MODEL-VALID)	State where authored model passes validation checks. Aliases:valid
`model loader` (FU-MODEL-LOADER)	Loads and normalizes architecture, catalog, requirements, and design documents. Aliases:bundle loader
`model updated event is received` (EVT-MODEL-UPDATED)	Event indicating architecture or requirement model source changed. Aliases:model change
`Open OTM document` (TERM-OPEN-OTM-DOCUMENT)	Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.
`open otm schema` (REF-OPEN-OTM-SCHEMA)	Open Threat Model schema used to validate OTM output artifacts. Aliases:otm schema
`OSCAL assessment results` (TERM-OSCAL-ASSESSMENT-RESULTS)	Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.
`oscal exporter` (FU-OSCAL-EXPORTER)	Exports the OSCAL system security plan, assessment results, and plan of action and milestones. Aliases:engoscal
`OSCAL POA&M` (TERM-OSCAL-POAM)	Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.
`OSCAL SSP` (TERM-OSCAL-SSP)	Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.
`oscal ssp artifact` (DATA-OSCAL-SSP)	Generated OSCAL system security plan JSON output. Aliases:ssp json
`path traversal in mcp calls` (AV-PATH-TRAVERSAL-IN-MCP)	Host file access attempts through untrusted path arguments in MCP tools. Aliases:path traversal
`POA&M item` (TERM-POAM-ITEM)	Plan of action and milestones item linked to a modeled risk and supporting artifacts.
`reference index` (TERM-REFERENCE-INDEX)	Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

Term	Definition
`referenced element` (TERM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
`repository index` (TERM-REPO-INDEX)	Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.
`requirement` (TERM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
`requirement trace is complete` (COND-REQUIREMENT-TRACE-COMPLETE)	Condition requiring requirements to map to implementation and verification evidence. Aliases:trace complete
`risk` (TERM-RISK)	Authored risk record with likelihood, impact, response, scope, controls, and evidence.
`runtime element` (TERM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
`schema supply chain tamper` (AV-SCHEMA-SUPPLY-CHAIN-TAMPER)	Tampering or drift in external schemas used for export validation. Aliases:schema tamper
`security context` (TERM-SECURITY-CONTEXT)	Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.
`source block` (TERM-SOURCE-BLOCK)	Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.
`state` (TERM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.
`Structurizr DSL` (TERM-STRUCTURIZR-DSL)	Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.
`structurizr dsl artifact` (DATA-STRUCTURIZR-DSL)	Generated Structurizr workspace DSL document. Aliases:dsl
`structurizr export is enabled` (FEAT-STRUCTURIZR-EXPORT)	Feature enabling Structurizr DSL generation and validation. Aliases:dsl export
`structurizr exporter` (FU-STRUCTURIZR-EXPORTER)	Exports Structurizr DSL and supports validation/static-site workflows. Aliases:engstruct
`structurizr validator` (REF-STRUCTURIZR-VALIDATOR)	Structurizr DSL validation runtime used to verify generated DSL. Aliases:structurizr
`system composition` (FU-SYSTEM-COMPOSITION)	Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model. Aliases:composition

Term	Definition
system composition is enabled (FEAT-SYSTEM-COMPOSITION)	Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems. Aliases:system of systems
threat assumption (TERM-THREAT-ASSUMPTION)	Authored security assumption that records scope, rationale, owner, status, and evidence.
Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)	Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.
threat dragon json artifact (DATA-THREAT-DRAGON-JSON)	Generated threat model artifact for OWASP Threat Dragon. Aliases:threat dragon output
threat dragon schemas (REF-THREAT-DRAGON-SCHEMAS)	External JSON schemas used to validate Threat Dragon exports. Aliases:threat dragon
threat export is enabled (FEAT-THREAT-EXPORT)	Feature enabling Threat Dragon and Open OTM exports. Aliases:threat modeling export
threat exporter (FU-THREAT-EXPORTER)	Exports threat model views to Threat Dragon and Open OTM representations. Aliases:engdragon
threat mitigation (TERM-THREAT-MITIGATION)	Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.
threat model export (TERM-THREAT-MODEL-EXPORT)	Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.
threat out-of-scope decision (TERM-THREAT-OUT-OF-SCOPE)	Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.
threat scenario (TERM-THREAT-SCENARIO)	Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.
trace marker (TERM-TRACE-MARKER)	Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.
traceability and compliance (FG-TRACEABILITY-COMPLIANCE)	Requirement trace exports and compliance-oriented output generation. Aliases:compliance
traceability export is enabled (FEAT-TRACEABILITY-EXPORT)	Feature enabling TRLC and LOBSTER generation flows. Aliases:trace export
traceability gap drift (AV-TRACEABILITY-GAP-DRIFT)	Requirement links or tests drifting from implementation over time. Aliases:trace drift
trlc exporter (FU-TRLC-EXPORTER)	Exports requirements and model structures to TRLC artifacts. Aliases:engtrlc
trlc package (DATA-TRLC-PACKAGE)	Generated TRLC requirements package and model metadata. Aliases:trlc output

Term	Definition
`TRLC package` (TERM-TRLC-PACKAGE)	Generated TRLC model and requirements package used for formal requirement traceability processing.
`trlc toolchain` (REF-TRLC-TOOLCHAIN)	TRLC parser and runtime used for requirements language validation. Aliases:trlc
`trust boundary` (TERM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
`upward linking` (TERM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
`validation` (TERM-VALIDATION)	Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.
`validation and analysis` (FG-VALIDATION-ANALYSIS)	Validation, inference, and evidence analysis over authored and inferred model content. Aliases:analysis
`validation diagnostic` (TERM-VALIDATION-DIAGNOSTIC)	Structured validation finding with code, severity, message, and optional source path.
`validation engine` (FU-VALIDATION-ENGINE)	Validates IDs, references, and model consistency across authored entities and mappings. Aliases:validator
`validation failed event is raised` (EVT-VALIDATION-FAILED)	Event indicating model or export validation failed. Aliases:validation failure
`verification check` (TERM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
`view` (TERM-VIEW)	Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.
`view projection` (FU-VIEW-PROJECTION)	Builds view-specific projections for architecture communication and traceability outputs. Aliases:projection

5. Architecture Decision Records

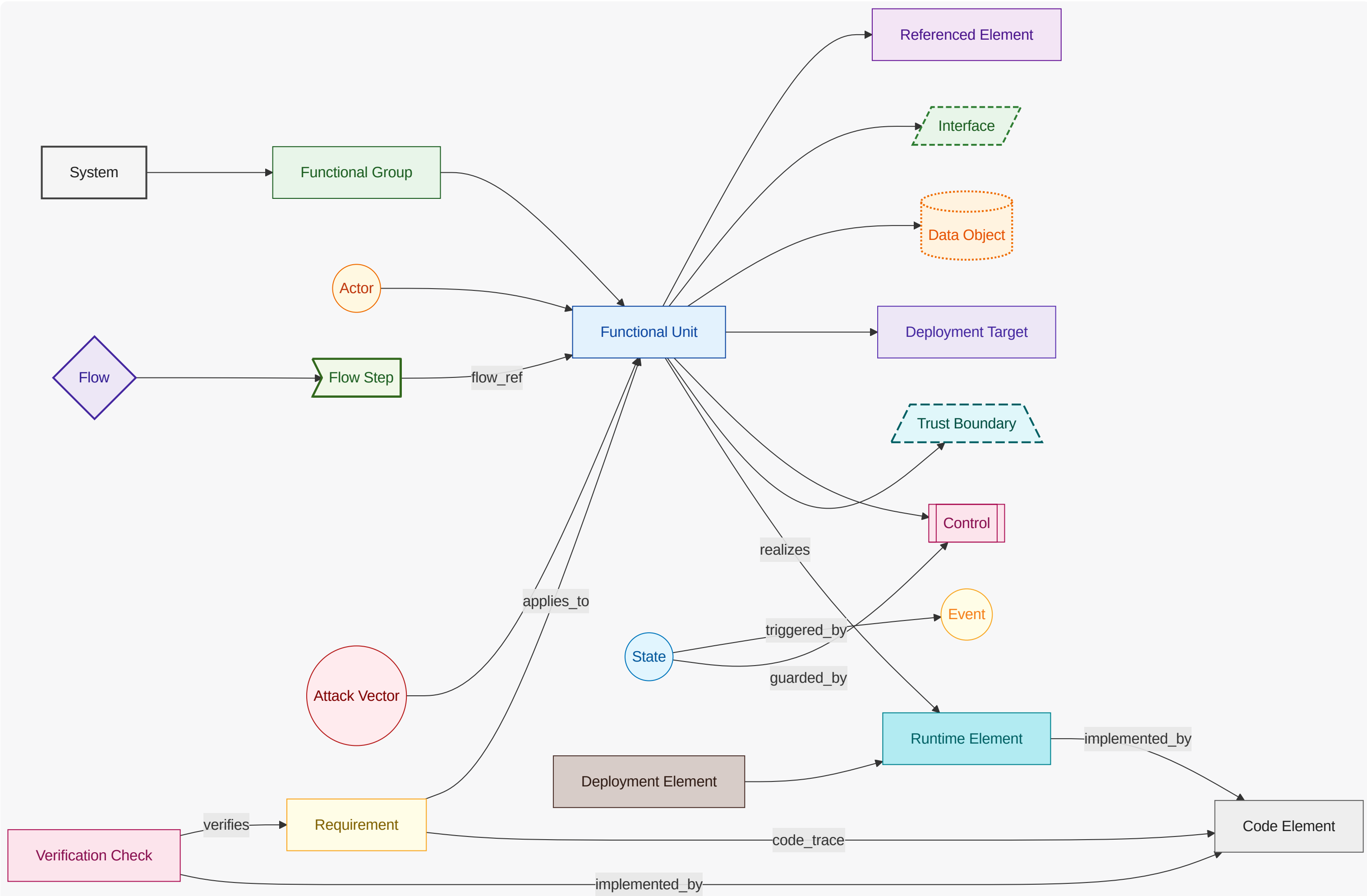
Architecture decision records are generated separately and linked here as a compact index.

ID	Title	Status	Date
ADR-EMG-001	Generate architecture decision records separately from the main architecture document	accepted	2026-04-29
ADR-EMG-002	Keep architecture decision records model-authored and separate from code links	accepted	2026-04-29

ID	Title	Status	Date
ADR-EMG-003	Render Gemara as a first-class output using the official go-gemara SDK types	accepted	2026-06-27
ADR-EMG-004	Source the Gemara evaluation log from authored control verifications, with pass and fail outcomes	accepted	2026-06-27
ADR-EMG-005	Keep hand-written OSCAL SSP/AR/POA&M; add Gemara-to-OSCAL only as an additive bridge	accepted	2026-06-27
ADR-EMG-006	Synthesize Gemara structure that the model does not author, with explicit fallbacks	accepted	2026-06-27
ADR-EMG-007	Attach trace and model-link markers to package-level var and const declarations	accepted	2026-06-27
ADR-EMG-008	Model multi-repository work as a recursive system-of-systems	accepted	2026-06-27
ADR-EMG-009	Subsystem references are downward-only with parent-owned allocation	accepted	2026-06-27
ADR-EMG-010	Resolve subsystem references from local subdirectories first	accepted	2026-06-27
ADR-EMG-011	Requirements carry no tiers; delegation is the link	accepted	2026-06-27
ADR-EMG-012	Trace links are existence-checked, scoped to the owning model, and exported as a matrix	accepted	2026-06-27
ADR-EMG-013	CI enforces the generation gates and artifact freshness	accepted	2026-06-28

5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



6. Architecture Intent View

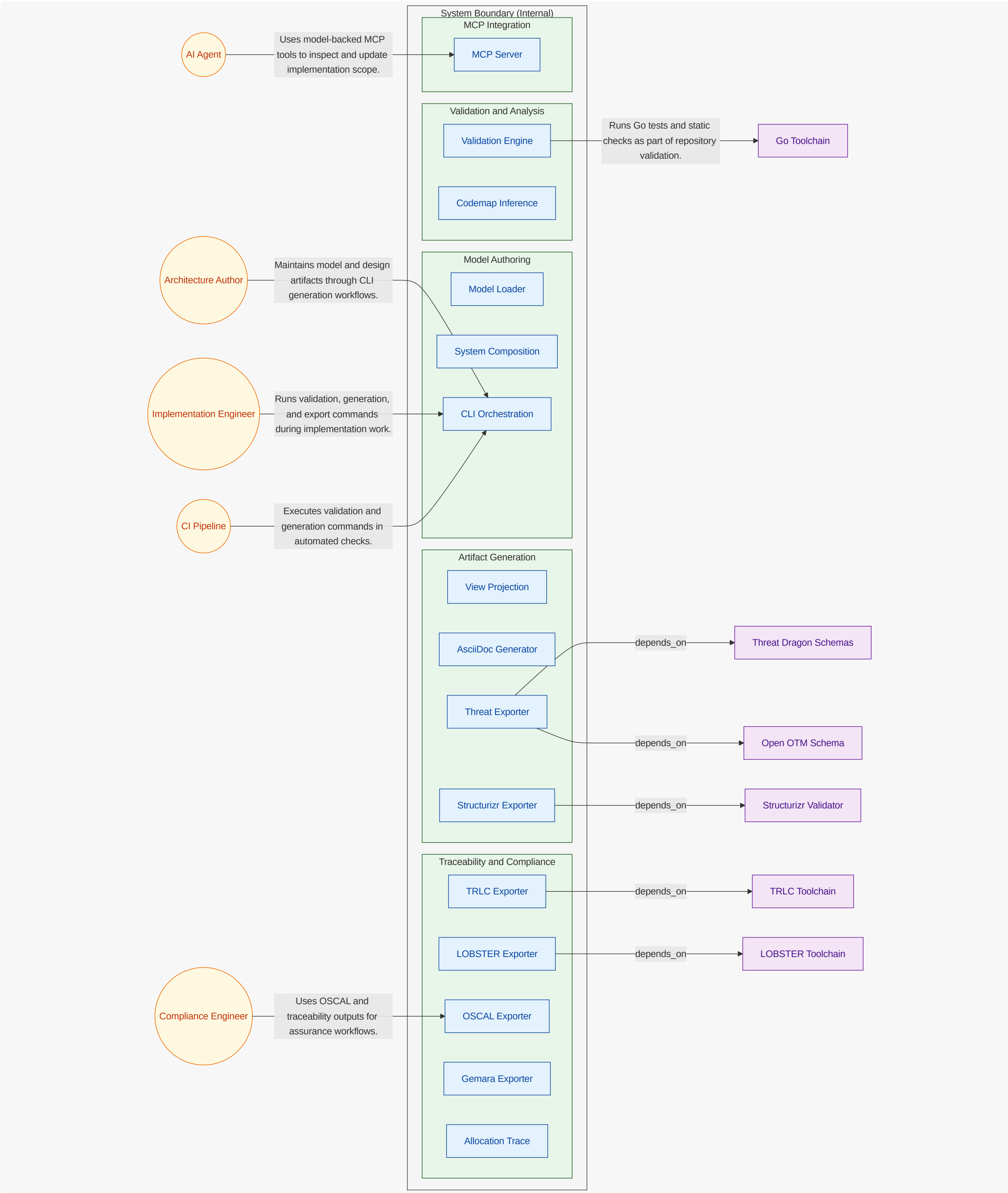
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

6.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

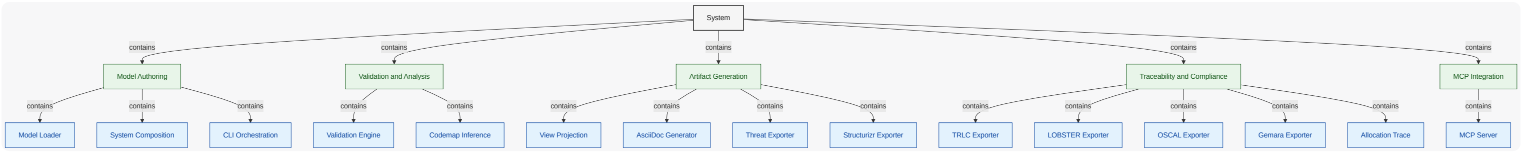
6.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



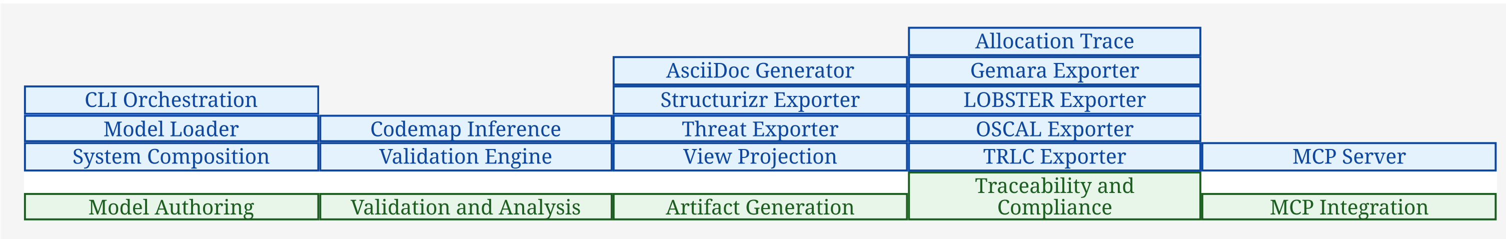
6.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.


System
contains
Model Authoring
Validation and Analysis
Artifact Generation
Traceability and Compliance
MCP Integration
Model Loader
System Composition
CLI Orchestration
Validation Engine
Codemap Inference
View Projection
AsciiDoc Generator
Threat Exporter
Structurizr Exporter
TRLC Exporter
LOBSTER Exporter
OSCAL Exporter
Gemara Exporter
Allocation Trace
MCP Server

6.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.



6.5. Functional Groups and Units (View Scoped)

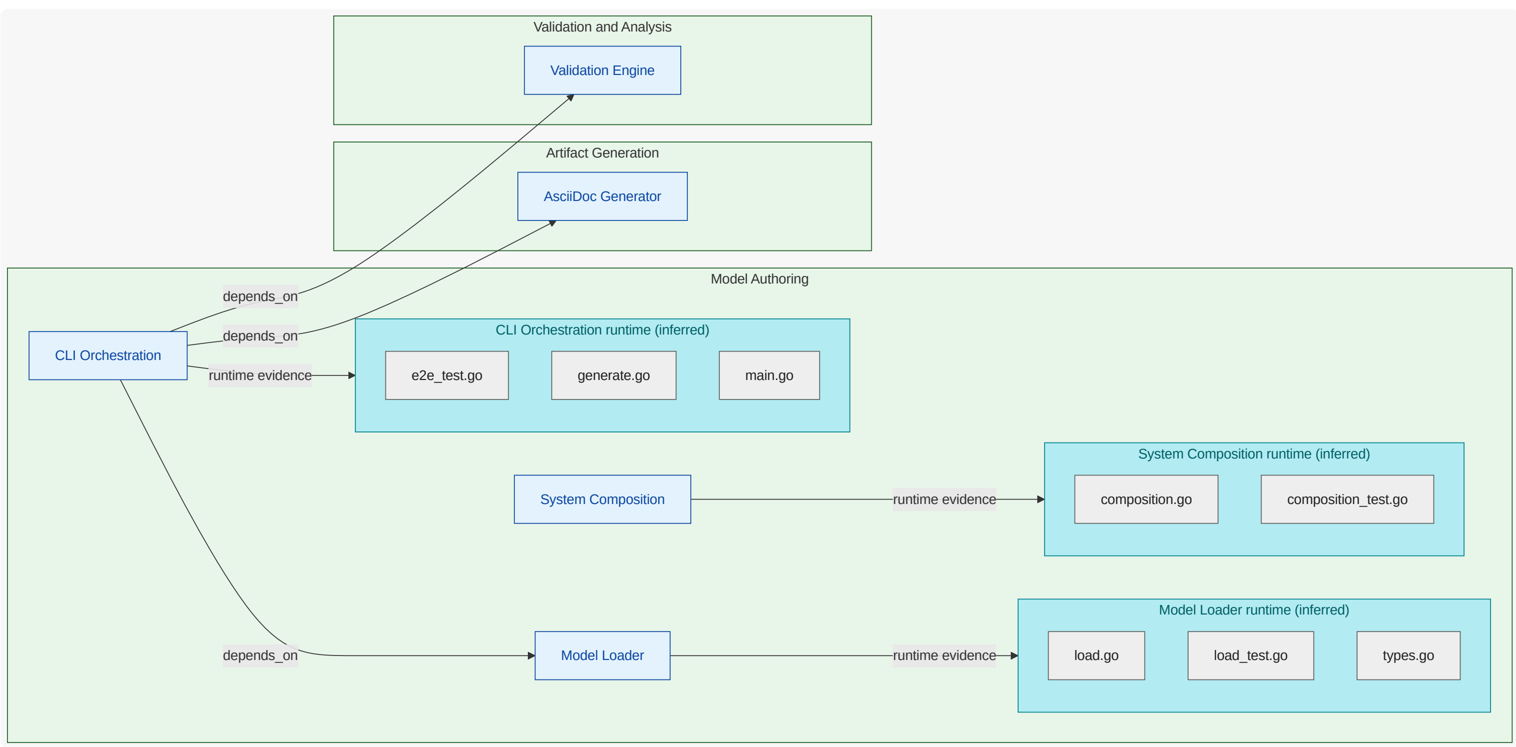
6.5.1. Model Authoring (FG-MODEL-AUTHORING)

Inputs and `model` loading responsibilities.

```
Maintains authored architecture, <<REF_IDX-ENGMODEL_TERM_REQUIREMENT,requirements>>, design, and <<REF_IDX-
ENGMODEL_TERM_CATALOG,catalog>> handling.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



CLI Orchestration (FU-CLI-ORCHESTRATION)

Command entrypoints orchestrating `model` load, `validation`, `generation`, and exports.

Owned Responsibility

functional responsibility for `cli orchestration`; includes decision and orchestration `flow` to `AsciiDoc Generator` and decision and orchestration `flow` to `Model Loader`

Key Collaborators

`AsciiDoc Generator`, `Model Loader`, `Validation Engine`

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-003`, `REQ-EMG-004`, `REQ-EMG-005`, `REQ-EMG-006`, `REQ-EMG-009`, `REQ-EMG-013`, `REQ-EMG-014`, `REQ-EMG-015`

Model Loader (FU-MODEL-LOADER)

Loads and normalizes architecture, `catalog`, `requirements`, and `design documents`.

Owned Responsibility

functional responsibility for `model loader`

Key Collaborators

none

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-016`, `REQ-EMG-027`

System Composition (FU-SYSTEM-COMPOSITION)

Resolves downward subsystem references from local subdirectories or external git repositories cloned into the .engmod/subsystems cache, and composes a federated system-of-systems `model`.

Owned Responsibility

functional responsibility for `system composition`

Key Collaborators

none

Linked Requirements

`REQ-EMG-016`, `REQ-EMG-017`, `REQ-EMG-018`, `REQ-EMG-019`, `REQ-EMG-027`

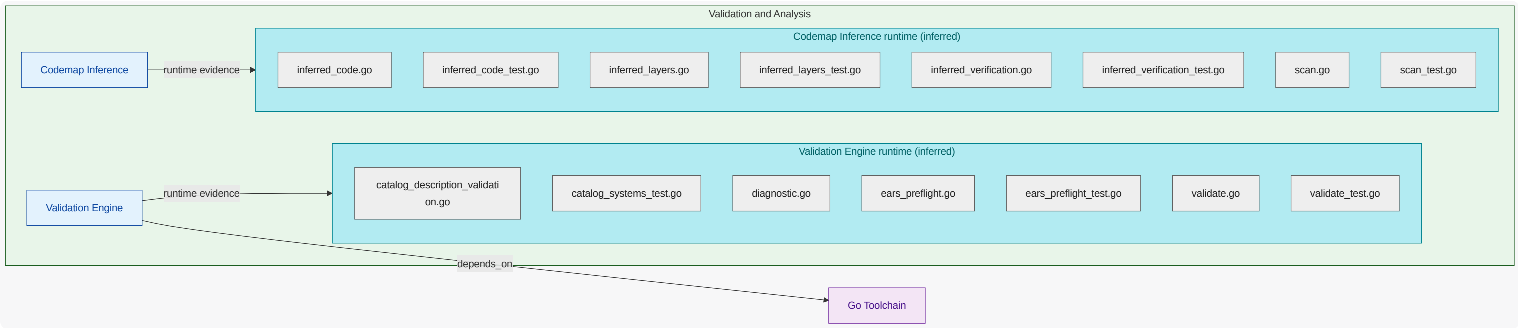
6.5.2. Validation and Analysis (FG-VALIDATION-ANALYSIS)

`Validation` and `inference` responsibilities.

```
Performs consistency checks and derives runtime, code, and verification <<REF_IDX-ENGMODEL_TERM_EVIDENCE,evidence>>.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Codemap Inference (FU-CODEMAP-INFERENCE)

Infers code/runtime ownership and verification links from source and tests.

Owned Responsibility

functional responsibility for `codemap inference`

Key Collaborators

none

Linked Requirements

`REQ-EMG-010`, `REQ-EMG-012`, `REQ-EMG-028`, `REQ-EMG-030`

Validation Engine (FU-VALIDATION-ENGINE)

Validates authored entities, IDs, references, and mapping consistency.

Owned Responsibility

functional responsibility for `validation engine`; includes decision and orchestration `flow` to `Go Toolchain`

Key Collaborators

`Go Toolchain`

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-009`, `REQ-EMG-011`, `REQ-EMG-018`, `REQ-EMG-021`, `REQ-EMG-022`, `REQ-EMG-023`, `REQ-EMG-026`, `REQ-EMG-028`, `REQ-EMG-029`

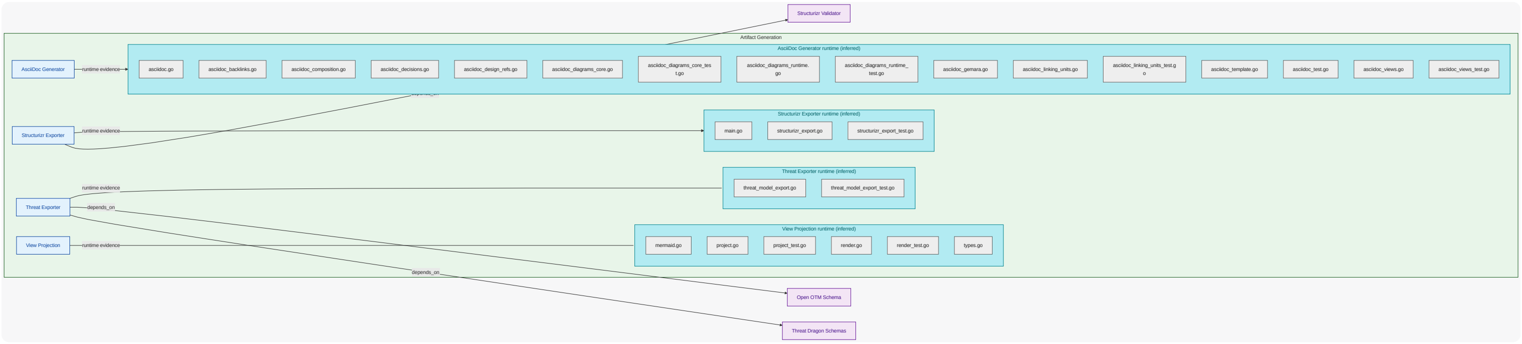
6.5.3. Artifact Generation (FG-ARTIFACT-GENERATION)

Publication and exchange `artifact generation`.

```
Produces AsciiDoc publication docs and external modeling outputs.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



AsciiDoc Generator (FU-ASCIIDOC-GENERATOR)

Renders architecture publication docs and `view` narratives for human consumption.

Owned Responsibility

functional responsibility for `asciidoc generator`

Key Collaborators

none

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-003`, `REQ-EMG-012`, `REQ-EMG-014`, `REQ-EMG-024`, `REQ-EMG-025`, `REQ-EMG-026`

Structurizr Exporter (FU-STRUCTURIZR-EXPORTER)

Emits `Structurizr DSL` and deployment-aware `model` views.

Owned Responsibility

functional responsibility for `structurizr exporter`; includes decision and orchestration `flow` to `Structurizr Validator`

Key Collaborators

`Structurizr Validator`

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-005`, `REQ-EMG-011`

Threat Exporter (FU-THREAT-EXPORTER)

Exports `Threat Dragon` and Open OTM `model` artifacts.

Owned Responsibility

functional responsibility for `threat exporter`; includes decision and orchestration `flow` to `Open OTM Schema` and decision and orchestration `flow` to `Threat Dragon Schemas`

Key Collaborators

`Open OTM Schema`, `Threat Dragon Schemas`

Linked Requirements

`REQ-EMG-001`, `REQ-EMG-004`, `REQ-EMG-011`

View Projection (FU-VIEW-PROJECTION)

Builds `projection` graphs for architecture, traceability, security, and `flow` views.

Owned Responsibility

functional responsibility for `view projection`

Key Collaborators

none

Linked Requirements

`REQ-EMG-003`, `REQ-EMG-024`

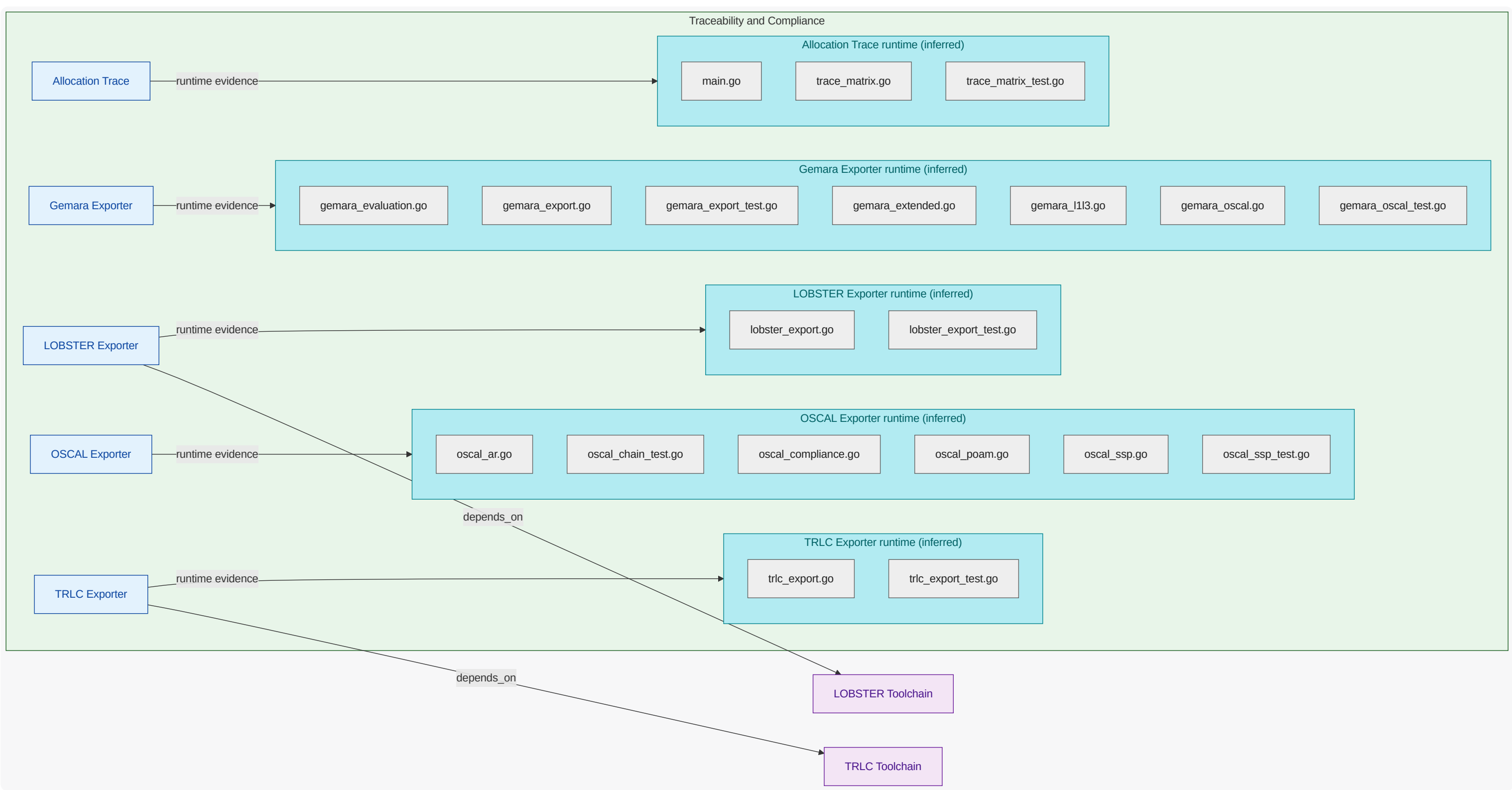
6.5.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)

Trace and `compliance` export responsibilities.

```
Produces <<REF_IDX-CATALOG_TRLC,TRLC>>, <<REF_IDX-CATALOG_LOBSTER,LOBSTER>>, and OSCAL outputs from modeled content.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Allocation Trace (FU-ALLOCATION-TRACE)

Materializes parent-to-subsystem `allocation` and bidirectional traceability across composed systems, without modifying subsystem models.

Owned Responsibility

functional responsibility for `allocation trace`

Key Collaborators

none

Linked Requirements

`REQ-EMG-012` , `REQ-EMG-019` , `REQ-EMG-020` , `REQ-EMG-021` , `REQ-EMG-022` , `REQ-EMG-023` , `REQ-EMG-025` , `REQ-EMG-026` , `REQ-EMG-028` , `REQ-EMG-029` , `REQ-EMG-030`

Gemara Exporter (FU-GEMARA-EXPORTER)

Renders OpenSSF Gemara L1-L7 catalogs and logs (vector, principle, guidance, `control` , capability, threat, `risk` , evaluation, enforcement, audit) with an optional OSCAL `catalog` and assessment-results bridge, validated by the Gemara SDK.

Owned Responsibility

functional responsibility for `gemara exporter`

Key Collaborators

none

Linked Requirements

`REQ-EMG-015`

LOBSTER Exporter (FU-LOBSTER-EXPORTER)

Generates `LOBSTER` traceability inputs and reports.

Owned Responsibility

functional responsibility for `lobster exporter` ; includes decision and orchestration `flow` to `LOBSTER Toolchain`

Key Collaborators

`LOBSTER Toolchain`

Linked Requirements

`REQ-EMG-006`

OSCAL Exporter (FU-OSCAL-EXPORTER)

Exports OSCAL System Security Plan (SSP), Assessment Results (AR), and Plan of Action and Milestones (POA&M).

Owned Responsibility

functional responsibility for `oscal exporter`

Key Collaborators

none

Linked Requirements

`REQ-EMG-001` , `REQ-EMG-013`

TRLC Exporter (FU-TRLC-EXPORTER)

Emits `TRLC` `model` and `requirement` artifacts.

Owned Responsibility

functional responsibility for `trlc exporter` ; includes decision and orchestration `flow` to `TRLC Toolchain`

Key Collaborators

`TRLC Toolchain`

Linked Requirements

`REQ-EMG-006`

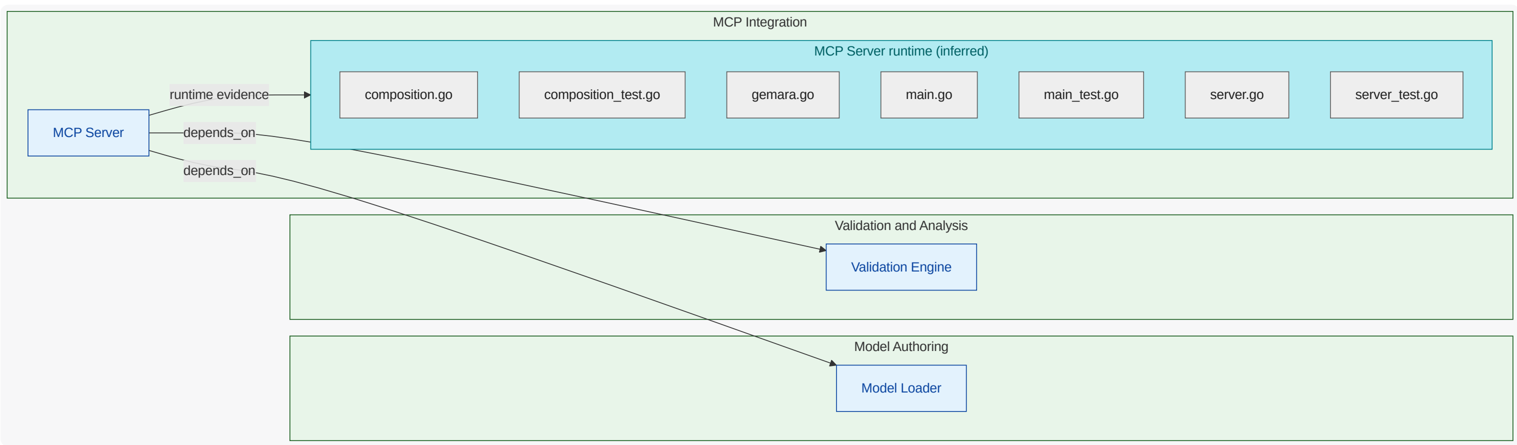
6.5.5. MCP Integration (FG-MCP-INTEGRATION)

AI-agent integration and runtime API responsibilities.

```
Exposes model-aware tools over MCP with strict input and error contracts.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



MCP Server (FU-MCP-SERVER)

Serves model-backed tools over MCP with path safety and structured errors.

Owned Responsibility

functional responsibility for `mcp server`; includes decision and orchestration `flow` to `Model Loader` and decision and orchestration `flow` to `Validation Engine`

Key Collaborators

`Model Loader`, `Validation Engine`

Linked Requirements

`REQ-EMG-007`, `REQ-EMG-008`, `REQ-EMG-015`

7. Security View

Show inferred exposure and dependency `risk` points aligned to unit boundaries, focused on attack paths and security `evidence`.

7.1. What This View Answers

- What are the main `trust boundaries` and attack paths?
- Which units are exposed and what security `evidence` exists?
- Where is the threat `model` incomplete?

7.2. Coverage Gaps

Coverage summary: 3/10 units have direct `evidence` coverage in this `view`.

- `AsciiDoc Generator` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `CLI Orchestration` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `LOBSTER Exporter` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `Structurizr Exporter` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `TRLC Exporter` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `Threat Exporter` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `Validation Engine` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

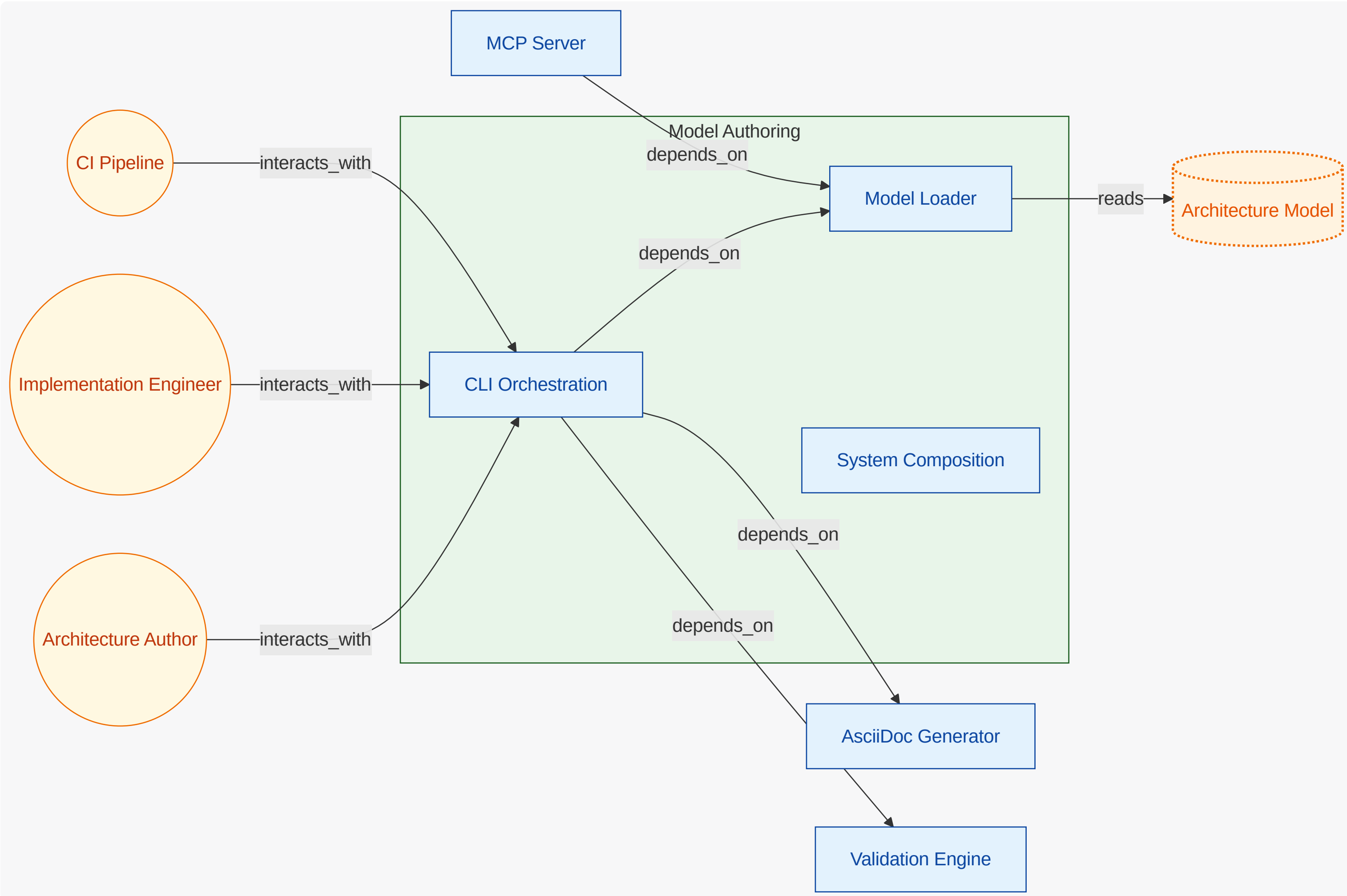
7.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/`events`) for unresolved observability `evidence`.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

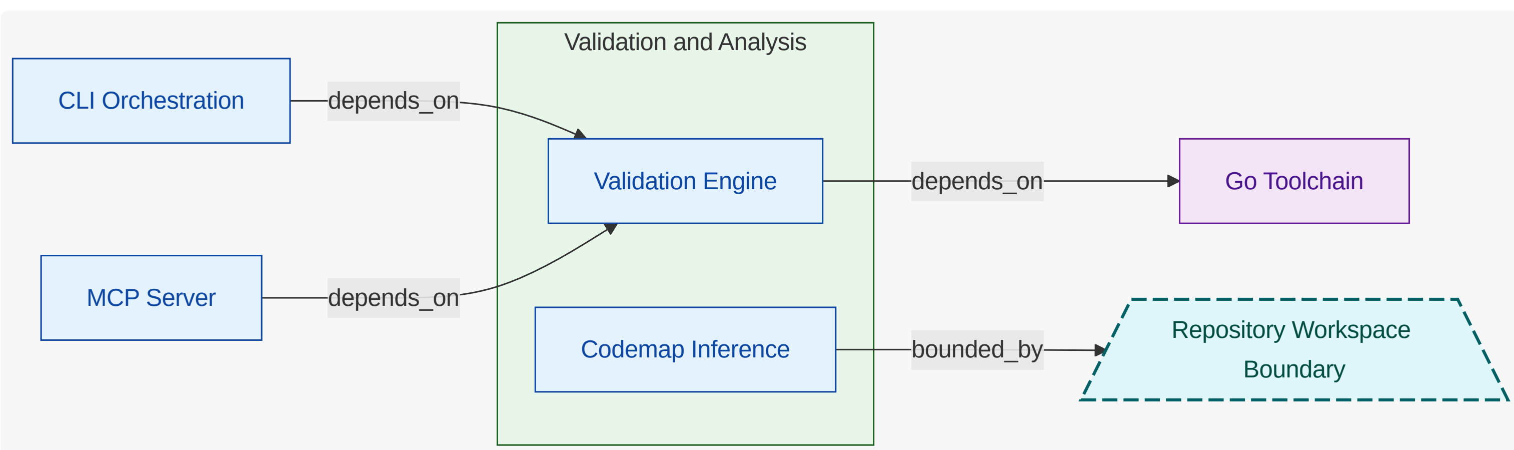
7.4. Threat Model Context Diagram

What these diagrams show: OWASP-style context views split by functional group. Each group diagram keeps owned functional units and owned security-relevant entities inside the functional-group container; connected actors, referenced elements, external functional units, and other external entities are rendered outside it.

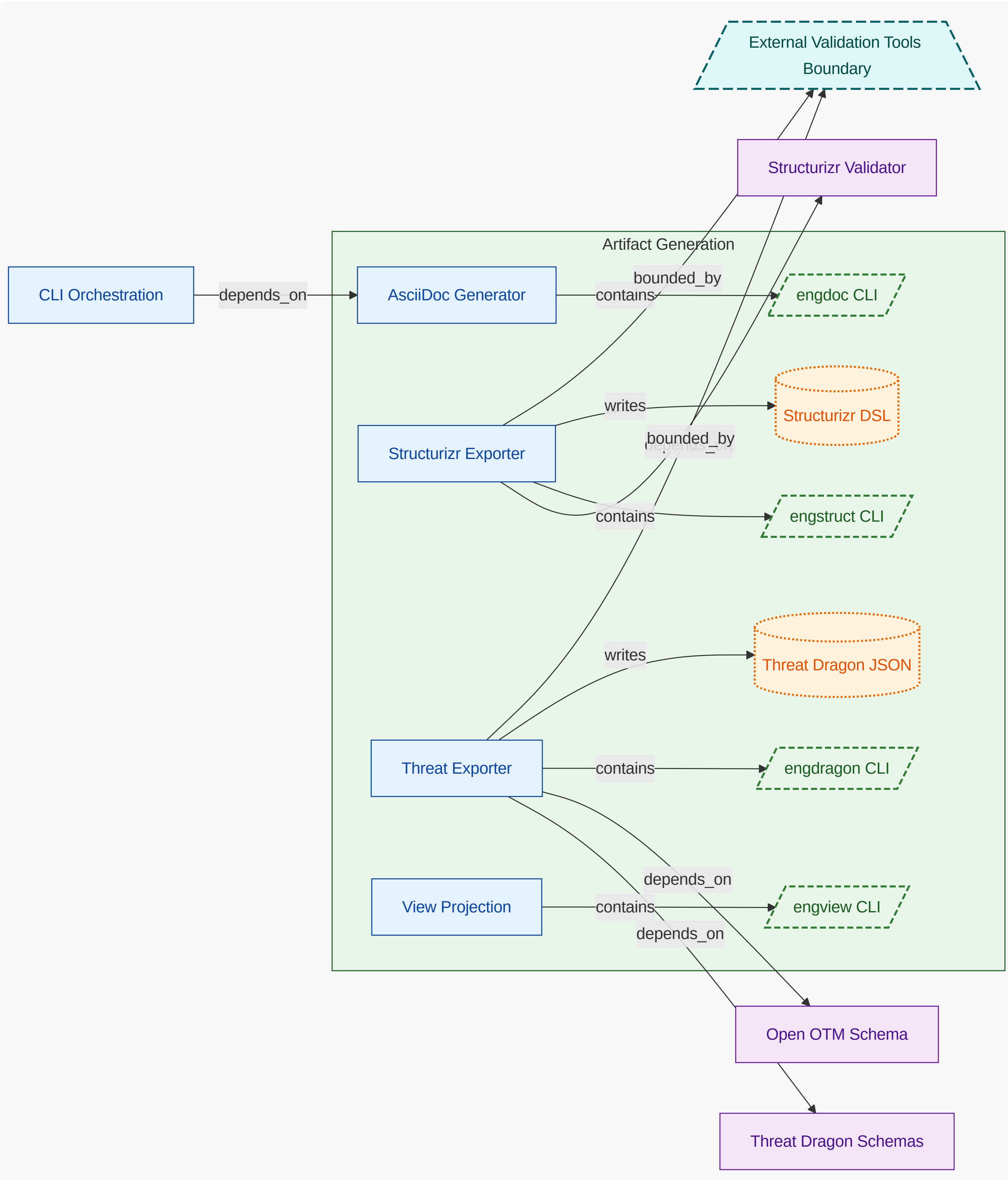
7.4.1. Model Authoring (FG-MODEL-AUTHORING)



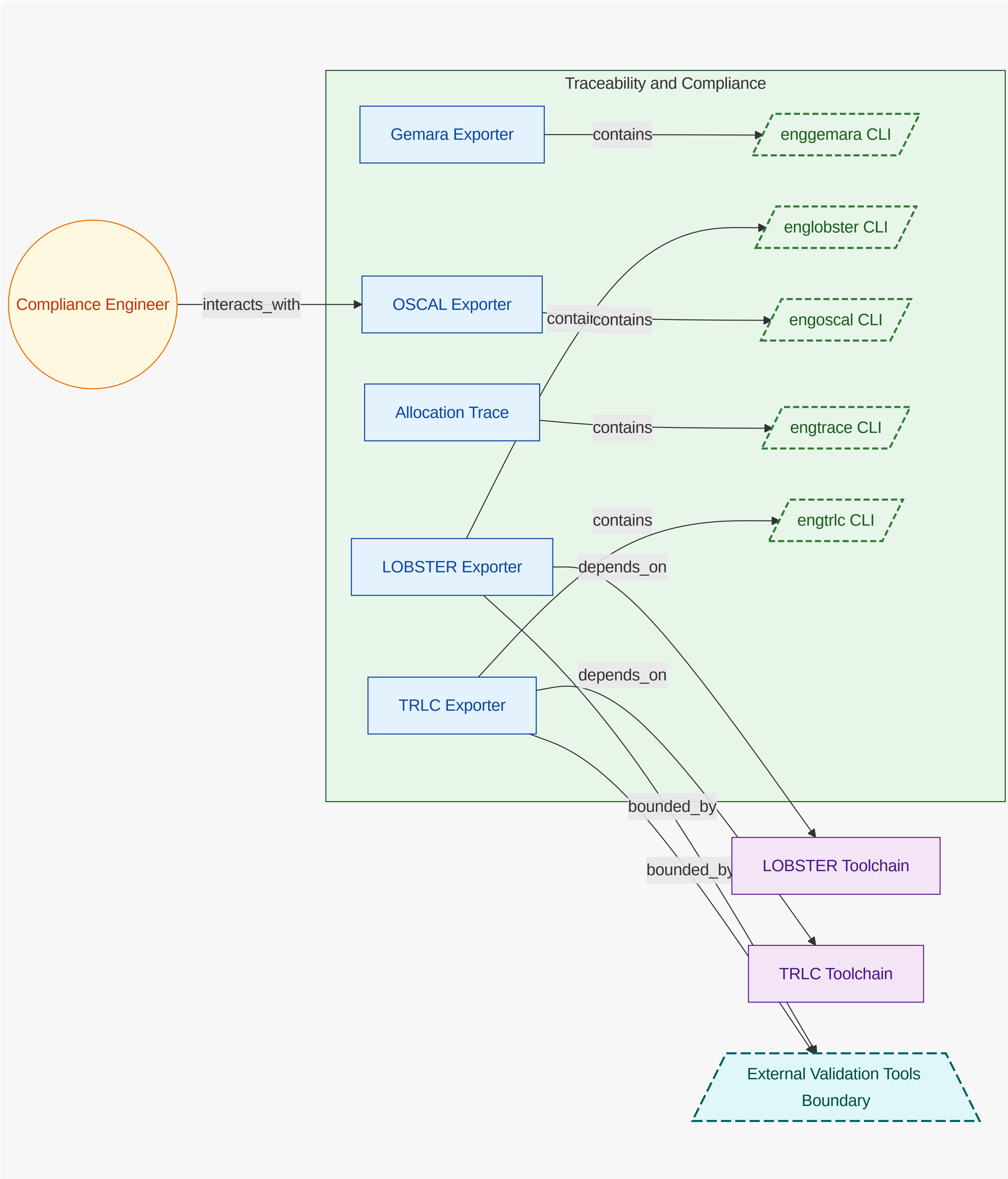
7.4.2. Validation and Analysis (FG-VALIDATION-ANALYSIS)



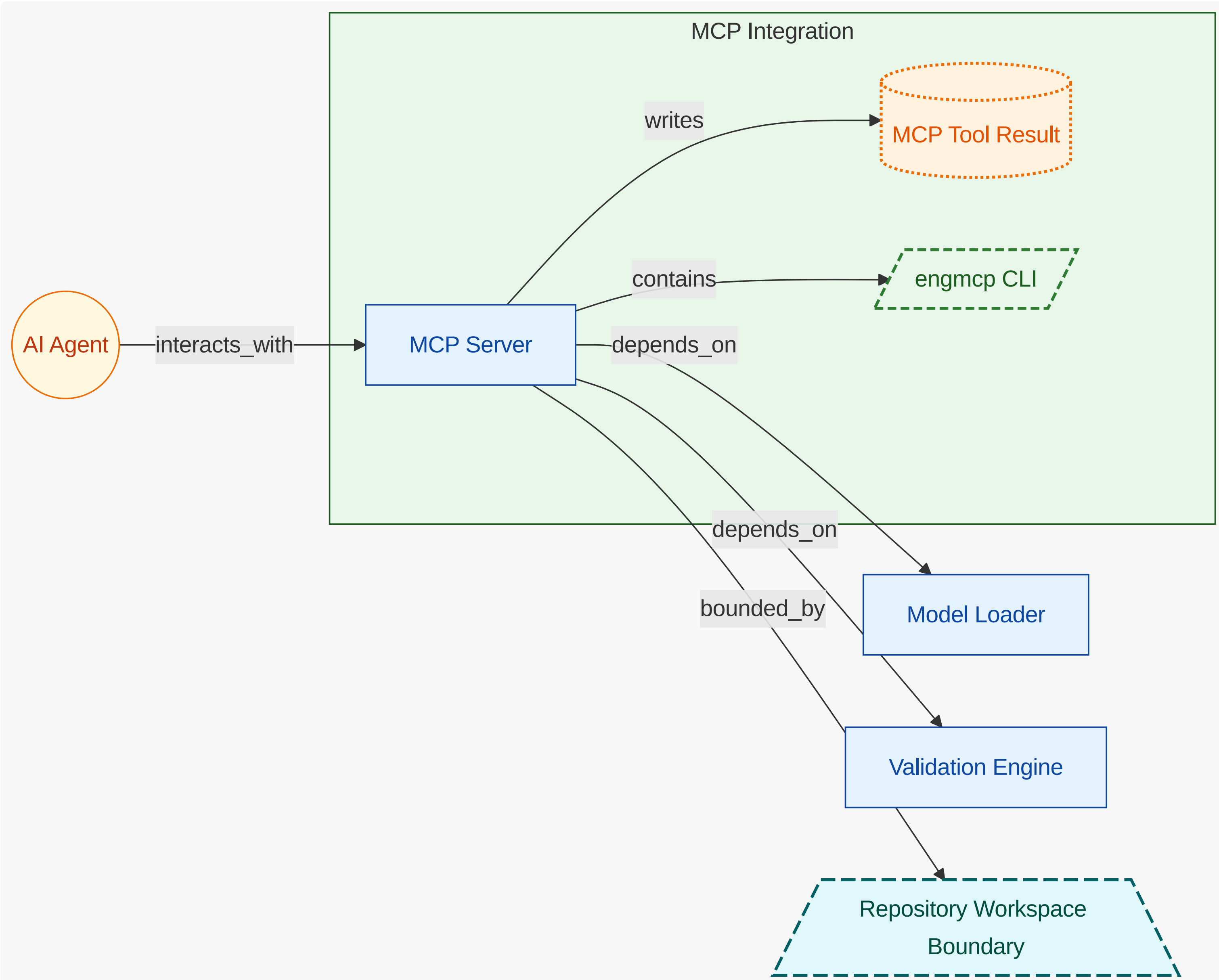
7.4.3. Artifact Generation (FG-ARTIFACT-GENERATION)



7.4.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)



7.4.5. MCP Integration (FG-MCP-INTEGRATION)



7.5. Threat Scenario Register

7.5.1. `Requirement` traceability drifts from implementation (TS-TRACEABILITY-DRIFT)

Code and tests change without corresponding `requirement` trace updates.

Field	Value
Status	mitigating
Owner	`Compliance Engineer`
Attack Vector	`Traceability Gap Drift`
Scope	`Codemap Inference`, `LOBSTER Exporter`, `TRLC Exporter`
Flows	none

Field	Value
Likelihood	medium
Impact	medium
Severity	medium
Risk	none
Controls	Requirement Traceability Coverage
Mitigations	none
Verifications	none
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

VER-TRACEABILITY-COVERAGE

Field	Value
Control	Requirement Traceability Coverage
Method	analysis
Status	partial
Owner	Compliance Engineer
Last Tested	unknown
Risks	none
Findings	Coverage depends on TRLC-LINKS completeness in source and tests.

7.5.2. MCP path traversal accesses non-repo files (TS-MCP-PATH-TRAVERSAL)

Untrusted MCP input attempts to escape repo root for sensitive file reads.

Field	Value
Status	mitigating
Owner	Architecture Author
Attack Vector	Path Traversal in MCP Calls
Scope	MCP Server
Flows	Model Change to Verified Artifacts Flow

Field	Value
Likelihood	medium
Impact	high
Severity	high
Risk	none
Controls	MCP Path Boundary Enforcement , Strict MCP Input Schemas
Mitigations	none
Verifications	none
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

VER-MCP-PATH-BOUNDARY

Field	Value
Control	MCP Path Boundary Enforcement
Method	test
Status	pass
Owner	Architecture Author
Last Tested	unknown
Risks	none
Findings	Path traversal rejected and error payload returned in MCP tests.

7.6. Threat Mitigation Coverage

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
TM-MCP-PATH-BOUNDARY	TS-MCP-PATH-TRAVERSAL	MCP Path Boundary Enforcement	implemented	high	Architecture Author	VER-MCP-PATH-BOUNDARY	
TM-TRACE-COVERAGE	TS-TRACEABILITY-DRIFT	Requirement Traceability Coverage	implemented	medium	Compliance Engineer	none	

7.7. Security Flow Semantics

ID	Title	Kind	Direction	Frequency	Source	Destination	Protocol
FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS	Model Change to Verified Artifacts Flow	control	internal	on-demand	Implementation Engineer	CLI Orchestration	cli

ID	Auth	Encryption	Integrity	Boundary Crossing	Trust Boundary	Threats	Data
FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS	repository-access	none	git-history	false	none	none	Architecture Model , MCP Tool Result , Structurizr DSL , Threat Dragon JSON

7.8. Security Logging and Observability

Signal	Layer	Owner	Evidence
application security telemetry hooks	code	FU-ALLOCATION-TRACE	code cmd/engtrace/main.go
application security telemetry hooks	code	FU-ALLOCATION-TRACE	code trace_matrix.go
application security telemetry hooks	code	FU-ALLOCATION-TRACE	code trace_matrix_test.go
application security telemetry hooks	code	FU-RISK-SCORING	code examples/payments-engineering-sample/src/support/audit_envelope.ts
application security telemetry hooks	code	FU-VALIDATION-ENGINE	code catalog_description_validation.go
application security telemetry hooks	code	FU-VALIDATION-ENGINE	code catalog_systems_test.go

Signal	Layer	Owner	Evidence
application security telemetry hooks	code	unresolved	code examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts
application security telemetry hooks	code	unresolved	code examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts
application security telemetry hooks	code	unresolved	code examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts
fraud and abuse detection code paths	code	FU-RISK-SCORING	code examples/payments-engineering-sample/src/risk_scorer.ts
fraud and abuse detection code paths	code	unresolved	code examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs

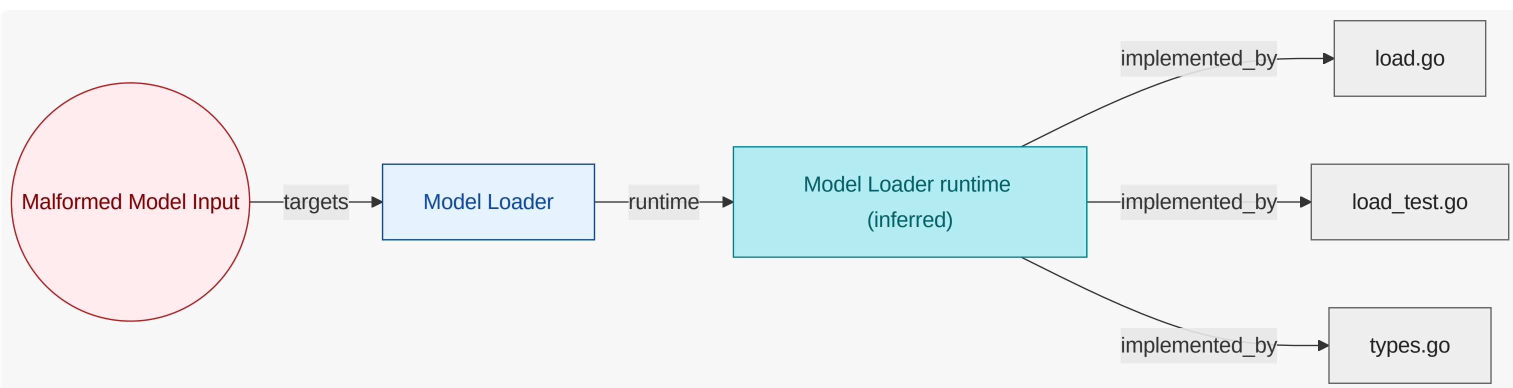
7.9. Attack Vector Chapters

7.9.1. `Malformed Model Input` (AV-MALFORMED-MODEL-INPUT)

`Invalid` or ambiguous `model` content causing incorrect parsing or graph interpretation.

Mitigated by: none

Trust boundaries: none



Model Loader (FU-MODEL-LOADER)

Rejects `invalid` parsing `states` and unresolved core `model` references.

Attack Surface

none

Security Evidence Present

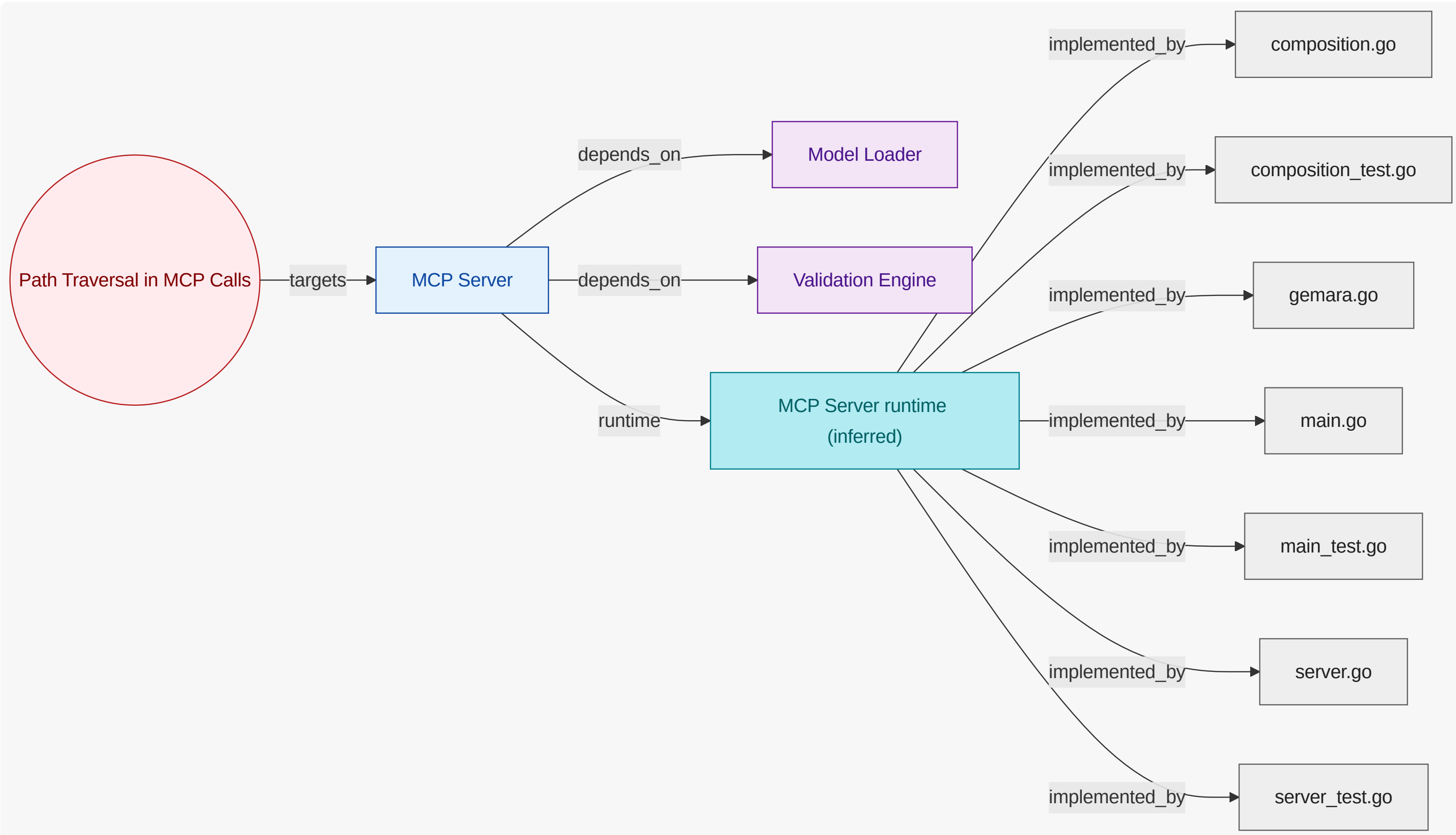
code modules: `model`

7.9.2. `Path Traversal in MCP Calls` (AV-PATH-TRAVERSAL-IN-MCP)

Attempts to read files outside repository root via MCP path arguments.

Mitigated by: MCP Path Boundary Enforcement, Strict MCP Input Schemas

Trust boundaries: Repository Workspace Boundary



MCP Server (FU-MCP-SERVER)

Enforces strict schemas, panic containment, and repo-root path constraints.

Attack Surface

`Model Loader`, `Validation Engine`

Security Evidence Present

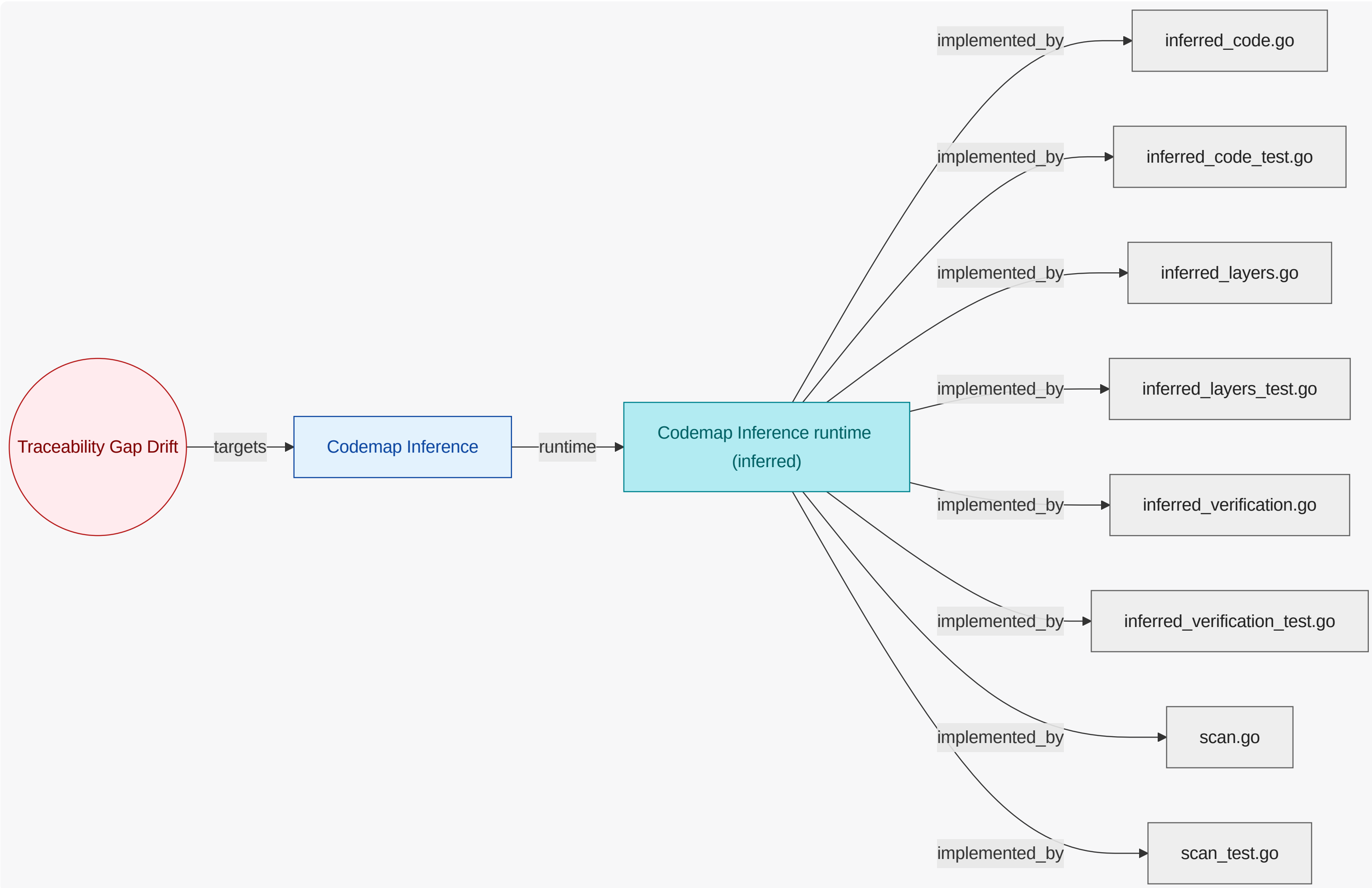
code modules: cmd/`engmcp`, mcp

7.9.3. `Traceability Gap Drift` (AV-TRACEABILITY-GAP-DRIFT)

`Requirement` and verification links diverge from implementation over time.

Mitigated by: Requirement Traceability Coverage

Trust boundaries: Repository Workspace Boundary



Codemap Inference (FU-CODEMAP-INFERENCE)

Limits `inference` scope to trusted repo paths and `evidence` formats.

Attack Surface

none

Security Evidence Present

code modules: codemap, inferred_code, inferred_code_test, inferred_layers, inferred_layers_test, inferred_verification, inferred_verification_test

8. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit `evidence` gaps.

8.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code `evidence` and which do not?
- Where are confidence or completeness gaps in trace coverage?

8.2. Coverage Gaps

Coverage summary: 8/8 units have direct `evidence` coverage in this `view`.

- `AsciiDoc Generator` has no direct inferred runtime `evidence` yet.
- `CLI Orchestration` has no direct inferred runtime `evidence` yet.
- `Codemap Inference` has no direct inferred runtime `evidence` yet.
- `LOBSTER Exporter` has no direct inferred runtime `evidence` yet.
- `MCP Server` has no direct inferred runtime `evidence` yet.
- `Model Loader` has no direct inferred runtime `evidence` yet.
- `TRLC Exporter` has no direct inferred runtime `evidence` yet.
- `Validation Engine` has no direct inferred runtime `evidence` yet.

8.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code `evidence`.

9. State Lifecycle View

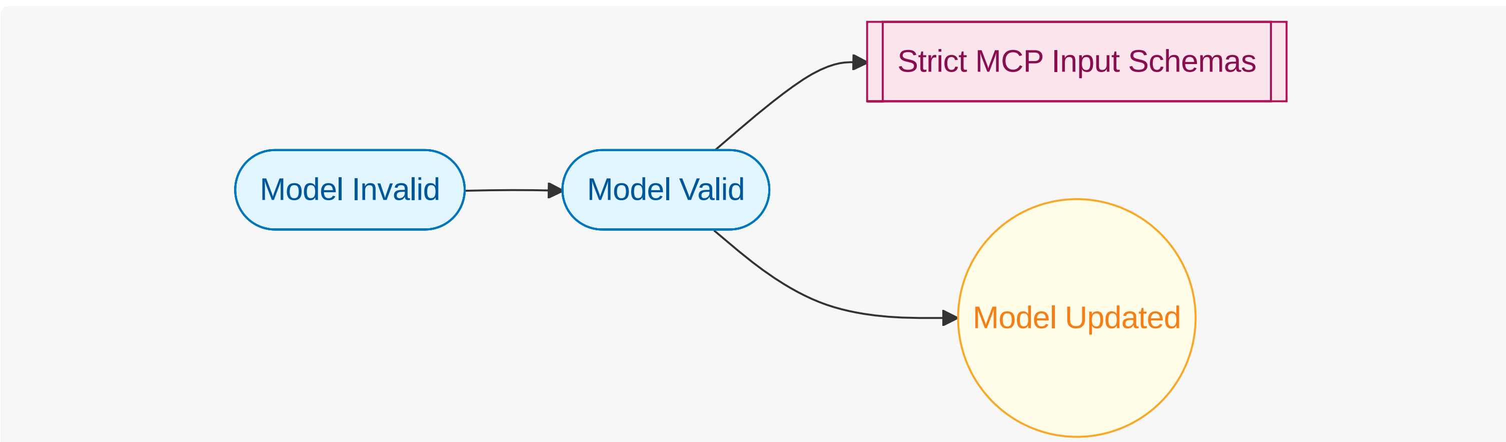
Show workflow `state` transitions, ownership of transitions, and key exceptional paths.

9.1. What This View Answers

- What `states` exist and which `events` trigger transitions?
- Which unit owns each `state` transition?
- Where do automatic and manual flows diverge?

9.2. State Lifecycle Diagram

What this diagram shows: authored lifecycle states, triggering events, guard controls, trust boundaries, and lifecycle transitions for this view.



9.3. Coverage Gaps

Coverage summary: No `functional units` are in scope for this `view`.

- No major coverage gaps detected from current authored and inferred inputs.

9.4. Recommended Next Evidence Additions

- No immediate `evidence` additions are required for this `view`.

9.5. Functional Groups and Units (View Scoped)

10. Interaction Flow View

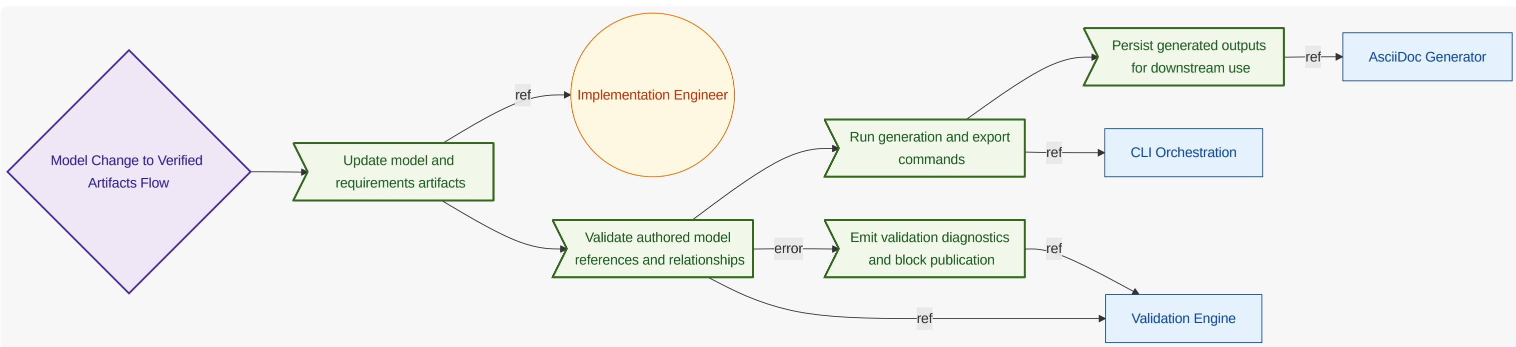
Show user-intent-to-outcome causal `flow`, including ordered steps, data movement, and error/async branches.

10.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

10.2. Interaction Flow Diagram

What this diagram shows: authored interaction flow steps, step ordering, async markers, and referenced actors/interfaces/units for this view.



10.3. Coverage Gaps

Coverage summary: 3/3 units have direct `evidence` coverage in this `view`.

- No major coverage gaps detected from current authored and inferred inputs.

10.4. Recommended Next Evidence Additions

- No immediate `evidence` additions are required for this `view`.

10.5. Functional Groups and Units (View Scoped)

10.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Functional Unit Columns 1-7

Requirement	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER	FU-MCP-SERVER
REQ-EMG-001		X	X				

Requirement	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER	FU-MCP-SERVER
REQ-EMG-003		X	X				
REQ-EMG-004			X				
REQ-EMG-005			X				
REQ-EMG-006			X			X	
REQ-EMG-007							X
REQ-EMG-008							X
REQ-EMG-009			X				
REQ-EMG-010				X			
REQ-EMG-011							
REQ-EMG-012	X	X		X			
REQ-EMG-013			X				
REQ-EMG-014		X	X				
REQ-EMG-015			X		X		X
REQ-EMG-016							
REQ-EMG-017							
REQ-EMG-018							
REQ-EMG-019	X						
REQ-EMG-020	X						
REQ-EMG-021	X						
REQ-EMG-022	X						
REQ-EMG-023	X						
REQ-EMG-024		X					
REQ-EMG-025	X	X					
REQ-EMG-026	X	X					
REQ-EMG-027							

Requirement	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER	FU-MCP-SERVER
REQ-EMG-028	X			X			
REQ-EMG-029	X						
REQ-EMG-030	X			X			

Functional Unit Columns 8-14

Requirement	FU-MODEL-LOADER	FU-OSCAL-EXPORTER	FU-STRUCTURIZR-EXPORTER	FU-SYSTEM-COMPOSITION	FU-THREAT-EXPORTER	FU-TRLC-EXPORTER	FU-VALIDATION-ENGINE
REQ-EMG-001	X	X	X		X		X
REQ-EMG-003							
REQ-EMG-004					X		
REQ-EMG-005			X				
REQ-EMG-006						X	
REQ-EMG-007							
REQ-EMG-008							
REQ-EMG-009							X
REQ-EMG-010							
REQ-EMG-011			X		X		X
REQ-EMG-012							
REQ-EMG-013		X					
REQ-EMG-014							
REQ-EMG-015							
REQ-EMG-016	X			X			
REQ-EMG-017				X			
REQ-EMG-018				X			X
REQ-EMG-019				X			

Requirement	FU-MODEL-LOADER	FU-OSCAL-EXPORTER	FU-STRUCTURIZR-EXPORTER	FU-SYSTEM-COMPOSITION	FU-THREAT-EXPORTER	FU-TRLC-EXPORTER	FU-VALIDATION-ENGINE
REQ-EMG-020							
REQ-EMG-021							X
REQ-EMG-022							X
REQ-EMG-023							X
REQ-EMG-024							
REQ-EMG-025							
REQ-EMG-026							X
REQ-EMG-027	X			X			
REQ-EMG-028							X
REQ-EMG-029							X
REQ-EMG-030							

Functional Unit Columns 15-15

Requirement	FU-VIEW-PROJECTION
REQ-EMG-001	
REQ-EMG-003	X
REQ-EMG-004	
REQ-EMG-005	
REQ-EMG-006	
REQ-EMG-007	
REQ-EMG-008	
REQ-EMG-009	
REQ-EMG-010	
REQ-EMG-011	
REQ-EMG-012	

Requirement	FU-VIEW-PROJECTION
REQ-EMG-013	
REQ-EMG-014	
REQ-EMG-015	
REQ-EMG-016	
REQ-EMG-017	
REQ-EMG-018	
REQ-EMG-019	
REQ-EMG-020	
REQ-EMG-021	
REQ-EMG-022	
REQ-EMG-023	
REQ-EMG-024	X
REQ-EMG-025	
REQ-EMG-026	
REQ-EMG-027	
REQ-EMG-028	
REQ-EMG-029	
REQ-EMG-030	

10.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2`	REQ-COF-001	not-run	none; No parsed test-result artifact for this `requirement` yet.
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8`	REQ-COF-002	not-run	none; No parsed test-result artifact for this `requirement` yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC	REQ-COF-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB	REQ-COF-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB	REQ-COF-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC	REQ-COF-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8	REQ-COF-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-E2E_TEST-BE56F8	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_TEST-F60765	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-PROJECT_TEST-2ECF72	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-RENDER_TEST-EECC23	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D	REQ-EMG-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374	REQ-EMG-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E	REQ-EMG-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRLC_EXPORT_TEST-DD1345	REQ-EMG-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-6B61AE	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-6B61AE	REQ-EMG-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_CODE_TEST-3CA82E	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-SCAN_TEST-B04F81	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D	REQ-EMG-011	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-011	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9	REQ-EMG-013	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_SSP_TEST-4B08EF	REQ-EMG-013	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_TEST-F60765	REQ-EMG-014	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_EXPORT_TEST-45C982	REQ-EMG-015	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA	REQ-EMG-015	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-016	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-016	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-017	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-020	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-022	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-023	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-025	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-027	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C	REQ-EMG-028	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C	REQ-EMG-029	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367	REQ-PAY-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163	REQ-PAY-002	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671	REQ-PAY-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD	REQ-PAY-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163	REQ-PAY-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4	REQ-PAY-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671	REQ-PAY-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD	REQ-PAY-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-002	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003	REQ-PRR-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F	REQ-PRR-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003	REQ-PRR-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4	REQ-PRR-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE	REQ-PRR-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCAN_TEST-B04F81	REQ-SAMPLE-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCAN_TEST-B04F81	REQ-SAMPLE-002	not-run	none; No parsed test-result artifact for this requirement yet.

11. Traceability Appendix

11.1. Requirement Details

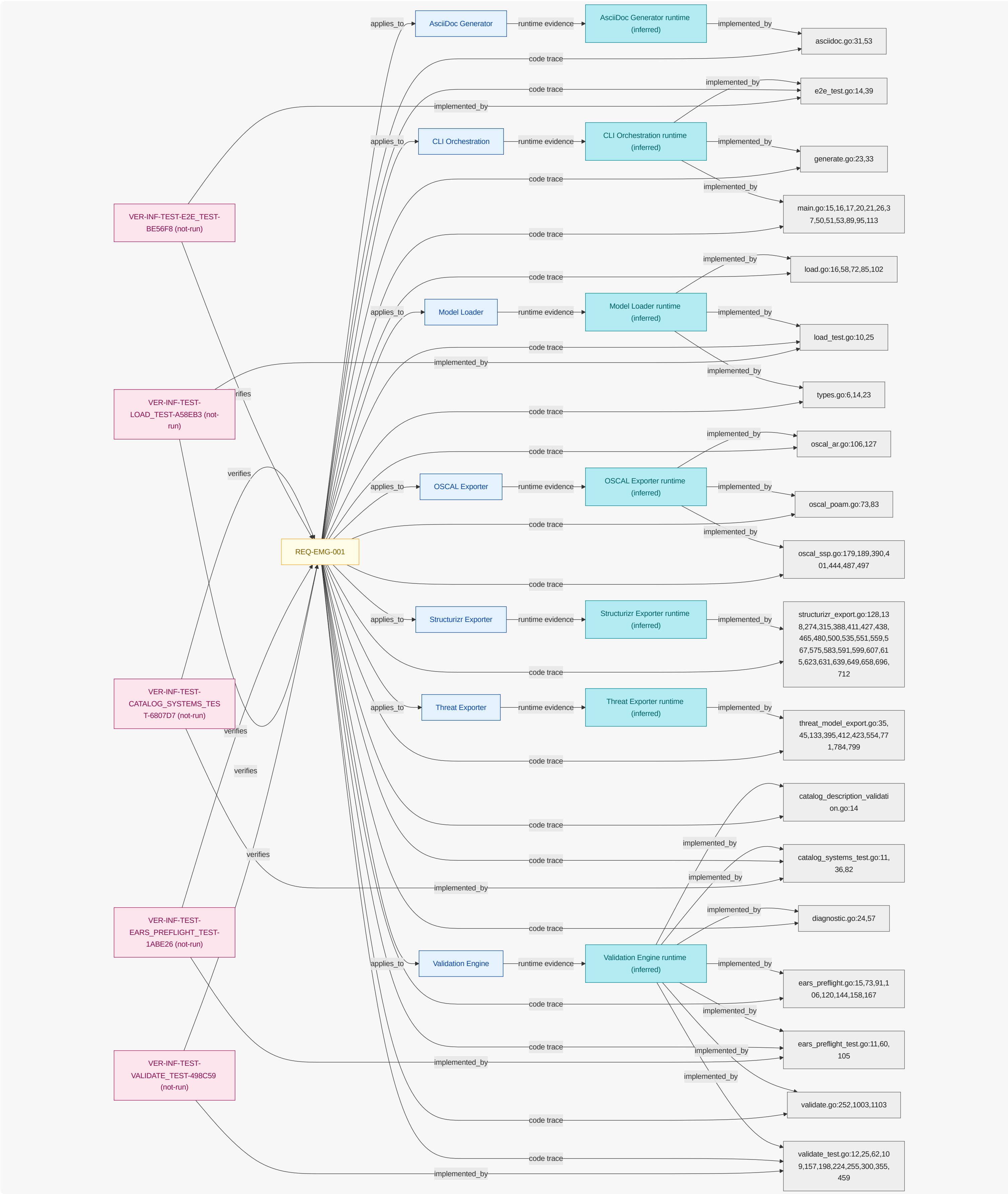
11.1.1. REQ-EMG-001

While `local development mode is active`, when `model updated event is received`, the `engineering model go` shall validate `architecture model` before `generation run is requested`.

> **NOTE** `Validation` must gate all `generation` flows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



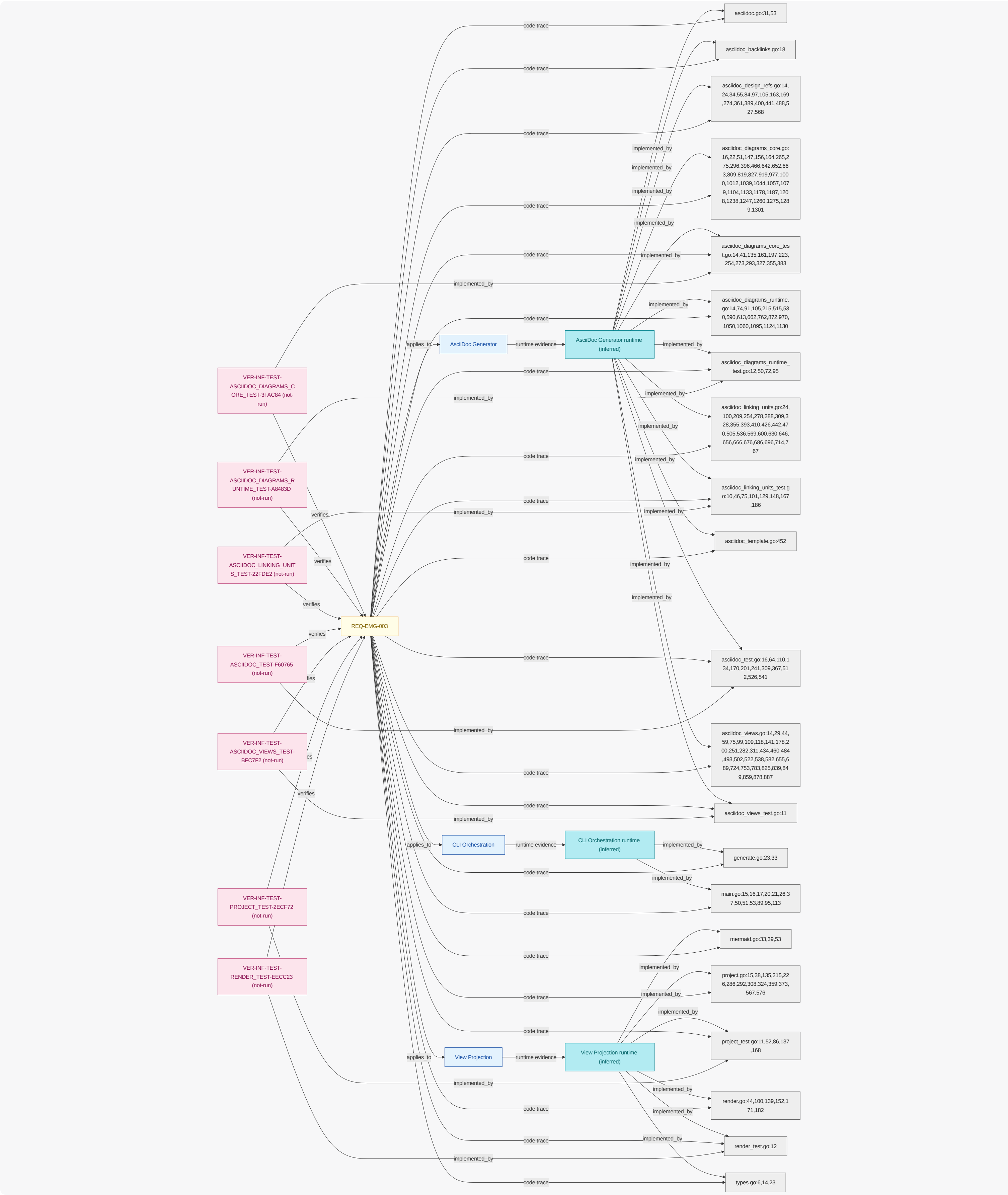
11.1.2. REQ-EMG-003

While `model is valid`, when `generation run is requested`, the `engineering model go` shall produce AsciiDoc architecture publication output.

NOTE	Human-readable architecture output remains required.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



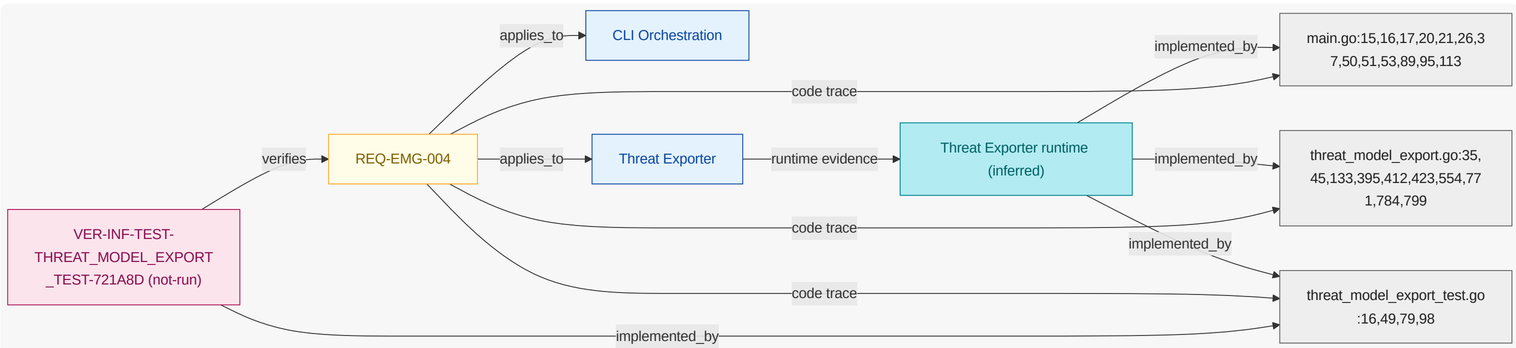
11.1.3. REQ-EMG-004

Where threat export is enabled, when generation run is requested, the engineering model go shall export threat dragon json artifact plus open threat model artifact.

NOTE Threat model interchange output contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



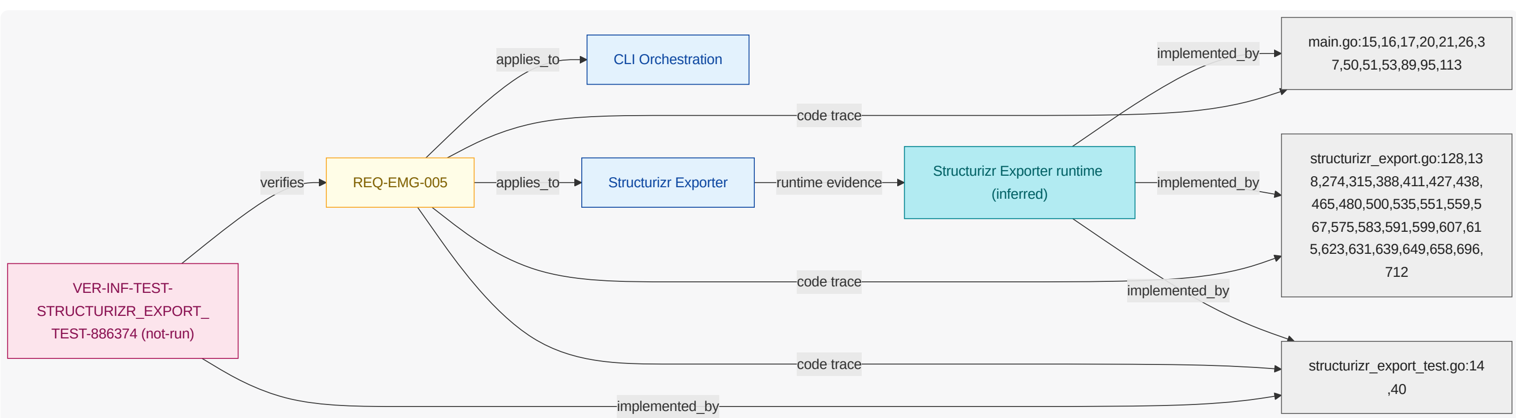
11.1.4. REQ-EMG-005

Where structurizr export is enabled, when generation run is requested, the engineering model go shall export structurizr dsl artifact.

NOTE Structurizr DSL is exported for C4 visualization; the exported DSL is validated best-effort by the validate-all gauntlet when a Structurizr validator is available.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



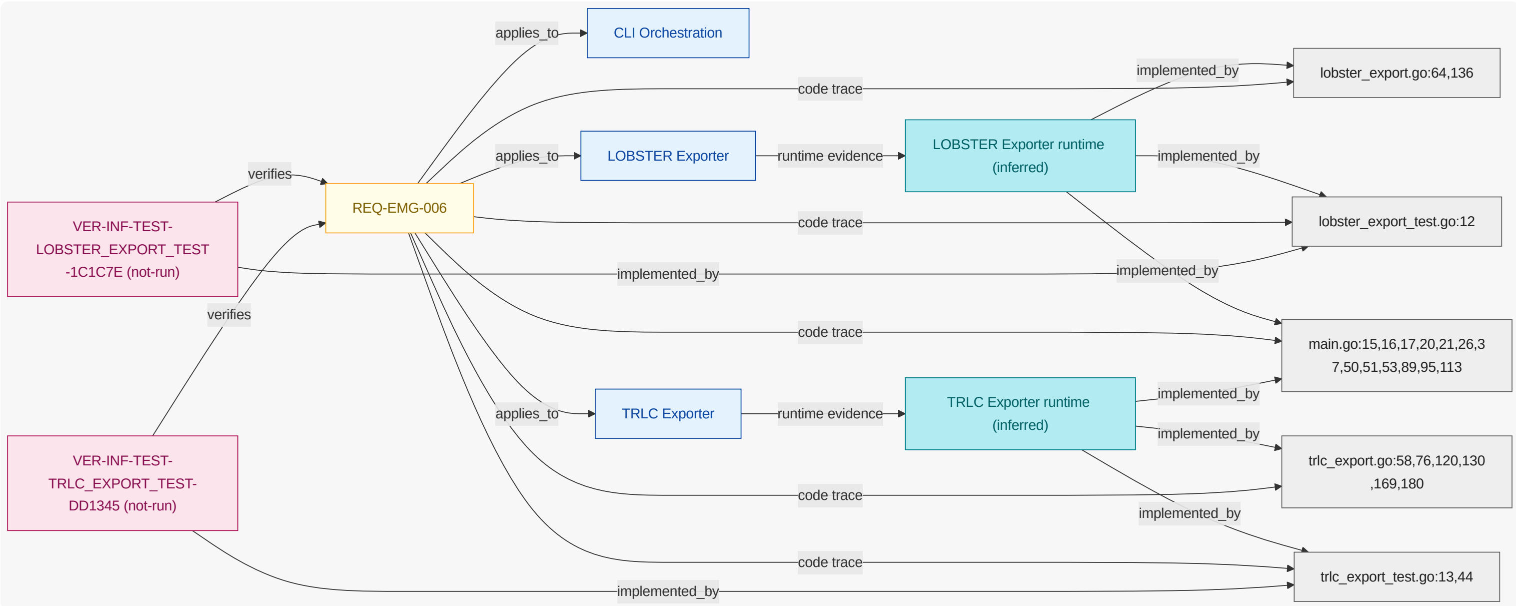
11.1.5. REQ-EMG-006

Where traceability export is enabled, when generation run is requested, the engineering model go shall produce trlc package and lobster trace report artifacts.

NOTE Traceability outputs required for assurance workflows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



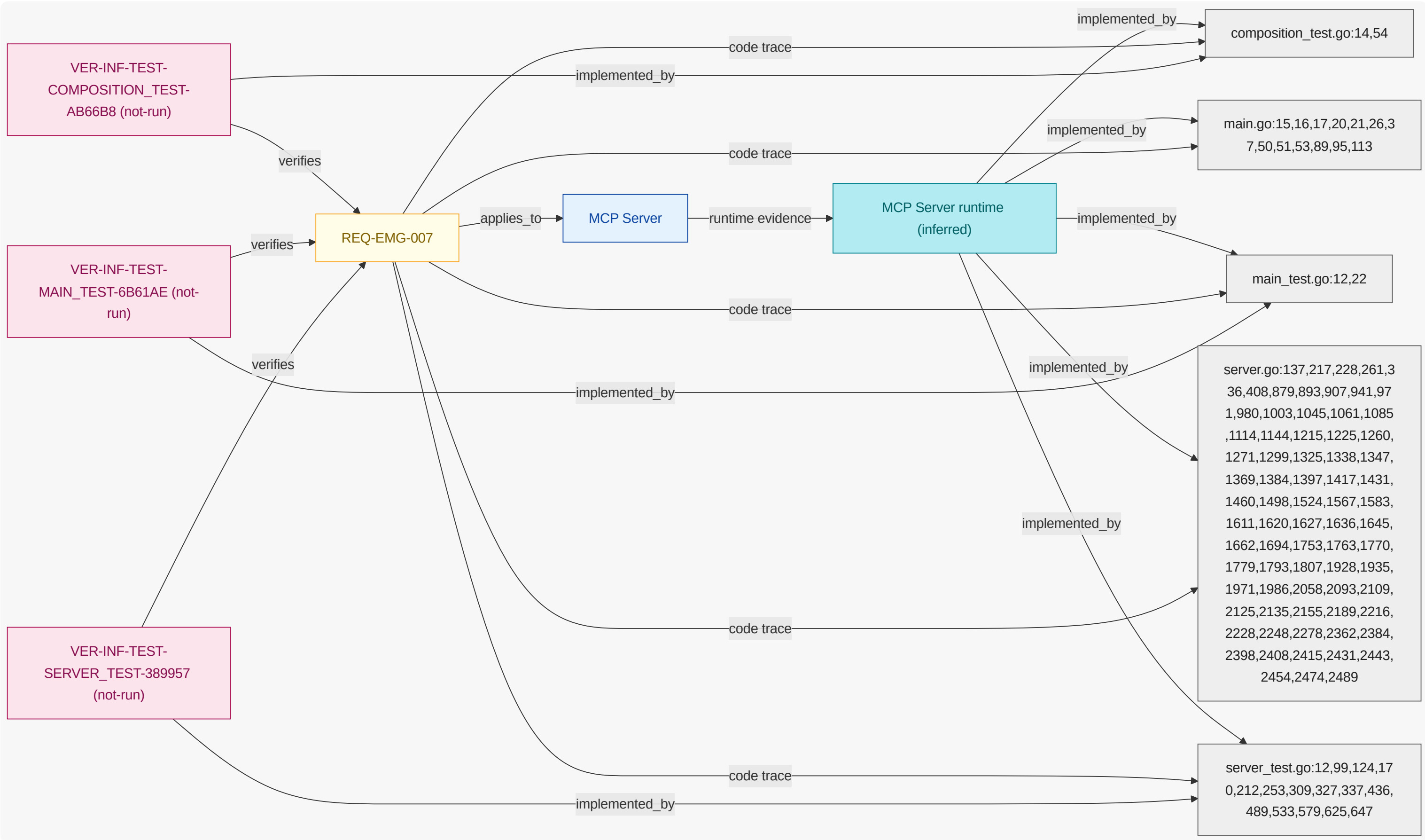
11.1.6. REQ-EMG-007

While `mcp server is enabled`, when `mcp tool call is received` and `catalog references resolve successfully`, the `engineering model go` shall return `mcp tool result` with schema version and structured error fields.

> **NOTE** MCP response contract stability `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



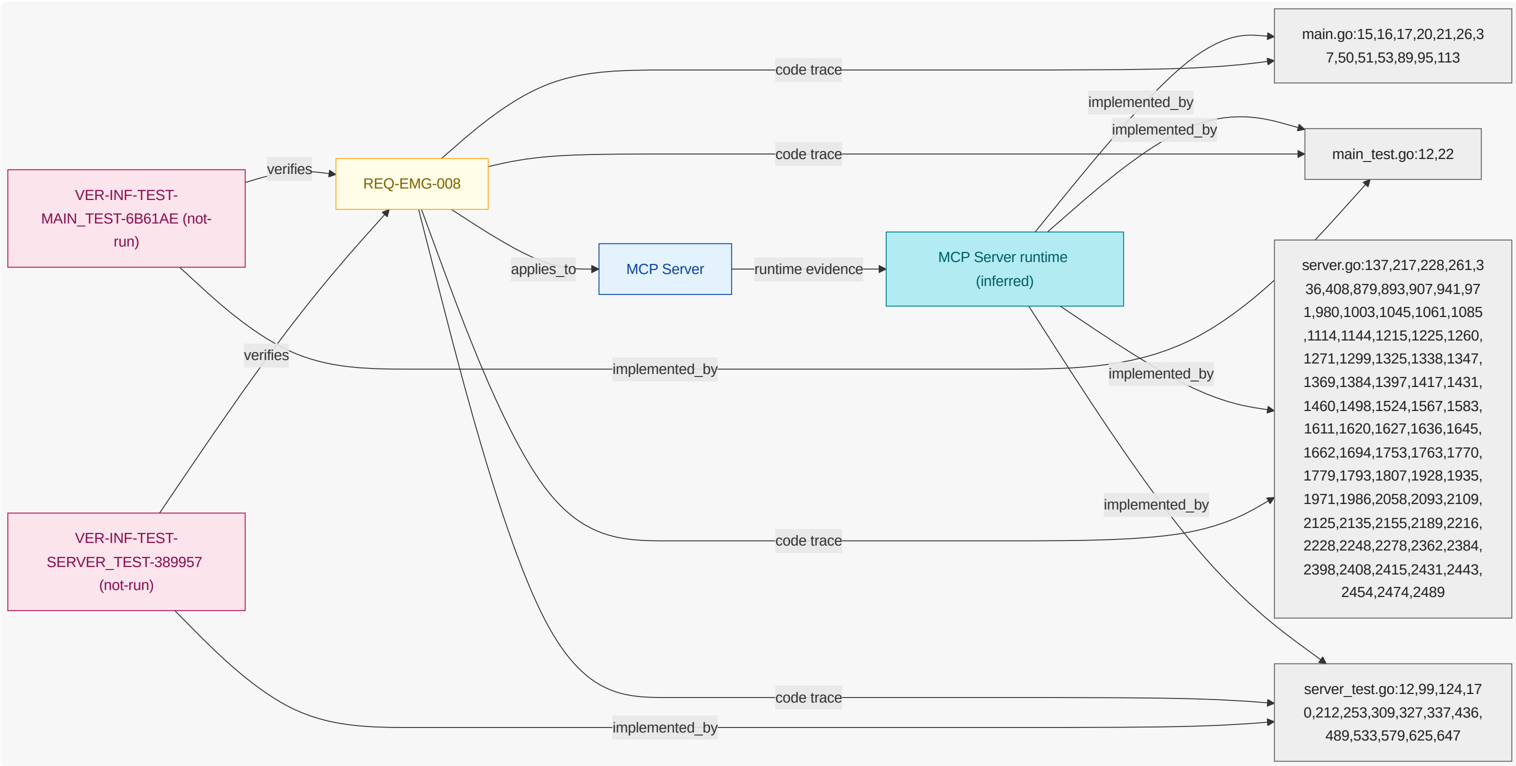
11.1.7. REQ-EMG-008

When `mcp tool call is received`, the `engineering model go` shall reject path arguments outside `repository workspace boundary`.

NOTE Filesystem `trust boundary` guardrail.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



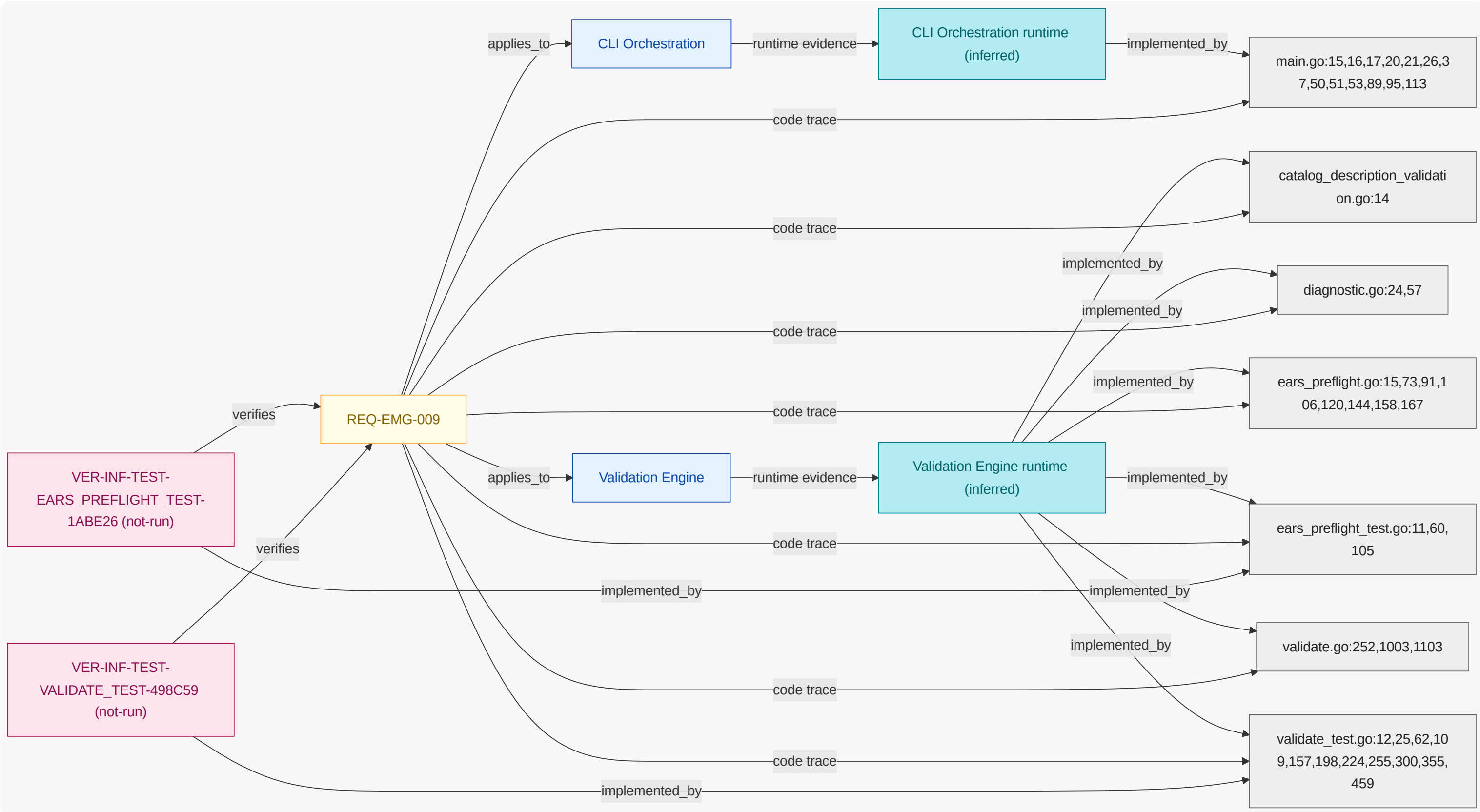
11.1.8. REQ-EMG-009

While `ci validation mode is active`, when `generation run is requested`, the `engineering model go` shall execute `validation` plus tests before `artifact generated event is raised`.

NOTE CI gate behavior for deterministic quality.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



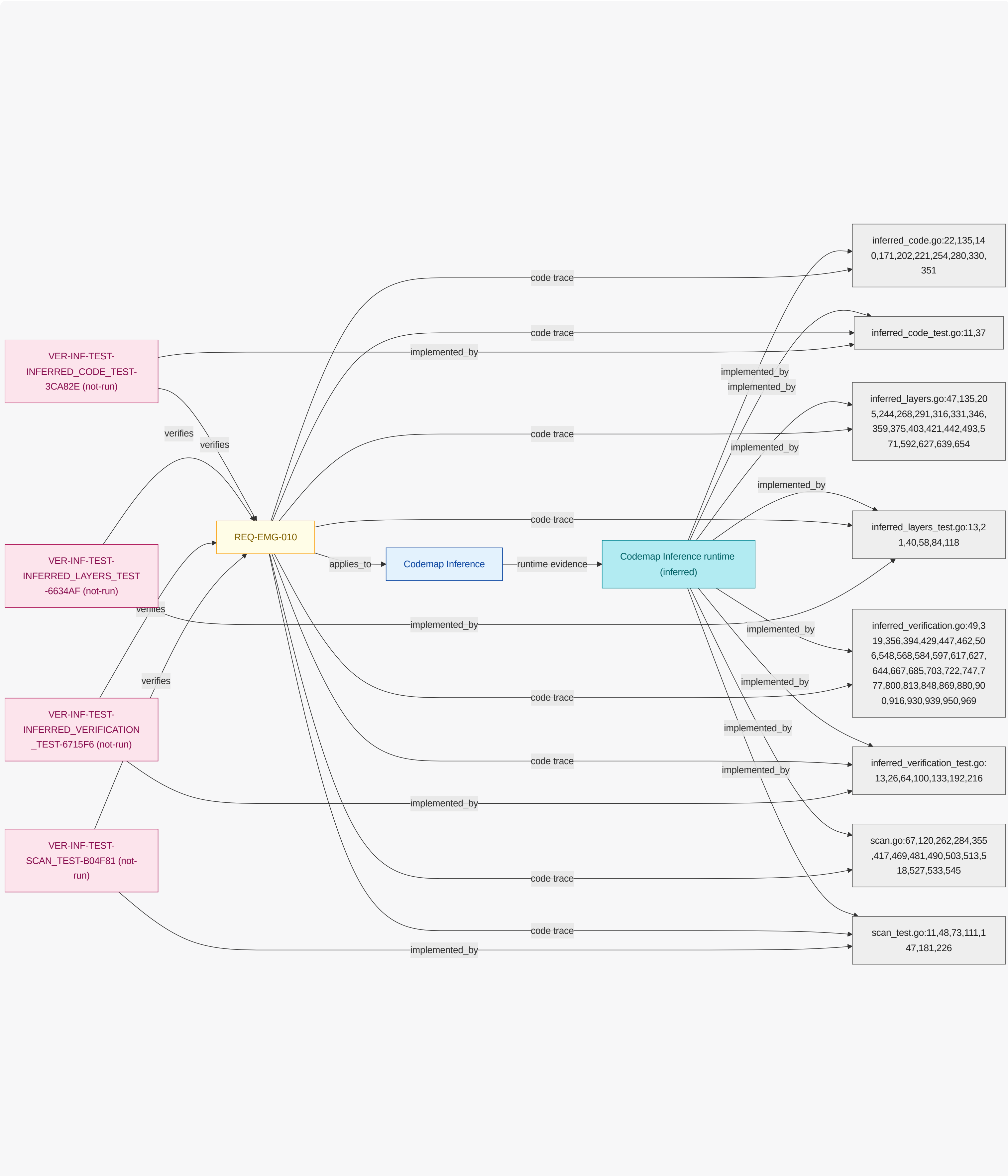
11.1.9. REQ-EMG-010

Where `requirement trace is complete`, when `model updated event is received`, the `engineering model go` shall infer traceability coverage from `trace markers`.

> **NOTE**
>
> Links runtime, code, and test markers to modeled `requirement` coverage. Code and metadata scanning excludes dependency caches (node_modules, vendor) and dot-directories (.git, .engmod, tool caches) so coverage reflects only committed first-party source and stays reproducible.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



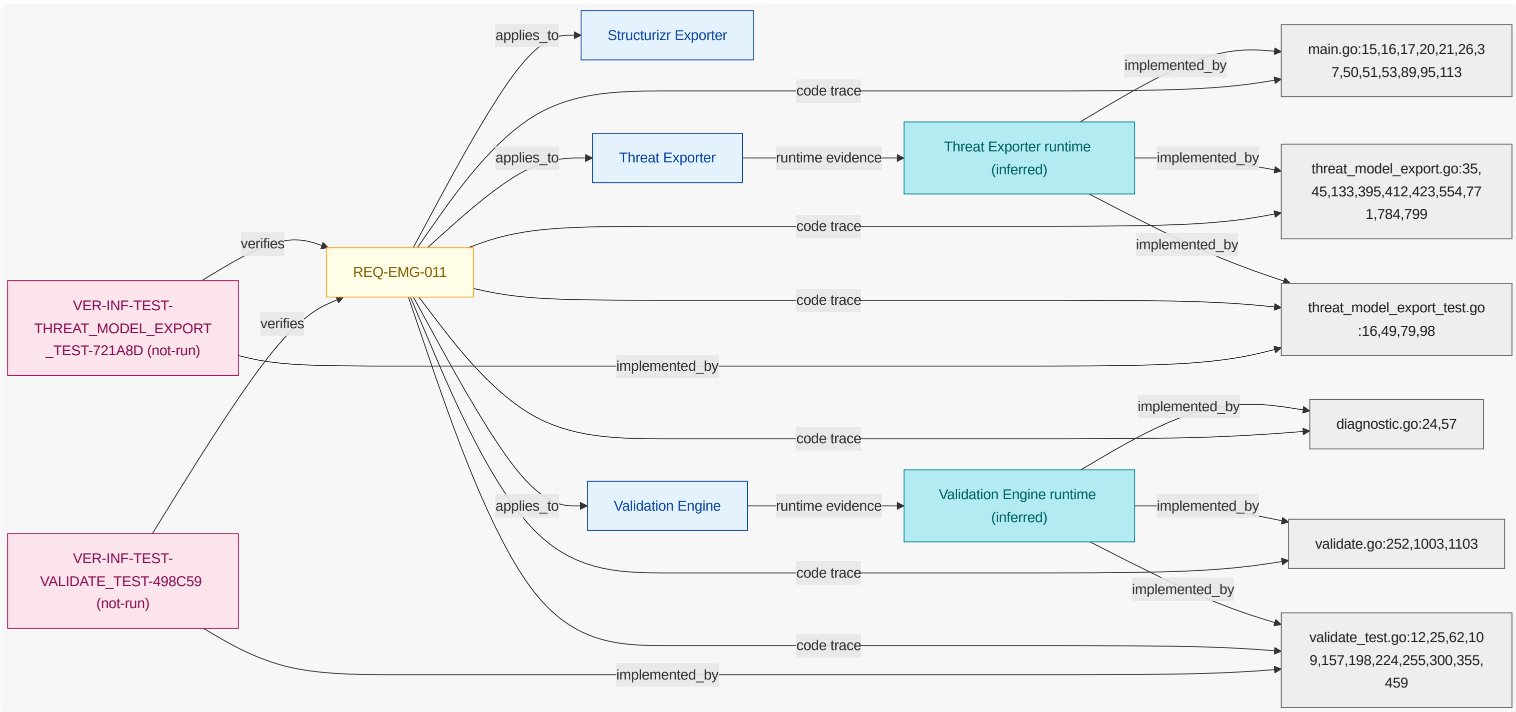
11.1.10. REQ-EMG-011

Where `external schema dependencies are available`, when `generation run is requested`, the `engineering model go` shall fail `validation` if schema checks do not match expected constraints.

NOTE Protect export integrity when schema trust fails.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



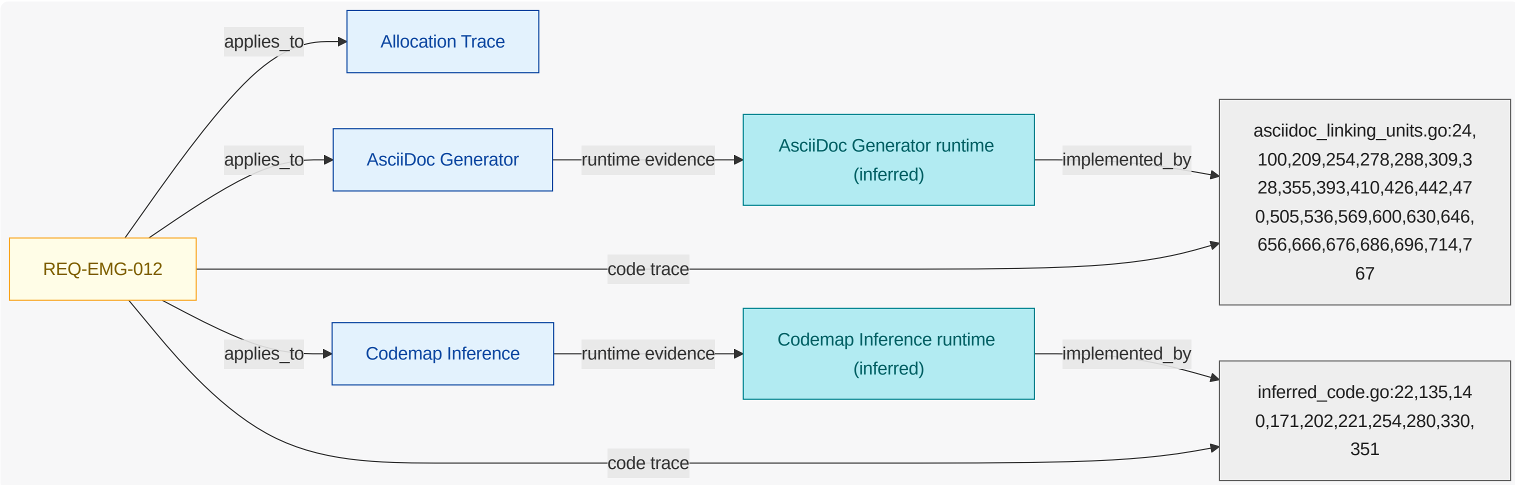
11.1.11. REQ-EMG-012

When `generation run is requested`, the `engineering model go` shall generate artifacts deterministically so that regeneration is reproducible.

NOTE `Generation` uses stable ordering (sorted diagnostics, link targets, scan results) and excludes environment-specific inputs, so regenerated documents are bit-for-bit reproducible and the CI freshness gate can detect drift.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



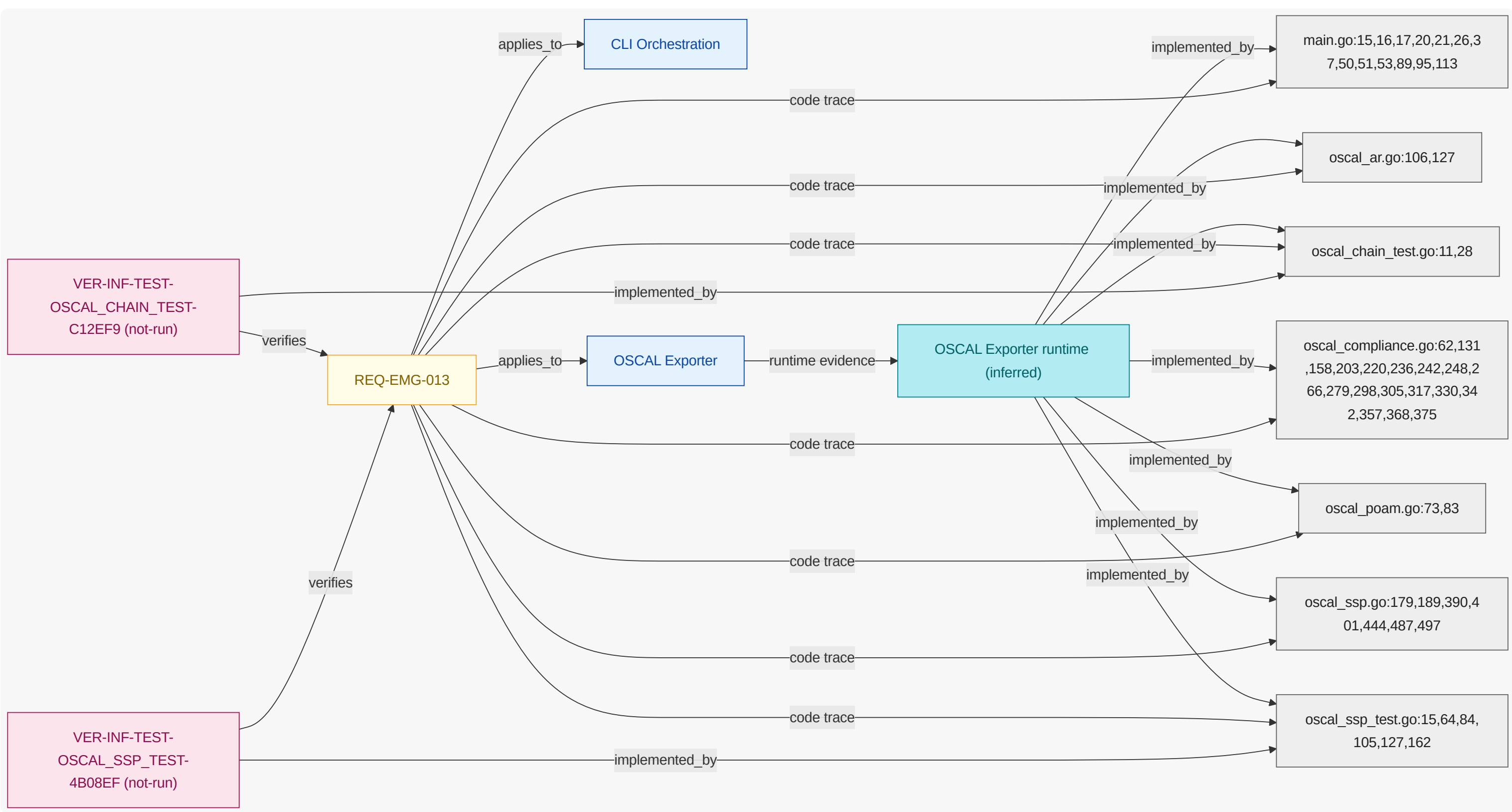
11.1.12. REQ-EMG-013

Where `traceability export is enabled`, when `generation run is requested`, the `engineering model go` shall export `OSCAL SSP` plus assessment results plus POA&M artifacts.

NOTE OSCAL export set for `compliance` workflows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



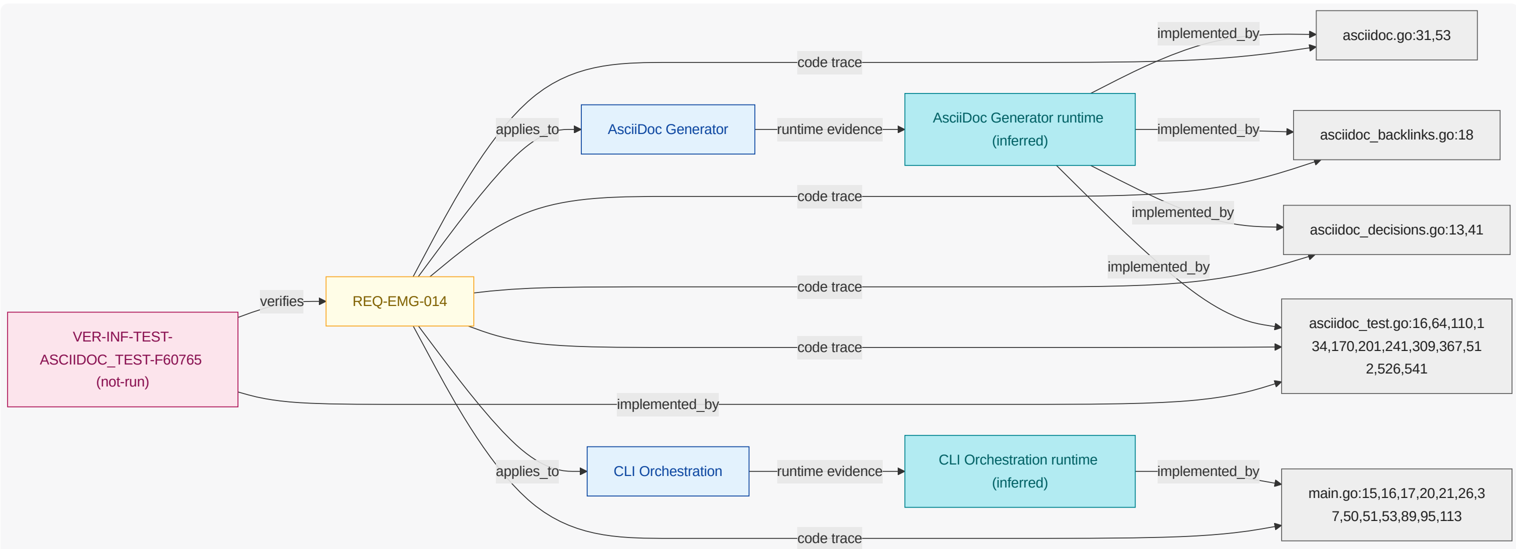
11.1.13. REQ-EMG-014

Where `architecture decision records are authored`, when `generation run is requested`, the `engineering model go` shall publish `architecture decision records document` linked from AsciiDoc architecture publication output.

NOTE	ADRs are first-class `model` content but are published separately from the main architecture narrative.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



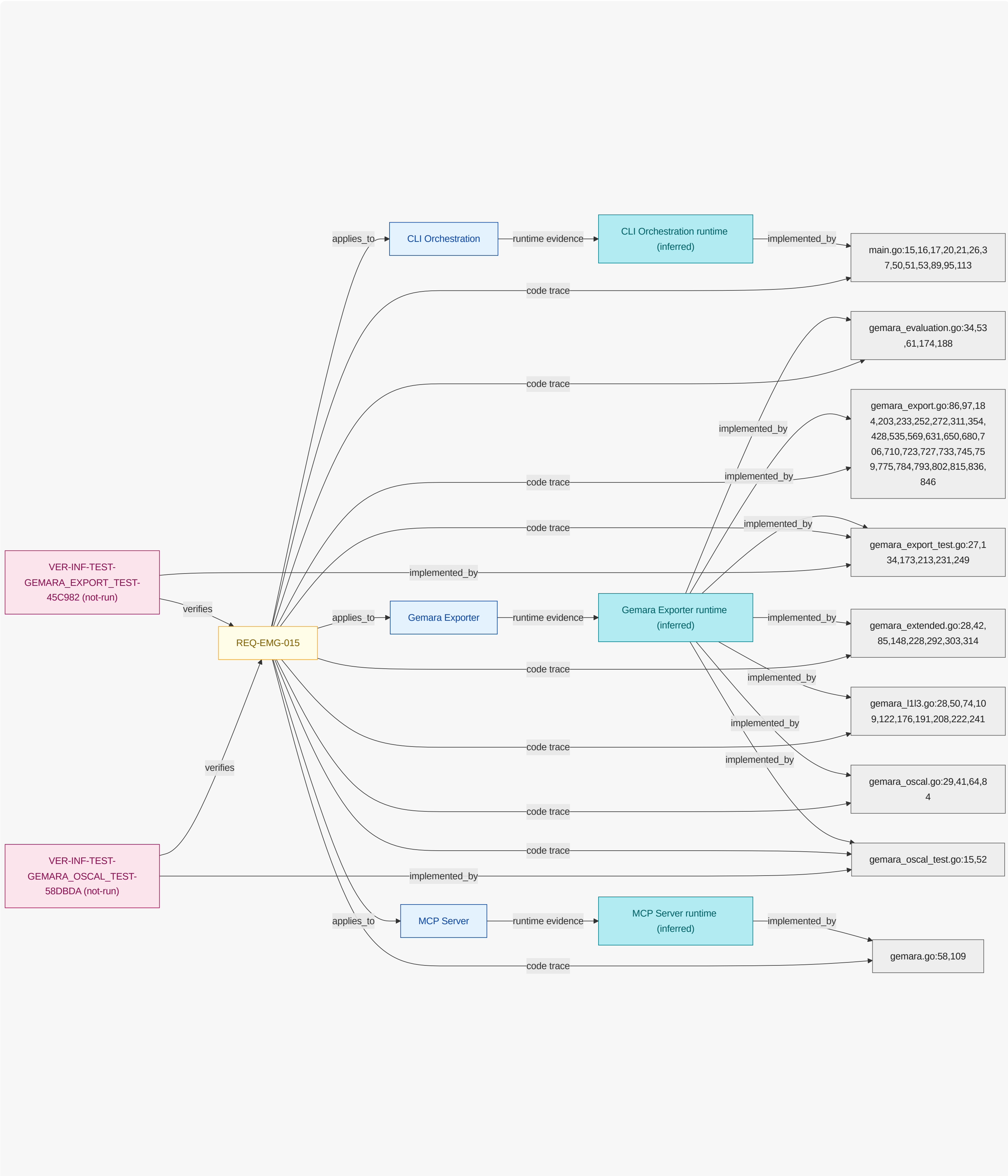
11.1.14. REQ-EMG-015

Where `gemara export is enabled`, when `generation run is requested`, the `engineering model go` shall render OpenSSF Gemara `model` artifacts.

NOTE	Gemara is a first-class GRC rendering of the `model`, validated against the published Gemara schemas via the go-gemara SDK.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



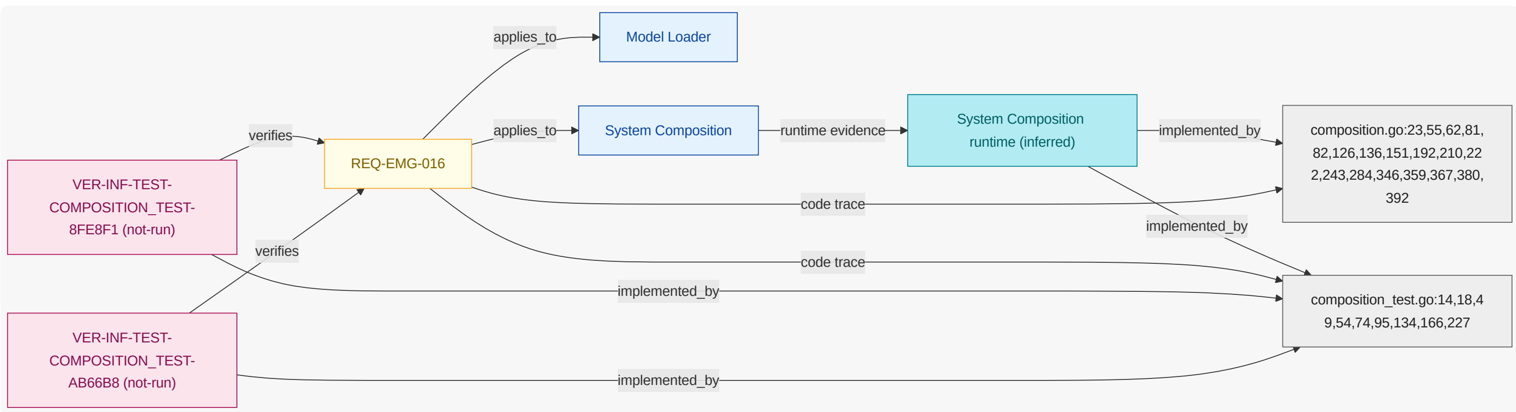
11.1.15. REQ-EMG-016

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

NOTE Recursive system-of-systems composition; local subdirectory references, with external git repositories cloned into the .engmod cache (REQ-EMG-027, ADR-EMG-008, ADR-EMG-010).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



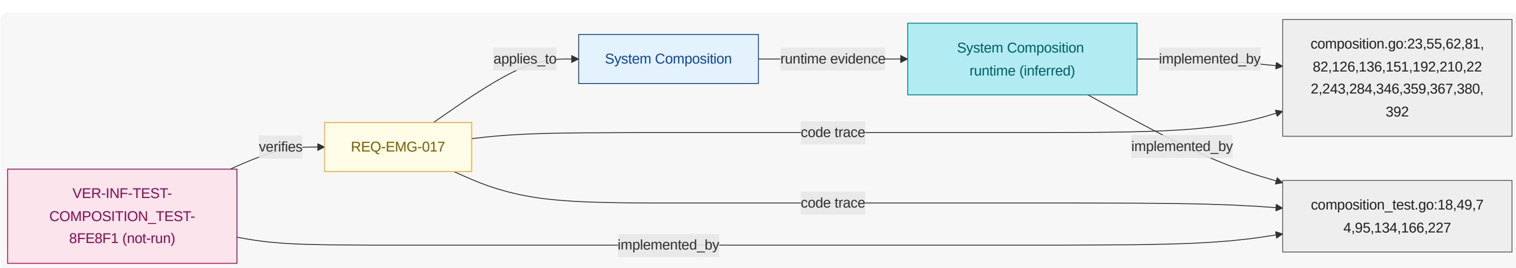
11.1.16. REQ-EMG-017

Where system composition is enabled, when generation run is requested, the engineering model go shall reject any subsystem reference that resolves outside the repository workspace boundary.

NOTE Downward-only references remain within the workspace; mirrors the MCP path-boundary guardrail.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



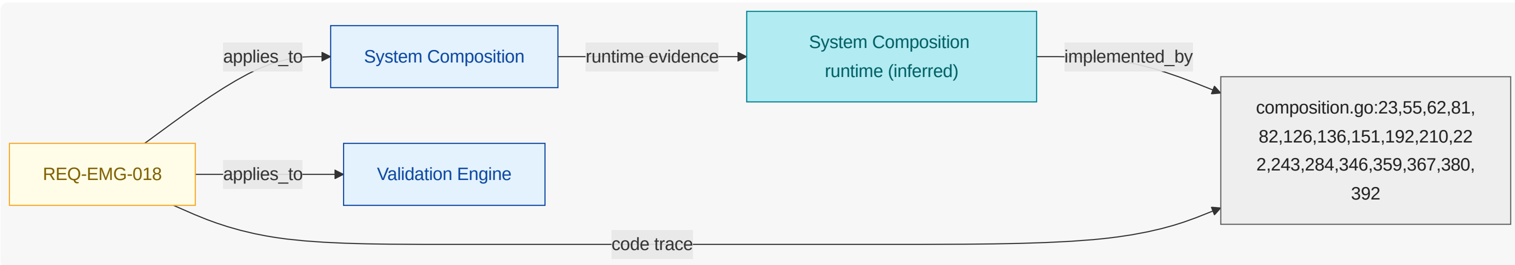
11.1.17. REQ-EMG-018

Where system composition is enabled, when generation run is requested, the engineering model go shall reject a subsystem reference graph that contains a cycle.

NOTE Subsystem references must form a directed acyclic graph (ADR-EMG-009).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



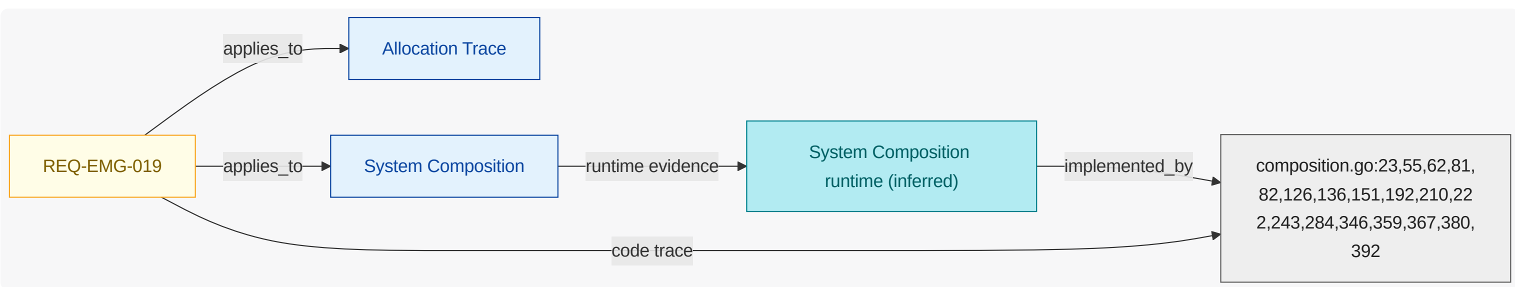
11.1.18. REQ-EMG-019

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall resolve cross-system references using subsystem-qualified identifiers.

> **NOTE** Identifiers are scoped per system; cross-system references are qualified by subsystem to avoid collisions.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



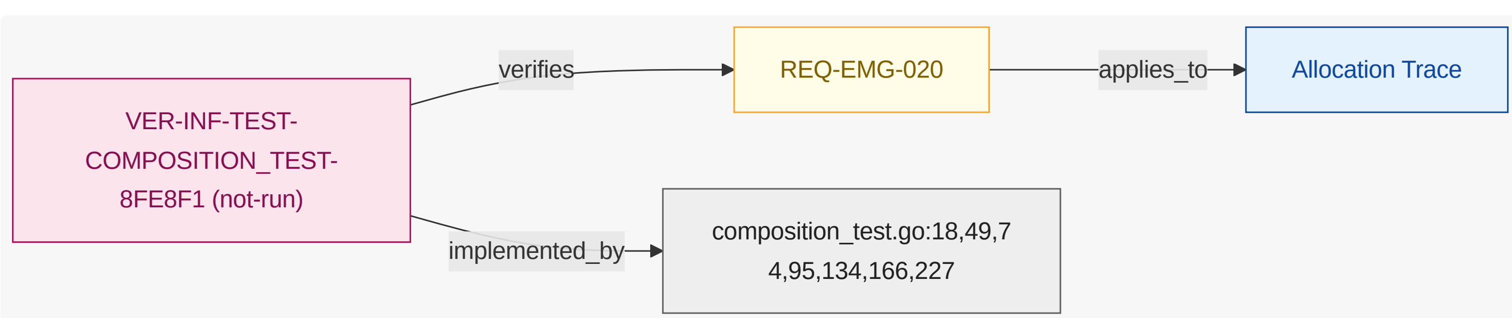
11.1.19. REQ-EMG-020

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall materialize parent-to-subsystem `allocation trace` ability in the composed output without modifying any subsystem `model`.

> **NOTE** A subsystem is parent-agnostic; parent `allocation` is materialized only in composed artifacts, never written back to the child (ADR-EMG-009, ADR-EMG-011).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



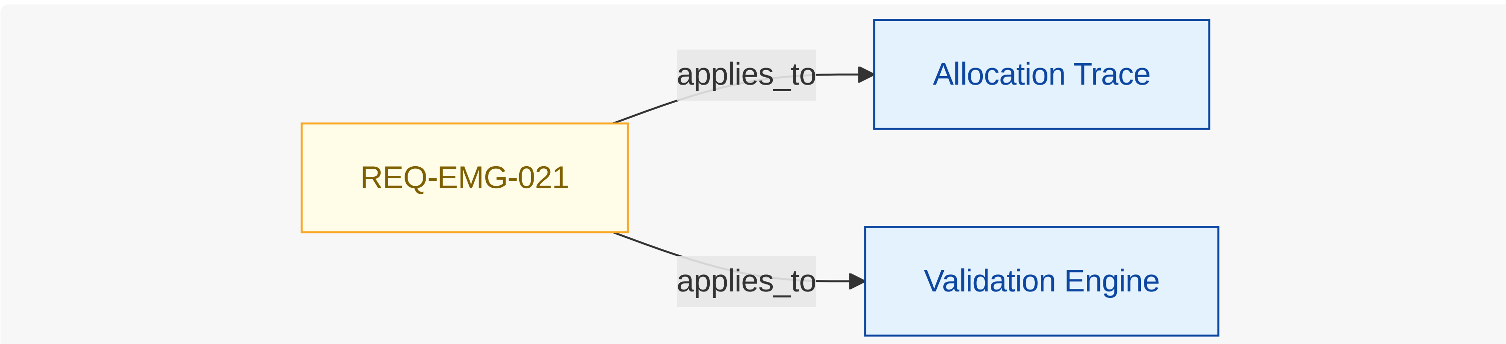
11.1.20. REQ-EMG-021

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall report each subsystem required `interface` that the parent system does not satisfy.

NOTE Subsystems publish a provides-and-requires contract; the parent must satisfy each required `interface`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



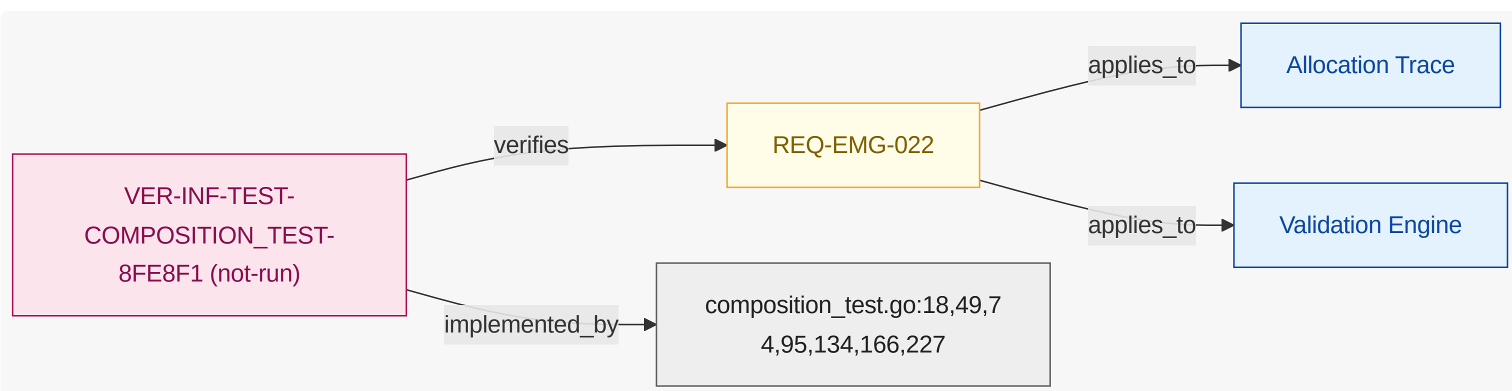
11.1.21. REQ-EMG-022

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall reject an `allocation` that targets a subsystem identifier not published in the subsystem contract.

NOTE Parents may bind only to a subsystem's public contract surface, not its internal identifiers.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



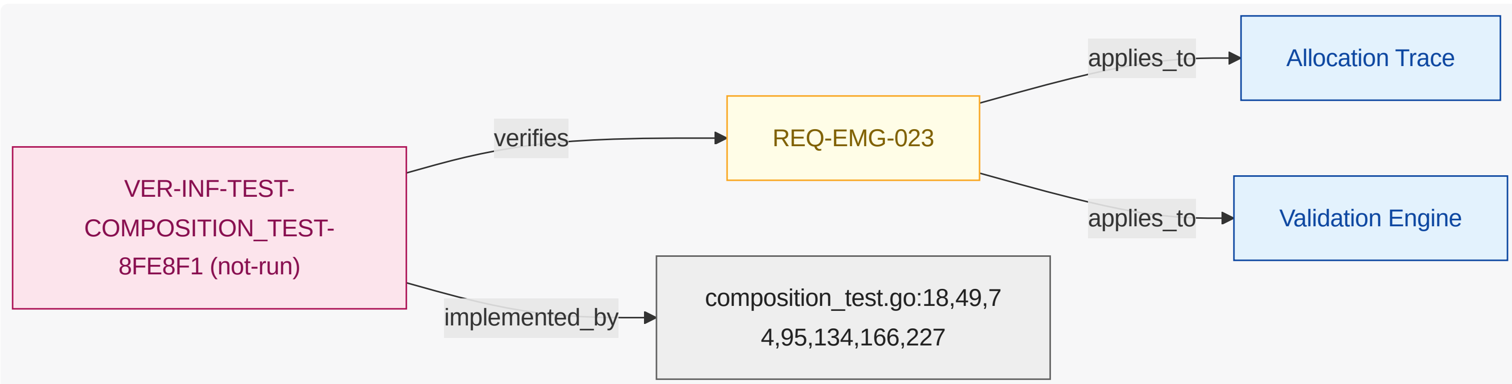
11.1.22. REQ-EMG-023

When `generation run is requested`, the `engineering model go` shall flag any `requirement` that links to another `requirement` in the same document.

NOTE Requirements carry no tiers. A requirement is either delegated to a subsystem through a downward composition allocation or implemented here, and the delegation link alone tells the two apart. Linking requirements to one another inside a document recreates tier coupling and is confusing across system boundaries, so it is flagged.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



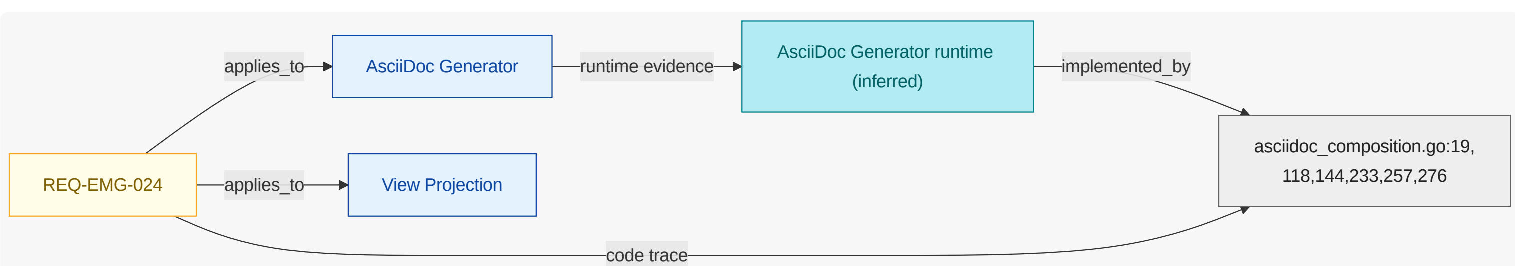
11.1.23. REQ-EMG-024

Where hardware interfaces are authored, when generation run is requested, the engineering model go shall render the hardware to software interface allocation view.

NOTE Hardware items and hardware interfaces capture the HW/SW interface (ICD/IRS) for the interface-control view.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



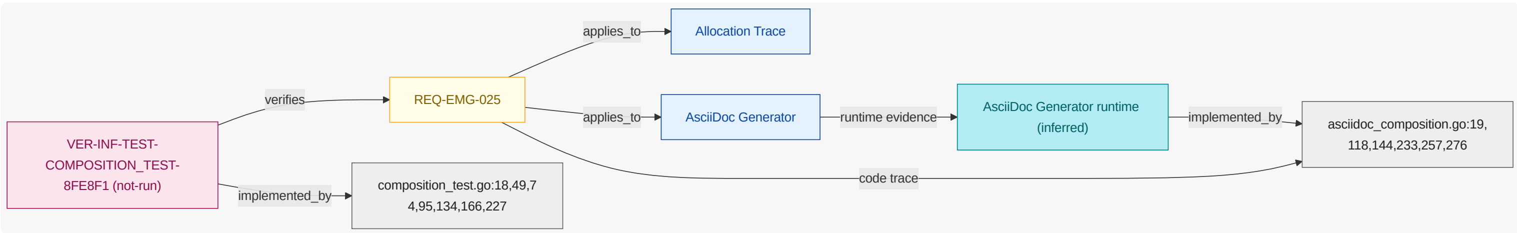
11.1.24. REQ-EMG-025

Where system composition is enabled, when generation run is requested, the engineering model go shall produce a system requirement allocation matrix aggregated across the composed subsystems.

NOTE Aggregate roll-up generation mode for the integrating system.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



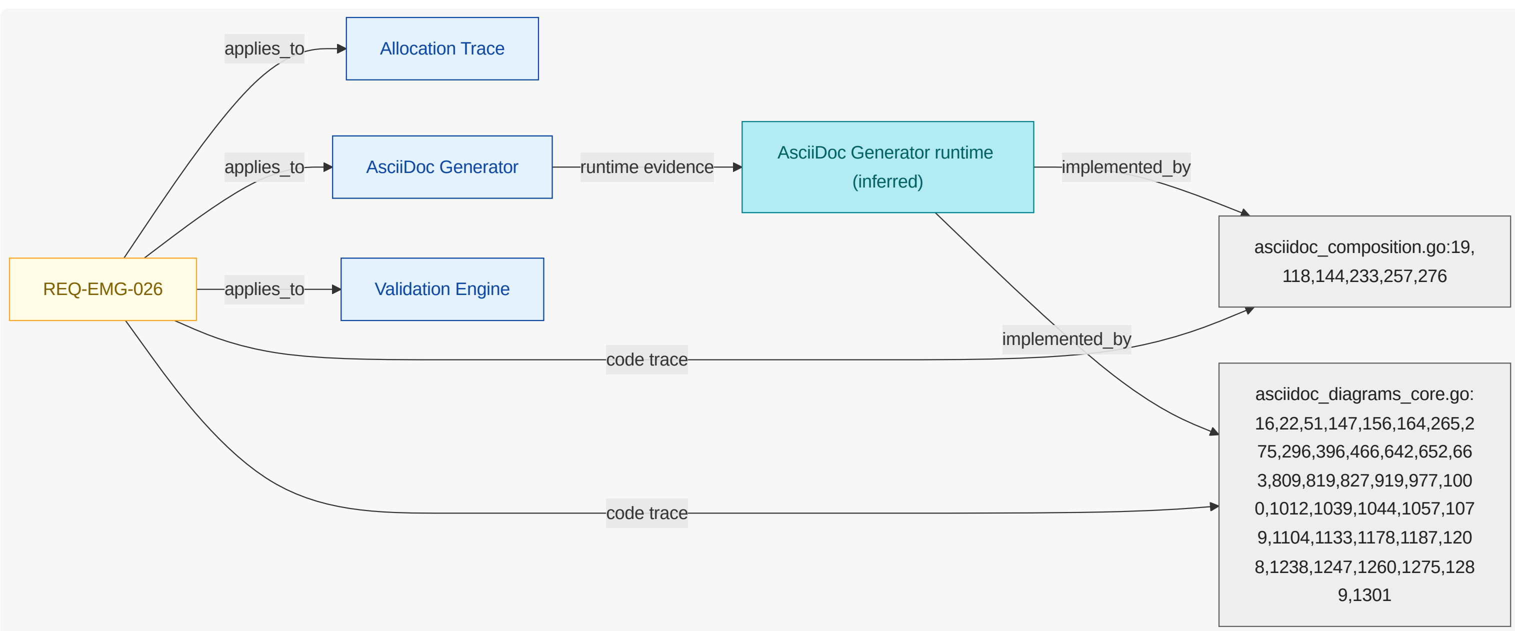
11.1.25. REQ-EMG-026

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall render in the `requirement` `view` the delegation link from each delegated `requirement` to the specific subsystem `requirement` that satisfies it.

> **NOTE**
>
> A provided contract entry names (ref) the subsystem `requirement` that realizes it, so a parent delegation traces to exactly the `requirement` that satisfies it. An allocated capability with no such ref is flagged as untraceable, and a ref to a `requirement` the subsystem does not define is rejected.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



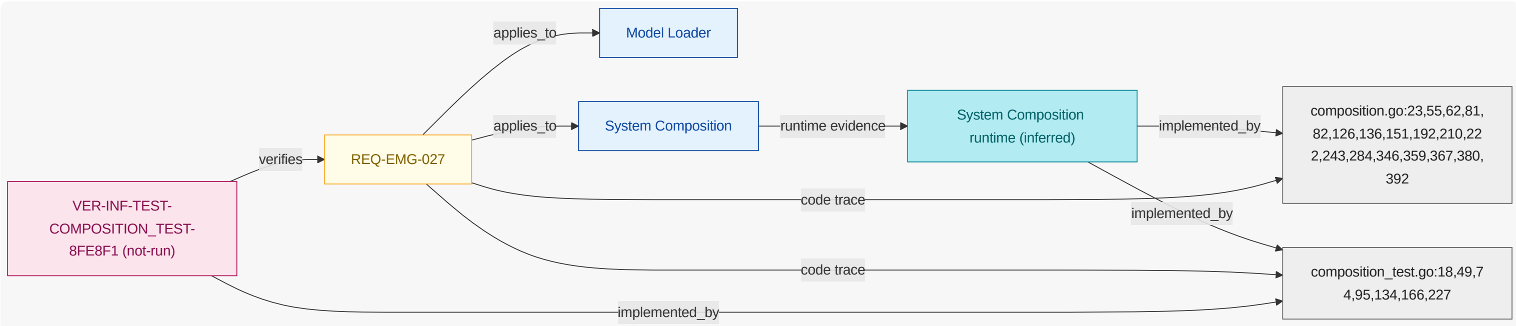
11.1.26. REQ-EMG-027

Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall resolve an external git subsystem reference by cloning the repository into the workspace .engmod cache.

> **NOTE**
>
> A subsystem may reference an external repository with git (and an optional rev and path). The `model` is cloned once into .engmod/subsystems/<id> — a gitignored, temporary cache reused while it points at the same repository — and resolved like a local subsystem, giving a full cross-repository composed `view`. References stay downward-only.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



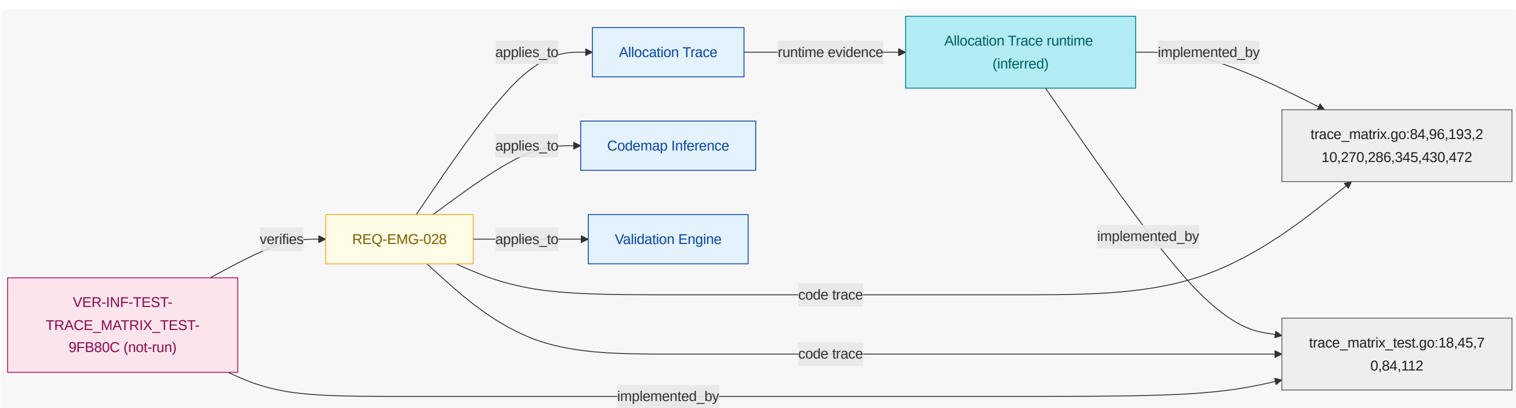
11.1.27. REQ-EMG-028

When `generation run is requested`, the `engineering model go` shall reject a code trace link that resolves to no `requirement` or `model` element.

> **NOTE** Code markers (TRLC-LINKS to a `requirement`, ENGMODEL-LINKS to a `model` element) are existence-checked against the `model`, not only shape-checked, so a link to a deleted or mistyped id is a dangling link and fails the build. A trace link is only checked for the `model` that owns its code (see `REQ-EMG-030` scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



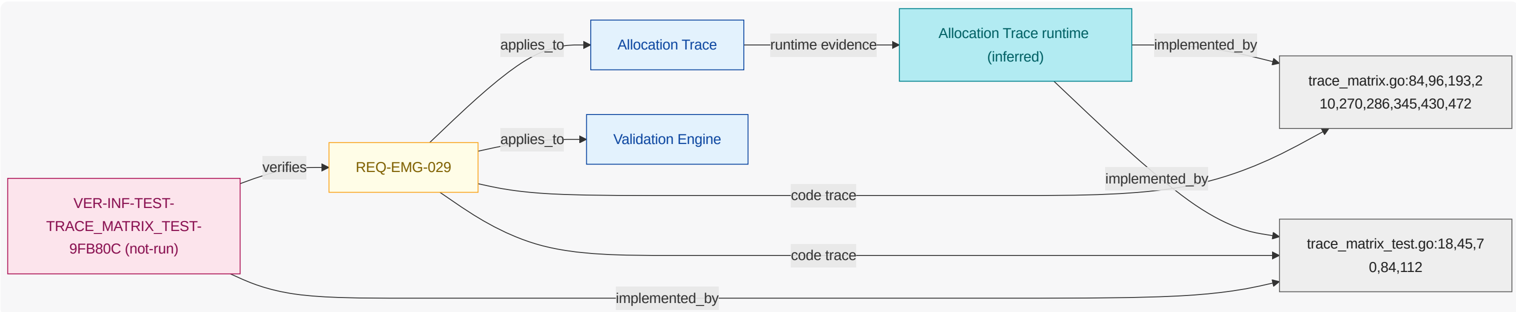
11.1.28. REQ-EMG-029

When `generation run is requested`, the `engineering model go` shall flag each `requirement` that is traced by neither code nor verification nor a downward delegation.

> **NOTE** An orphan `requirement` has no implementing code symbol, no verification, and is not delegated to a subsystem; it is a coverage gap surfaced as a warning so it is visible without breaking the build.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



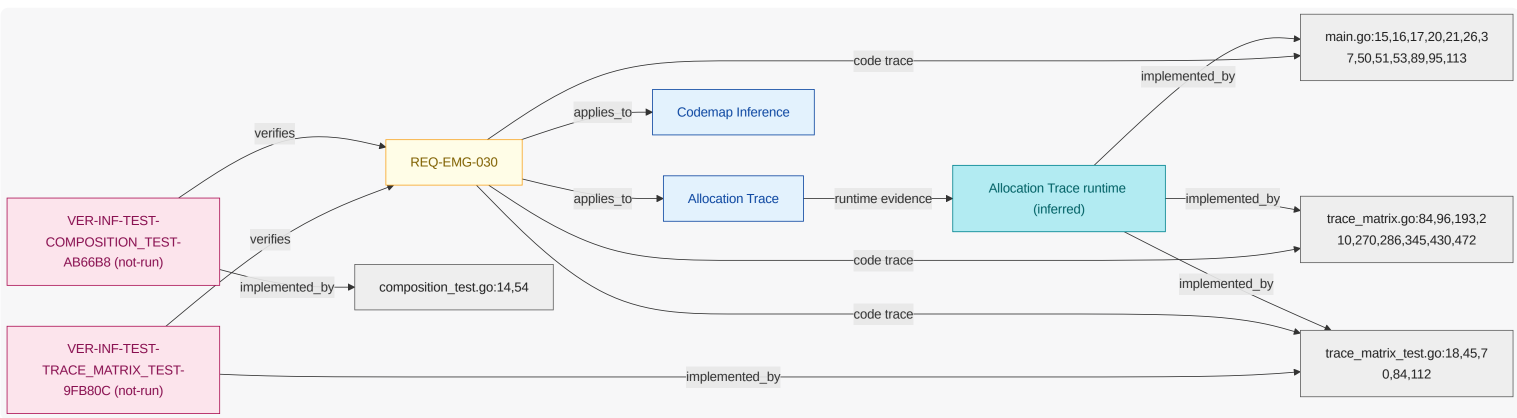
11.1.29. REQ-EMG-030

When `generation run is requested`, the `engineering model go` shall export a machine-readable traceability matrix that links each `requirement` to its implementing code and verification and delegated subsystem `requirement`.

> **NOTE** One consolidated matrix (JSON) replaces having to read it out of prose and diagrams. Code is attributed to the `model` whose architecture.yml is its nearest enclosing root, so a parent `model` never validates or counts a child `model`'s trace links (monorepo scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

11.2.1. Authorized Checkout Flow

Field	Value
ID	`VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367`
Kind	e2e
Status	not-run
Verifies Requirements	REQ-PAY-001

Field	Value
Test Code Elements	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Notes	executes `authorized checkout flow` from session start through payment authorization success

11.2.2. Ota Rollout Flow

Field	Value
ID	`VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC`
Kind	e2e
Status	not-run
Verifies Requirements	REQ-COF-003, REQ-COF-005
Test Code Elements	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Notes	runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

11.2.3. Pr Opened Review Flow

Field	Value
ID	`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6`
Kind	e2e
Status	not-run
Verifies Requirements	REQ-PRR-001, REQ-PRR-002, REQ-PRR-003, REQ-PRR-005
Test Code Elements	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Derived Functional Owners	none

Field	Value
Result Summary	not-run:4
Evidence	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Notes	exercises end-to-end pull request opened `flow` from webhook ingest to published review

11.2.4. Audit Record Persistence Test

Field	Value
ID	`VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4`
Kind	integration
Status	not-run
Verifies Requirements	REQ-PAY-005
Test Code Elements	audit_record_persistence_test.ts:5
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts
Notes	checks audit record persistence includes payment id and computed `risk` score

11.2.5. Audit Redaction Test

Field	Value
ID	`VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4`
Kind	integration
Status	not-run
Verifies Requirements	REQ-PRR-007
Test Code Elements	audit_redaction_test.ts:3
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts
Notes	checks audit `evidence` redaction for secrets and sensitive payload fields

11.2.6. Fraud Gate Test

Field	Value
ID	`VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163`
Kind	integration
Status	not-run
Verifies Requirements	REQ-PAY-002, REQ-PAY-004
Test Code Elements	fraud_gate_test.rs:6
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs
Notes	validates fraud gate escalation and block decisions before authorization completion

11.2.7. Logging Pipeline Test

Field	Value
ID	`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8`
Kind	integration
Status	not-run
Verifies Requirements	REQ-COF-002, REQ-COF-007
Test Code Elements	logging_pipeline_test.ts:4
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts
Notes	validates telemetry logging pipeline persistence and operational reporting signals

11.2.8. Policy Only Fallback Test

Field	Value
ID	`VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003`
Kind	integration

Field	Value
Status	not-run
Verifies Requirements	REQ-PRR-004, REQ-PRR-006
Test Code Elements	policy_only_fallback_test.rs:5
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs
Notes	verifies policy-only fallback when Bedrock `inference` is unavailable

11.2.9. Asciidoc Diagrams Core Test

Field	Value
ID	`VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-003`
Test Code Elements	asciidoc_diagrams_core_test.go:14,41,135,161,197,223,254,273,293,327,355,383
Derived Functional Owners	`FU-ASCIIDOC-GENERATOR`, `FU-CLI-ORCHESTRATION`, `FU-VIEW-PROJECTION`
Result Summary	not-run:1
Evidence	asciidoc_diagrams_core_test.go
Notes	Inferred from test source artifact.

11.2.10. Asciidoc Diagrams Runtime Test

Field	Value
ID	`VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-003`
Test Code Elements	asciidoc_diagrams_runtime_test.go:12,50,72,95

Field	Value
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_diagrams_runtime_test.go
Notes	Inferred from test source artifact.

11.2.11. Asciidoc Linking Units Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_linking_units_test.go:10,46,75,101,129,148,167,186
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_linking_units_test.go
Notes	Inferred from test source artifact.

11.2.12. Asciidoc Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_TEST-F60765
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003 , REQ-EMG-014
Test Code Elements	asciidoc_test.go:16,64,110,134,170,201,241,309,367,512,526,541
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:2

Field	Value
Evidence	asciidoc_test.go
Notes	Inferred from test source artifact.

11.2.13. Asciidoc Views Test

Field	Value
ID	`VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-003`
Test Code Elements	asciidoc_views_test.go:11
Derived Functional Owners	`FU-ASCIIDOC-GENERATOR`, `FU-CLI-ORCHESTRATION`, `FU-VIEW-PROJECTION`
Result Summary	not-run:1
Evidence	asciidoc_views_test.go
Notes	Inferred from test source artifact.

11.2.14. Catalog Systems Test

Field	Value
ID	`VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-001`
Test Code Elements	catalog_systems_test.go:11,36,82
Derived Functional Owners	`FU-ASCIIDOC-GENERATOR`, `FU-CLI-ORCHESTRATION`, `FU-MODEL-LOADER`, `FU-OSCAL-EXPORTER`, `FU-STRUCTURIZR-EXPORTER`, `FU-THREAT-EXPORTER`, `FU-VALIDATION-ENGINE`
Result Summary	not-run:1
Evidence	catalog_systems_test.go
Notes	Inferred from test source artifact.

11.2.15. Checkout Decline Response Test

Field	Value
ID	VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671
Kind	test
Status	not-run
Verifies Requirements	REQ-PAY-003, REQ-PAY-006
Test Code Elements	checkout_decline_response_test.go:7
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go
Notes	validates checkout decline response contract includes reason and retry guidance

11.2.16. Composition Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_TEST-8FE8F1
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-016, REQ-EMG-017, REQ-EMG-020, REQ-EMG-022, REQ-EMG-023, REQ-EMG-025, REQ-EMG-027
Test Code Elements	composition_test.go:18,49,74,95,134,166,227
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-ASCIIDOC-GENERATOR, FU-MODEL-LOADER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE
Result Summary	not-run:7
Evidence	composition_test.go
Notes	Inferred from test source artifact.

11.2.17. Composition Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_TEST-AB66B8
Kind	test

Field	Value
Status	not-run
Verifies Requirements	REQ-EMG-007, REQ-EMG-016, REQ-EMG-030
Test Code Elements	composition_test.go:14,54
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-CODEMAP-INFERENCE, FU-MCP-SERVER, FU-MODEL-LOADER, FU-SYSTEM-COMPOSITION
Result Summary	not-run:3
Evidence	mcp/composition_test.go
Notes	Inferred from test source artifact.

11.2.18. E2e Test

Field	Value
ID	VER-INF-TEST-E2E_TEST-BE56F8
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001
Test Code Elements	e2e_test.go:14,39
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	e2e_test.go
Notes	Inferred from test source artifact.

11.2.19. Ears Preflight Test

Field	Value
ID	VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001, REQ-EMG-009
Test Code Elements	ears_preflight_test.go:11,60,105

Field	Value
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	ears_preflight_test.go
Notes	Inferred from test source artifact.

11.2.20. Gemara Export Test

Field	Value
ID	VER-INF-TEST-GEMARA_EXPORT_TEST-45C982
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-015
Test Code Elements	gemara_export_test.go:27,134,173,213,231,249
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-GEMARA-EXPORTER, FU-MCP-SERVER
Result Summary	not-run:1
Evidence	gemara_export_test.go
Notes	Inferred from test source artifact.

11.2.21. Gemara Oscal Test

Field	Value
ID	VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-015
Test Code Elements	gemara_oscal_test.go:15,52
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-GEMARA-EXPORTER, FU-MCP-SERVER
Result Summary	not-run:1

Field	Value
Evidence	gemara_oscal_test.go
Notes	Inferred from test source artifact.

11.2.22. Github Check Run Contract Test

Field	Value
ID	`VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F`
Kind	test
Status	not-run
Verifies Requirements	REQ-PRR-005
Test Code Elements	github_check_run_contract_test.go:8
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go
Notes	validates GitHub check-run contract shape and required review result fields

11.2.23. Inferred Code Test

Field	Value
ID	`VER-INF-TEST-INFERRED_CODE_TEST-3CA82E`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-010`
Test Code Elements	inferred_code_test.go:11,37
Derived Functional Owners	`FU-CODEMAP-INFERENCE`
Result Summary	not-run:1
Evidence	inferred_code_test.go
Notes	Inferred from test source artifact.

11.2.24. Inferred Layers Test

Field	Value
ID	`VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-010`
Test Code Elements	inferred_layers_test.go:13,21,40,58,84,118
Derived Functional Owners	`FU-CODEMAP-INFERENCE`
Result Summary	not-run:1
Evidence	inferred_layers_test.go
Notes	Inferred from test source artifact.

11.2.25. Inferred Verification Test

Field	Value
ID	`VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6`
Kind	test
Status	not-run
Verifies Requirements	`REQ-EMG-010`
Test Code Elements	inferred_verification_test.go:13,26,64,100,133,192,216
Derived Functional Owners	`FU-CODEMAP-INFERENCE`
Result Summary	not-run:1
Evidence	inferred_verification_test.go
Notes	Inferred from test source artifact.

11.2.26. Load Test

Field	Value
ID	`VER-INF-TEST-LOAD_TEST-A58EB3`
Kind	test
Status	not-run

Field	Value
Verifies Requirements	REQ-EMG-001
Test Code Elements	load_test.go:10,25
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	model /load_test.go
Notes	Inferred from test source artifact.

11.2.27. Lobster Export Test

Field	Value
ID	VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-006
Test Code Elements	lobster_export_test.go:12
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-LOBSTER-EXPORTER, FU-TRLC-EXPORTER
Result Summary	not-run:1
Evidence	lobster_export_test.go
Notes	Inferred from test source artifact.

11.2.28. Main Test

Field	Value
ID	VER-INF-TEST-MAIN_TEST-6B61AE
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007, REQ-EMG-008
Test Code Elements	main_test.go:12,22

Field	Value
Derived Functional Owners	FU-MCP-SERVER
Result Summary	not-run:2
Evidence	cmd/engmcp/main_test.go
Notes	Inferred from test source artifact.

11.2.29. Oscal Chain Test

Field	Value
ID	VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-013
Test Code Elements	oscal_chain_test.go:11,28
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-OSCAL-EXPORTER
Result Summary	not-run:1
Evidence	oscal_chain_test.go
Notes	Inferred from test source artifact.

11.2.30. Oscal Ssp Test

Field	Value
ID	VER-INF-TEST-OSCAL_SSP_TEST-4B08EF
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-013
Test Code Elements	oscal_ssp_test.go:15,64,84,105,127,162
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-OSCAL-EXPORTER
Result Summary	not-run:1

Field	Value
Evidence	oscal_ssp_test.go
Notes	Inferred from test source artifact.

11.2.31. Project Test

Field	Value
ID	VER-INF-TEST-PROJECT_TEST-2ECF72
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	project_test.go:11,52,86,137,168
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	view /project_test.go
Notes	Inferred from test source artifact.

11.2.32. Render Test

Field	Value
ID	VER-INF-TEST-RENDER_TEST-EECC23
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	render_test.go:12
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	render/mermaid/render_test.go
Notes	Inferred from test source artifact.

11.2.33. Scan Test

Field	Value
ID	VER-INF-TEST-SCAN_TEST-B04F81
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010 , REQ-SAMPLE-001, REQ-SAMPLE-002
Test Code Elements	scan_test.go:11,48,73,111,147,181,226
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:3
Evidence	codemap/scan_test.go
Notes	Inferred from test source artifact.

11.2.34. Server Test

Field	Value
ID	VER-INF-TEST-SERVER_TEST-389957
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007 , REQ-EMG-008
Test Code Elements	server_test.go:12,99,124,170,212,253,309,327,337,436,489,533,579,625,647
Derived Functional Owners	FU-MCP-SERVER
Result Summary	not-run:2
Evidence	mcp/server_test.go
Notes	Inferred from test source artifact.

11.2.35. Structurizr Export Test

Field	Value
ID	VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374
Kind	test
Status	not-run

Field	Value
Verifies Requirements	REQ-EMG-005
Test Code Elements	structurizr_export_test.go:14,40
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-STRUCTURIZR-EXPORTER
Result Summary	not-run:1
Evidence	structurizr_export_test.go
Notes	Inferred from test source artifact.

11.2.36. Telemetry Ingest Contract Test

Field	Value
ID	VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2
Kind	test
Status	not-run
Verifies Requirements	REQ-COF-001
Test Code Elements	telemetry_ingest_contract_test.go:7
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go
Notes	validates telemetry ingest contract fields and accepted request semantics

11.2.37. Threat Model Export Test

Field	Value
ID	VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-004, REQ-EMG-011
Test Code Elements	threat_model_export_test.go:16,49,79,98

Field	Value
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	threat_model_export_test.go
Notes	Inferred from test source artifact.

11.2.38. Trace Matrix Test

Field	Value
ID	VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-028, REQ-EMG-029, REQ-EMG-030
Test Code Elements	trace_matrix_test.go:18,45,70,84,112
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-CODEMAP-INFERENCE, FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	trace_matrix_test.go
Notes	Inferred from test source artifact.

11.2.39. Trlc Export Test

Field	Value
ID	VER-INF-TEST-TRLC_EXPORT_TEST-DD1345
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-006
Test Code Elements	trlc_export_test.go:13,44
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-LOBSTER-EXPORTER, FU-TRLC-EXPORTER
Result Summary	not-run:1

Field	Value
Evidence	trlc_export_test.go
Notes	Inferred from test source artifact.

11.2.40. Validate Test

Field	Value
ID	VER-INF-TEST-VALIDATE_TEST-498C59
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-011
Test Code Elements	validate_test.go:12,25,62,109,157,198,224,255,300,355,459
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	validate/validate_test.go
Notes	Inferred from test source artifact.

11.2.41. Checkout Decline Reason Test

Field	Value
ID	VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD
Kind	unit
Status	not-run
Verifies Requirements	REQ-PAY-003, REQ-PAY-008
Test Code Elements	checkout_decline_reason_test.go:7,15,20
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go
Notes	validates checkout decline reason normalization and retry policy messaging

11.2.42. Ota Campaign Test

Field	Value
ID	`VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB`
Kind	unit
Status	not-run
Verifies Requirements	REQ-COF-003, REQ-COF-004
Test Code Elements	ota_campaign_test.go:8
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go
Notes	verifies OTA campaign signature `validation` and rollout trigger behavior

11.2.43. Rate Limit Deferral Test

Field	Value
ID	`VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE`
Kind	unit
Status	not-run
Verifies Requirements	REQ-PRR-008
Test Code Elements	rate_limit_deferral_test.go:8
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go
Notes	verifies rate-limit deferral behavior for pull request review processing

12. Gemara GRC Model

This chapter renders the model as OpenSSF Gemara (https://gemara.openssf.org) documents. Each artifact is built with the official `go-gemara` SDK types and validated against the published Gemara CUE schemas. Generate the full YAML set with `enggemara`; query it via the `gemara.*` MCP tools.

Gemara Layer	Artifact	Entries
L1	Vector Catalog	4
L1	Principle Catalog	5
L1	Guidance Catalog	2
L2	Capability Catalog	16
L2	Control Catalog	5
L2	Threat Catalog	2
L3	Risk Catalog	3
L3	Policy	2
-	Control→Threat Mapping	3
-	Lexicon	59
L7	Audit Log	5
L6	Enforcement Log	2

12.1. Controls (L2)

Control	Objective	Group	Assessment Requirements
CTRL-MCP-PATH-BOUNDARY	Enforce repo-root path constraints and reject traversal paths in MCP tools.	input-validation	1
CTRL-STRICT-MCP-INPUT-SCHEMA	Enforce per-tool input schema and reject unknown arguments.	input-validation	1
CTRL-TRACEABILITY-COVERAGE	Require requirement-linked verification evidence for modeled behavior.	assurance	1
CTRL-TRACE-LINK-INTEGRITY	Reject code trace links (TRLC-LINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability…	assurance	1

Control	Objective	Group	Assessment Requirements
CTRL-ARTIFACT-FRESHNESS-GATE	Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts.	assurance	1

12.2. Threats (L2)

Threat	Title	Group
TS-MCP-PATH-TRAVERSAL	MCP path traversal accesses non-repo files	stride-tampering
TS-TRACEABILITY-DRIFT	Requirement traceability drifts from implementation	stride-repudiation

12.3. Risks (L3)

Risk	Title	Severity
RISK-MCP-PATH-ESCAPE	MCP path traversal exposes non-repository files	High
RISK-TRACEABILITY-DRIFT	Traceability drift weakens assurance evidence	Medium
RISK-STALE-ARTIFACTS	Stale generated artifacts misrepresent the system	Low

13. Reference Index

13.1. Authored References

CTRL-ARTIFACT-FRESHNESS-GATE

category=assurance; description=Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts.

Key	Value
Kind	Control
Name	Generated Artifact Freshness Gate
Mentioned In	n/a

CTRL-MCP-PATH-BOUNDARY

category=input-validation; description=Enforce repo-root path constraints and reject traversal paths in MCP tools.

Key	Value
Kind	Control
Name	MCP Path Boundary Enforcement
Mentioned In	Security View

CTRL-STRICT-MCP-INPUT-SCHEMA

category=input-validation; description=Enforce per-tool input schema and reject unknown arguments.

Key	Value
Kind	Control
Name	Strict MCP Input Schemas
Mentioned In	Security View

CTRL-TRACE-LINK-INTEGRITY

category=assurance; description=Reject code trace links (TRLC-LINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability matrix with an implemented, verified, delegated, and orphan rollup.

Key	Value
Kind	Control
Name	Trace Link Integrity

Key	Value
Mentioned In	n/a

CTRL-TRACEABILITY-COVERAGE

category=assurance; description=Require requirement-linked verification evidence for modeled behavior.

Key	Value
Kind	Control
Name	Requirement Traceability Coverage
Mentioned In	Security View

DO-ARCHITECTURE-MODEL

termRef=DATA-ARCHITECTURE-MODEL; schemaRef=architecture.yml; sensitivity=internal

Key	Value
Kind	Data Object
Name	Architecture Model
Mentioned In	Security View , REQ-EMG-001

DO-MCP-TOOL-RESULT

termRef=DATA-MCP-TOOL-RESULT; schemaRef=mcp.tool-response.v1; sensitivity=internal

Key	Value
Kind	Data Object
Name	MCP Tool Result
Mentioned In	Security View , REQ-EMG-007

DO-STRUCTURIZR-DSL

termRef=DATA-STRUCTURIZR-DSL; schemaRef=generated/STRUCTURIZR.dsl; sensitivity=internal

Key	Value
Kind	Data Object
Name	Structurizr DSL
Mentioned In	Architecture Intent View , Security View , REQ-EMG-005

DO-THREAT-DRAGON-JSON

termRef=DATA-THREAT-DRAGON-JSON; schemaRef=generated/threat-dragon-v2.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Threat Dragon JSON
Mentioned In	Security View

DEP-CI-PIPELINE

environment=ci; region=hosted; account=github-actions; cluster=shared-runner; namespace=engineering-model-go; trustZone=automation

Key	Value
Kind	Deployment Target
Name	CI Pipeline Runner
Mentioned In	n/a

DEP-LOCAL-WORKSPACE

environment=dev; region=local; account=workstation; cluster=local-shell; namespace=engineering-model-go; trustZone=developer

Key	Value
Kind	Deployment Target
Name	Local Workspace
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS

entry=edit-model; exits=publish-artifacts; steps=5

Key	Value
Kind	Flow
Name	Model Change to Verified Artifacts Flow
Mentioned In	Security View

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::edit-model

kind=user_action; ref=ACT-IMPLEMENTATION-ENGINEER; action=Update model and requirements artifacts; dataIn=; dataOut=architecture_update; async=false

Key	Value
Kind	Flow Step
Name	Update model and requirements artifacts
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::fail-validation

kind=system_action; ref=FU-VALIDATION-ENGINE; action=Emit validation diagnostics and block publication; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Emit validation diagnostics and block publication
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::generate-artifacts

kind=system_action; ref=FU-CLI-ORCHESTRATION; action=Run generation and export commands; dataIn=; dataOut=generated_artifacts; async=false

Key	Value
Kind	Flow Step
Name	Run generation and export commands
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::publish-artifacts

kind=system_action; ref=FU-ASCIIDOC-GENERATOR; action=Persist generated outputs for downstream use; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Persist generated outputs for downstream use
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::validate-model

kind=system_action; ref=FU-VALIDATION-ENGINE; action=Validate authored model references and relationships; dataIn=architecture_update; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Validate authored model references and relationships
Mentioned In	n/a

IF-CLI-ENGDOC

protocol=cli; endpoint=cmd/engdoc; schemaRef=docs/asciidoc-architecture-generator-design.md; owner=FU-ASCIIDOC-GENERATOR

Key	Value
Kind	Interface
Name	engdoc CLI
Mentioned In	n/a

IF-CLI-ENGDRAGON

protocol=cli; endpoint=cmd/engdragon; schemaRef=threat_model_export.go; owner=FU-THREAT-EXPORTER

Key	Value
Kind	Interface
Name	engdragon CLI
Mentioned In	n/a

IF-CLI-ENGGEMARA

protocol=cli; endpoint=cmd/enggemara; schemaRef=gemara_export.go; owner=FU-GEMARA-EXPORTER

Key	Value
Kind	Interface
Name	enggemara CLI
Mentioned In	n/a

IF-CLI-ENGLOBSTER

protocol=cli; endpoint=cmd/englobster; schemaRef=lobster_export.go; owner=FU-LOBSTER-EXPORTER

Key	Value
Kind	Interface
Name	englobster CLI
Mentioned In	n/a

IF-CLI-ENGMCP

protocol=stdio-jsonrpc; endpoint=cmd/engmcp; schemaRef=mcp/server.go; owner=FU-MCP-SERVER

Key	Value
Kind	Interface
Name	engmcp CLI
Mentioned In	n/a

IF-CLI-ENGOSCAL

protocol=cli; endpoint=cmd/engoscal; schemaRef=oscal_ssp.go; owner=FU-OSCAL-EXPORTER

Key	Value
Kind	Interface
Name	engoscal CLI
Mentioned In	n/a

IF-CLI-ENGSTRUCT

protocol=cli; endpoint=cmd/engstruct; schemaRef=structurizr_export.go; owner=FU-STRUCTURIZR-EXPORTER

Key	Value
Kind	Interface
Name	engstruct CLI
Mentioned In	n/a

IF-CLI-ENGTRACE

protocol=cli; endpoint=cmd/engtrace; schemaRef=trace_matrix.go; owner=FU-ALLOCATION-TRACE

Key	Value
Kind	Interface

Key	Value
Name	engtrace CLI
Mentioned In	n/a

IF-CLI-ENGTRLC

protocol=cli; endpoint=cmd/engtrlc; schemaRef=trlc_export.go; owner=FU-TRLC-EXPORTER

Key	Value
Kind	Interface
Name	engtrlc CLI
Mentioned In	n/a

IF-CLI-ENGVIEW

protocol=cli; endpoint=cmd/engview; schemaRef=view/project.go; owner=FU-VIEW-PROJECTION

Key	Value
Kind	Interface
Name	engview CLI
Mentioned In	n/a

REQ-EMG-001

While local development mode is active, when model updated event is received, the engineering model go shall validate architecture model before generation run is requested.

Key	Value
Kind	Requirement
Name	REQ-EMG-001
Mentioned In	Architecture Intent View , Interaction Flow View , Catalog Systems Test , E2e Test , Ears Preflight Test , Load Test , Validate Test

REQ-EMG-003

While model is valid, when generation run is requested, the engineering model go shall produce AsciiDoc architecture publication output.

Key	Value
Kind	Requirement

Key	Value
Name	REQ-EMG-003
Mentioned In	Architecture Intent View , Interaction Flow View , Asciidoc Diagrams Core Test , Asciidoc Diagrams Runtime Test , Asciidoc Linking Units Test , Asciidoc Test , Asciidoc Views Test , Project Test , Render Test

REQ-EMG-004

Where threat export is enabled, when generation run is requested, the engineering model go shall export threat dragon json artifact plus open threat model artifact.

Key	Value
Kind	Requirement
Name	REQ-EMG-004
Mentioned In	Architecture Intent View , Interaction Flow View , Threat Model Export Test

REQ-EMG-005

Where structurizr export is enabled, when generation run is requested, the engineering model go shall export structurizr dsl artifact.

Key	Value
Kind	Requirement
Name	REQ-EMG-005
Mentioned In	Architecture Intent View , Interaction Flow View , Structurizr Export Test

REQ-EMG-006

Where traceability export is enabled, when generation run is requested, the engineering model go shall produce trlc package and lobster trace report artifacts.

Key	Value
Kind	Requirement
Name	REQ-EMG-006
Mentioned In	Architecture Intent View , Interaction Flow View , Lobster Export Test , Trlc Export Test

REQ-EMG-007

While mcp server is enabled, when mcp tool call is received and catalog references resolve successfully, the engineering model go shall return mcp tool result with schema version and structured error fields.

Key	Value
Kind	Requirement
Name	REQ-EMG-007
Mentioned In	Architecture Intent View, Interaction Flow View, Composition Test, Main Test, Server Test

REQ-EMG-008

When mcp tool call is received, the engineering model go shall reject path arguments outside repository workspace boundary.

Key	Value
Kind	Requirement
Name	REQ-EMG-008
Mentioned In	Architecture Intent View, Interaction Flow View, Main Test, Server Test

REQ-EMG-009

While ci validation mode is active, when generation run is requested, the engineering model go shall execute validation plus tests before artifact generated event is raised.

Key	Value
Kind	Requirement
Name	REQ-EMG-009
Mentioned In	Architecture Intent View, Interaction Flow View, Ears Preflight Test, Validate Test

REQ-EMG-010

Where requirement trace is complete, when model updated event is received, the engineering model go shall infer traceability coverage from trace markers.

Key	Value
Kind	Requirement
Name	REQ-EMG-010
Mentioned In	Architecture Intent View, Interaction Flow View, Inferred Code Test, Inferred Layers Test, Inferred Verification Test, Scan Test

REQ-EMG-011

Where external schema dependencies are available, when generation run is requested, the engineering model go shall fail validation if schema checks do not match expected constraints.

Key	Value
Kind	Requirement
Name	REQ-EMG-011
Mentioned In	Architecture Intent View, Interaction Flow View, Threat Model Export Test, Validate Test

REQ-EMG-012

When generation run is requested, the engineering model go shall generate artifacts deterministically so that regeneration is reproducible.

Key	Value
Kind	Requirement
Name	REQ-EMG-012
Mentioned In	Architecture Intent View

REQ-EMG-013

Where traceability export is enabled, when generation run is requested, the engineering model go shall export OSCAL SSP plus assessment results plus POA&M artifacts.

Key	Value
Kind	Requirement
Name	REQ-EMG-013
Mentioned In	Architecture Intent View, Interaction Flow View, Oscal Chain Test, Oscal Ssp Test

REQ-EMG-014

Where architecture decision records are authored, when generation run is requested, the engineering model go shall publish architecture decision records document linked from AsciiDoc architecture publication output.

Key	Value
Kind	Requirement
Name	REQ-EMG-014
Mentioned In	Architecture Intent View, Interaction Flow View, Asciidoc Test

REQ-EMG-015

Where gemara export is enabled, when generation run is requested, the engineering model go shall render OpenSSF Gemara model artifacts.

Key	Value
Kind	Requirement
Name	REQ-EMG-015
Mentioned In	Architecture Intent View, Interaction Flow View, Gemara Export Test, Gemara Oscal Test

REQ-EMG-016

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

Key	Value
Kind	Requirement
Name	REQ-EMG-016
Mentioned In	Architecture Intent View, Interaction Flow View, Composition Test, Composition Test

REQ-EMG-017

Where system composition is enabled, when generation run is requested, the engineering model go shall reject any subsystem reference that resolves outside the repository workspace boundary.

Key	Value
Kind	Requirement
Name	REQ-EMG-017
Mentioned In	Architecture Intent View, Interaction Flow View, Composition Test

REQ-EMG-018

Where system composition is enabled, when generation run is requested, the engineering model go shall reject a subsystem reference graph that contains a cycle.

Key	Value
Kind	Requirement
Name	REQ-EMG-018
Mentioned In	Architecture Intent View

REQ-EMG-019

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve cross-system references using subsystem-qualified identifiers.

Key	Value
Kind	Requirement
Name	REQ-EMG-019
Mentioned In	Architecture Intent View

REQ-EMG-020

Where system composition is enabled, when generation run is requested, the engineering model go shall materialize parent-to-subsystem allocation traceability in the composed output without modifying any subsystem model.

Key	Value
Kind	Requirement
Name	REQ-EMG-020
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test

REQ-EMG-021

Where system composition is enabled, when generation run is requested, the engineering model go shall report each subsystem required interface that the parent system does not satisfy.

Key	Value
Kind	Requirement
Name	REQ-EMG-021
Mentioned In	Architecture Intent View

REQ-EMG-022

Where system composition is enabled, when generation run is requested, the engineering model go shall reject an allocation that targets a subsystem identifier not published in the subsystem contract.

Key	Value
Kind	Requirement
Name	REQ-EMG-022
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test

REQ-EMG-023

When generation run is requested, the engineering model go shall flag any requirement that links to another requirement in the same document.

Key	Value
Kind	Requirement
Name	REQ-EMG-023
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test

REQ-EMG-024

Where hardware interfaces are authored, when generation run is requested, the engineering model go shall render the hardware to software interface allocation view.

Key	Value
Kind	Requirement
Name	REQ-EMG-024
Mentioned In	Architecture Intent View

REQ-EMG-025

Where system composition is enabled, when generation run is requested, the engineering model go shall produce a system requirement allocation matrix aggregated across the composed subsystems.

Key	Value
Kind	Requirement
Name	REQ-EMG-025
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test

REQ-EMG-026

Where system composition is enabled, when generation run is requested, the engineering model go shall render in the requirement view the delegation link from each delegated requirement to the specific subsystem requirement that satisfies it.

Key	Value
Kind	Requirement
Name	REQ-EMG-026
Mentioned In	Architecture Intent View

REQ-EMG-027

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve an external git subsystem reference by cloning the repository into the workspace .engmod cache.

Key	Value
Kind	Requirement
Name	REQ-EMG-027
Mentioned In	Architecture Intent View , Interaction Flow View , REQ-EMG-016 , Composition Test

REQ-EMG-028

When generation run is requested, the engineering model go shall reject a code trace link that resolves to no requirement or model element.

Key	Value
Kind	Requirement
Name	REQ-EMG-028
Mentioned In	Architecture Intent View , Interaction Flow View , Trace Matrix Test

REQ-EMG-029

When generation run is requested, the engineering model go shall flag each requirement that is traced by neither code nor verification nor a downward delegation.

Key	Value
Kind	Requirement
Name	REQ-EMG-029
Mentioned In	Architecture Intent View , Interaction Flow View , Trace Matrix Test

REQ-EMG-030

When generation run is requested, the engineering model go shall export a machine-readable traceability matrix that links each requirement to its implementing code and verification and delegated subsystem requirement.

Key	Value
Kind	Requirement
Name	REQ-EMG-030
Mentioned In	Architecture Intent View , Interaction Flow View , REQ-EMG-028 , Composition Test , Trace Matrix Test

TB-EXTERNAL-VALIDATION-TOOLS

Boundary for external validators, schemas, and compliance toolchains.

Key	Value
Kind	Trust Boundary
Name	External Validation Tools Boundary
Mentioned In	n/a

TB-REPO-WORKSPACE

Boundary limiting file operations to repository-root owned paths.

Key	Value
Kind	Trust Boundary
Name	Repository Workspace Boundary
Mentioned In	REQ-EMG-008, REQ-EMG-017

13.2. Catalog References

ACT-AI-AGENT

Consumes MCP outputs to plan and execute model-guided edits.

Key	Value
Kind	Catalog Actor
Name	ai agent
Mentioned In	n/a

ACT-ARCHITECTURE-AUTHOR

Maintains model structure, mappings, and architecture intent.

Key	Value
Kind	Catalog Actor
Name	architecture author
Mentioned In	Security View

ACT-CI-PIPELINE

Runs automated validation and export checks for pull requests.

Key	Value
Kind	Catalog Actor
Name	ci pipeline
Mentioned In	n/a

ACT-COMPLIANCE-ENGINEER

Reviews controls, risks, and traceability outputs for compliance needs.

Key	Value
Kind	Catalog Actor
Name	compliance engineer
Mentioned In	Security View, REQ-EMG-006

ACT-IMPLEMENTATION-ENGINEER

Implements code changes guided by model requirements and support paths.

Key	Value
Kind	Catalog Actor
Name	implementation engineer
Mentioned In	Security View

adr document

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Alias
Name	adr document
Mentioned In	n/a

adrs authored

Feature indicating the architecture model includes first-class ADR entries.

Key	Value
Kind	Catalog Alias

Key	Value
Name	adrs authored
Mentioned In	n/a

allocation

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

Key	Value
Kind	Catalog Alias
Name	allocation
Mentioned In	Architecture Intent View, REQ-EMG-020, REQ-EMG-022, REQ-EMG-023, REQ-EMG-024, REQ-EMG-025

analysis

Validation, inference, and evidence analysis over authored and inferred model content.

Key	Value
Kind	Catalog Alias
Name	analysis
Mentioned In	Security View

architecture yml

Authored architecture source describing functional intent and mappings.

Key	Value
Kind	Catalog Alias
Name	architecture yml
Mentioned In	n/a

artifact generated

Event indicating one or more generated outputs were produced successfully.

Key	Value
Kind	Catalog Alias
Name	artifact generated

Key	Value
Mentioned In	n/a

assurance

Reviews controls, risks, and traceability outputs for compliance needs.

Key	Value
Kind	Catalog Alias
Name	assurance
Mentioned In	n/a

authoring

Authoring and loading of architecture, catalog, requirements, and design inputs.

Key	Value
Kind	Catalog Alias
Name	authoring
Mentioned In	n/a

bundle loader

Loads and normalizes architecture, catalog, requirements, and design documents.

Key	Value
Kind	Catalog Alias
Name	bundle loader
Mentioned In	n/a

catalog resolved

Condition requiring all authored references to match known catalog entries.

Key	Value
Kind	Catalog Alias
Name	catalog resolved
Mentioned In	n/a

ci

Runs automated validation and export checks for pull requests.

Key	Value
Kind	Catalog Alias
Name	ci
Mentioned In	n/a

ci mode

Automated pipeline mode focused on deterministic checks and artifacts.

Key	Value
Kind	Catalog Alias
Name	ci mode
Mentioned In	n/a

cmd layer

Command entrypoints that orchestrate loading, validation, generation, and export flows.

Key	Value
Kind	Catalog Alias
Name	cmd layer
Mentioned In	n/a

coding agent

Consumes MCP outputs to plan and execute model-guided edits.

Key	Value
Kind	Catalog Alias
Name	coding agent
Mentioned In	n/a

compliance

Requirement trace exports and compliance-oriented output generation.

Key	Value
Kind	Catalog Alias
Name	compliance
Mentioned In	Architecture Intent View, REQ-EMG-013

composition

Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model.

Key	Value
Kind	Catalog Alias
Name	composition
Mentioned In	REQ-EMG-016, REQ-EMG-023

decisions document

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Alias
Name	decisions document
Mentioned In	n/a

dsl

Generated Structurizr workspace DSL document.

Key	Value
Kind	Catalog Alias
Name	dsl
Mentioned In	n/a

dsl export

Feature enabling Structurizr DSL generation and validation.

Key	Value
Kind	Catalog Alias

Key	Value
Name	dsl export
Mentioned In	n/a

engdoc

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

Key	Value
Kind	Catalog Alias
Name	engdoc
Mentioned In	n/a

engdragon

Exports threat model views to Threat Dragon and Open OTM representations.

Key	Value
Kind	Catalog Alias
Name	engdragon
Mentioned In	n/a

enggemara

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

Key	Value
Kind	Catalog Alias
Name	enggemara
Mentioned In	n/a

engineer

Implements code changes guided by model requirements and support paths.

Key	Value
Kind	Catalog Alias
Name	engineer

Key	Value
Mentioned In	n/a

engineering model toolchain

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog Alias
Name	engineering model toolchain
Mentioned In	n/a

englobster

Generates LOBSTER traceability inputs and reports from model and verification links.

Key	Value
Kind	Catalog Alias
Name	englobster
Mentioned In	n/a

engmcp

Serves model-backed MCP tools with validated inputs and structured responses.

Key	Value
Kind	Catalog Alias
Name	engmcp
Mentioned In	Security View , Main Test

engmodel

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog Alias
Name	engmodel
Mentioned In	n/a

engoscal

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

Key	Value
Kind	Catalog Alias
Name	engoscal
Mentioned In	n/a

engstruct

Exports Structurizr DSL and supports validation/static-site workflows.

Key	Value
Kind	Catalog Alias
Name	engstruct
Mentioned In	n/a

engtrlc

Exports requirements and model structures to TRLC artifacts.

Key	Value
Kind	Catalog Alias
Name	engtrlc
Mentioned In	n/a

gemara generation

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

Key	Value
Kind	Catalog Alias
Name	gemara generation
Mentioned In	n/a

generation

Generation of AsciiDoc and exchange artifacts from model inputs.

Key	Value
Kind	Catalog Alias
Name	generation
Mentioned In	Architecture Intent View, REQ-EMG-001, REQ-EMG-012, REQ-EMG-025

generation requested

Event indicating documentation or export generation execution starts.

Key	Value
Kind	Catalog Alias
Name	generation requested
Mentioned In	n/a

go

Go compiler, test runner, and module tooling used by the repository.

Key	Value
Kind	Catalog Alias
Name	go
Mentioned In	n/a

hw interfaces

Feature enabling hardware items and hardware/software interface modeling.

Key	Value
Kind	Catalog Alias
Name	hw interfaces
Mentioned In	n/a

inference

Infers runtime, code, and verification evidence from repository sources and test markers.

Key	Value
Kind	Catalog Alias

Key	Value
Name	inference
Mentioned In	Architecture Intent View , Security View , Policy Only Fallback Test

invalid

State where validation diagnostics include one or more errors.

Key	Value
Kind	Catalog Alias
Name	invalid
Mentioned In	Security View

invalid model

Invalid or ambiguous model content that causes incorrect graph construction.

Key	Value
Kind	Catalog Alias
Name	invalid model
Mentioned In	n/a

lobster

LOBSTER core and TRLC integration tools used for traceability reporting.

Key	Value
Kind	Catalog Alias
Name	lobster
Mentioned In	Architecture Intent View

lobster report

Traceability report output generated from TRLC and test links.

Key	Value
Kind	Catalog Alias
Name	lobster report

Key	Value
Mentioned In	n/a

local mode

Local developer mode for iterative model and code updates.

Key	Value
Kind	Catalog Alias
Name	local mode
Mentioned In	n/a

mcp

MCP server interfaces that expose model-aware tooling for AI agents.

Key	Value
Kind	Catalog Alias
Name	mcp
Mentioned In	n/a

mcp enabled

Feature enabling MCP runtime APIs for model-aware tooling.

Key	Value
Kind	Catalog Alias
Name	mcp enabled
Mentioned In	n/a

model author

Maintains model structure, mappings, and architecture intent.

Key	Value
Kind	Catalog Alias
Name	model author
Mentioned In	n/a

model change

Event indicating architecture or requirement model source changed.

Key	Value
Kind	Catalog Alias
Name	model change
Mentioned In	n/a

otm schema

Open Threat Model schema used to validate OTM output artifacts.

Key	Value
Kind	Catalog Alias
Name	otm schema
Mentioned In	n/a

path traversal

Host file access attempts through untrusted path arguments in MCP tools.

Key	Value
Kind	Catalog Alias
Name	path traversal
Mentioned In	Security View

projection

Builds view-specific projections for architecture communication and traceability outputs.

Key	Value
Kind	Catalog Alias
Name	projection
Mentioned In	Architecture Intent View

schema available

Condition requiring external schema artifacts to be present for validation.

Key	Value
Kind	Catalog Alias
Name	schema available
Mentioned In	n/a

schema tamper

Tampering or drift in external schemas used for export validation.

Key	Value
Kind	Catalog Alias
Name	schema tamper
Mentioned In	n/a

ssp json

Generated OSCAL system security plan JSON output.

Key	Value
Kind	Catalog Alias
Name	ssp json
Mentioned In	n/a

structurizr

Structurizr DSL validation runtime used to verify generated DSL.

Key	Value
Kind	Catalog Alias
Name	structurizr
Mentioned In	n/a

system of systems

Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems.

Key	Value
Kind	Catalog Alias

Key	Value
Name	system of systems
Mentioned In	n/a

threat dragon

External JSON schemas used to validate Threat Dragon exports.

Key	Value
Kind	Catalog Alias
Name	threat dragon
Mentioned In	Architecture Intent View

threat dragon output

Generated threat model artifact for OWASP Threat Dragon.

Key	Value
Kind	Catalog Alias
Name	threat dragon output
Mentioned In	n/a

threat modeling export

Feature enabling Threat Dragon and Open OTM exports.

Key	Value
Kind	Catalog Alias
Name	threat modeling export
Mentioned In	n/a

tool call

Event indicating the MCP server received a tool invocation request.

Key	Value
Kind	Catalog Alias
Name	tool call

Key	Value
Mentioned In	n/a

tool result

Structured MCP tool response payload carrying model-aware query results.

Key	Value
Kind	Catalog Alias
Name	tool result
Mentioned In	n/a

trace complete

Condition requiring requirements to map to implementation and verification evidence.

Key	Value
Kind	Catalog Alias
Name	trace complete
Mentioned In	n/a

trace drift

Requirement links or tests drifting from implementation over time.

Key	Value
Kind	Catalog Alias
Name	trace drift
Mentioned In	n/a

trace export

Feature enabling TRLC and LOBSTER generation flows.

Key	Value
Kind	Catalog Alias
Name	trace export
Mentioned In	n/a

trlc

TRLC parser and runtime used for requirements language validation.

Key	Value
Kind	Catalog Alias
Name	trlc
Mentioned In	Architecture Intent View

trlc output

Generated TRLC requirements package and model metadata.

Key	Value
Kind	Catalog Alias
Name	trlc output
Mentioned In	n/a

up to date

State where generated outputs reflect the latest authored model inputs.

Key	Value
Kind	Catalog Alias
Name	up to date
Mentioned In	n/a

valid

State where authored model passes validation checks.

Key	Value
Kind	Catalog Alias
Name	valid
Mentioned In	n/a

validation failure

Event indicating model or export validation failed.

Key	Value
Kind	Catalog Alias
Name	validation failure
Mentioned In	n/a

validator

Validates IDs, references, and model consistency across authored entities and mappings.

Key	Value
Kind	Catalog Alias
Name	validator
Mentioned In	n/a

AV-MALFORMED-MODEL-INPUT

Invalid or ambiguous model content that causes incorrect graph construction.

Key	Value
Kind	Catalog Attack Vector
Name	malformed model input
Mentioned In	n/a

AV-PATH-TRAVERSAL-IN-MCP

Host file access attempts through untrusted path arguments in MCP tools.

Key	Value
Kind	Catalog Attack Vector
Name	path traversal in mcp calls
Mentioned In	Security View

AV-SCHEMA-SUPPLY-CHAIN-TAMPER

Tampering or drift in external schemas used for export validation.

Key	Value
Kind	Catalog Attack Vector

Key	Value
Name	schema supply chain tamper
Mentioned In	n/a

AV-TRACEABILITY-GAP-DRIFT

Requirement links or tests drifting from implementation over time.

Key	Value
Kind	Catalog Attack Vector
Name	traceability gap drift
Mentioned In	Security View

COND-CATALOG-REFERENCES-RESOLVE

Condition requiring all authored references to match known catalog entries.

Key	Value
Kind	Catalog Condition
Name	catalog references resolve successfully
Mentioned In	REQ-EMG-007

COND-EXTERNAL-SCHEMA-AVAILABLE

Condition requiring external schema artifacts to be present for validation.

Key	Value
Kind	Catalog Condition
Name	external schema dependencies are available
Mentioned In	REQ-EMG-011

COND-REQUIREMENT-TRACE-COMPLETE

Condition requiring requirements to map to implementation and verification evidence.

Key	Value
Kind	Catalog Condition
Name	requirement trace is complete

Key	Value
Mentioned In	`REQ-EMG-010`

DATA-ARCHITECTURE-DECISION-RECORDS

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Data Term
Name	architecture decision records document
Mentioned In	`REQ-EMG-014`

DATA-ARCHITECTURE-MODEL

Authored architecture source describing functional intent and mappings.

Key	Value
Kind	Catalog Data Term
Name	architecture model
Mentioned In	n/a

DATA-LOBSTER-TRACE-REPORT

Traceability report output generated from TRLC and test links.

Key	Value
Kind	Catalog Data Term
Name	lobster trace report
Mentioned In	`REQ-EMG-006`

DATA-MCP-TOOL-RESULT

Structured MCP tool response payload carrying model-aware query results.

Key	Value
Kind	Catalog Data Term
Name	mcp tool result
Mentioned In	n/a

DATA-OSCAL-SSP

Generated OSCAL system security plan JSON output.

Key	Value
Kind	Catalog Data Term
Name	oscal ssp artifact
Mentioned In	n/a

DATA-STRUCTURIZR-DSL

Generated Structurizr workspace DSL document.

Key	Value
Kind	Catalog Data Term
Name	structurizr dsl artifact
Mentioned In	`REQ-EMG-005`

DATA-THREAT-DRAGON-JSON

Generated threat model artifact for OWASP Threat Dragon.

Key	Value
Kind	Catalog Data Term
Name	threat dragon json artifact
Mentioned In	`REQ-EMG-004`

DATA-TRLC-PACKAGE

Generated TRLC requirements package and model metadata.

Key	Value
Kind	Catalog Data Term
Name	trlc package
Mentioned In	n/a

EVT-ARTIFACT-GENERATED

Event indicating one or more generated outputs were produced successfully.

Key	Value
Kind	Catalog Event
Name	artifact generated event is raised
Mentioned In	REQ-EMG-009

EVT-GENERATION-RUN-REQUESTED

Event indicating documentation or export generation execution starts.

Key	Value
Kind	Catalog Event
Name	generation run is requested
Mentioned In	REQ-EMG-001, REQ-EMG-003, REQ-EMG-004, REQ-EMG-005, REQ-EMG-006, REQ-EMG-009, REQ-EMG-011, REQ-EMG-012, REQ-EMG-013, REQ-EMG-014, REQ-EMG-015, REQ-EMG-016, REQ-EMG-017, REQ-EMG-018, REQ-EMG-019, REQ-EMG-020, REQ-EMG-021, REQ-EMG-022, REQ-EMG-023, REQ-EMG-024, REQ-EMG-025, REQ-EMG-026, REQ-EMG-027, REQ-EMG-028, REQ-EMG-029, REQ-EMG-030

EVT-MCP-TOOL-CALL-RECEIVED

Event indicating the MCP server received a tool invocation request.

Key	Value
Kind	Catalog Event
Name	mcp tool call is received
Mentioned In	REQ-EMG-007, REQ-EMG-008

EVT-MODEL-UPDATED

Event indicating architecture or requirement model source changed.

Key	Value
Kind	Catalog Event
Name	model updated event is received
Mentioned In	REQ-EMG-001, REQ-EMG-010

EVT-VALIDATION-FAILED

Event indicating model or export validation failed.

Key	Value
Kind	Catalog Event
Name	validation failed event is raised
Mentioned In	n/a

FEAT-ARCHITECTURE-DECISION-RECORDS

Feature indicating the architecture model includes first-class ADR entries.

Key	Value
Kind	Catalog Feature
Name	architecture decision records are authored
Mentioned In	`REQ-EMG-014`

FEAT-GEMARA-EXPORT

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

Key	Value
Kind	Catalog Feature
Name	gemara export is enabled
Mentioned In	`REQ-EMG-015`

FEAT-HARDWARE-INTERFACES

Feature enabling hardware items and hardware/software interface modeling.

Key	Value
Kind	Catalog Feature
Name	hardware interfaces are authored
Mentioned In	`REQ-EMG-024`

FEAT-MCP-SERVER

Feature enabling MCP runtime APIs for model-aware tooling.

Key	Value
Kind	Catalog Feature

Key	Value
Name	mcp server is enabled
Mentioned In	REQ-EMG-007

FEAT-STRUCTURIZR-EXPORT

Feature enabling Structurizr DSL generation and validation.

Key	Value
Kind	Catalog Feature
Name	structurizr export is enabled
Mentioned In	REQ-EMG-005

FEAT-SYSTEM-COMPOSITION

Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems.

Key	Value
Kind	Catalog Feature
Name	system composition is enabled
Mentioned In	REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027

FEAT-THREAT-EXPORT

Feature enabling Threat Dragon and Open OTM exports.

Key	Value
Kind	Catalog Feature
Name	threat export is enabled
Mentioned In	REQ-EMG-004

FEAT-TRACEABILITY-EXPORT

Feature enabling TRLC and LOBSTER generation flows.

Key	Value
Kind	Catalog Feature
Name	traceability export is enabled

Key	Value
Mentioned In	REQ-EMG-006 , REQ-EMG-013

FG-ARTIFACT-GENERATION

Generation of AsciiDoc and exchange artifacts from model inputs.

Key	Value
Kind	Catalog Functional Group
Name	artifact generation
Mentioned In	Architecture Intent View

FG-MCP-INTEGRATION

MCP server interfaces that expose model-aware tooling for AI agents.

Key	Value
Kind	Catalog Functional Group
Name	mcp integration
Mentioned In	n/a

FG-MODEL-AUTHORING

Authoring and loading of architecture, catalog, requirements, and design inputs.

Key	Value
Kind	Catalog Functional Group
Name	model authoring
Mentioned In	n/a

FG-TRACEABILITY-COMPLIANCE

Requirement trace exports and compliance-oriented output generation.

Key	Value
Kind	Catalog Functional Group
Name	traceability and compliance
Mentioned In	n/a

FG-VALIDATION-ANALYSIS

Validation, inference, and evidence analysis over authored and inferred model content.

Key	Value
Kind	Catalog Functional Group
Name	validation and analysis
Mentioned In	n/a

FU-ALLOCATION-TRACE

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

Key	Value
Kind	Catalog Functional Unit
Name	allocation trace
Mentioned In	Architecture Intent View , REQ-EMG-020 , Composition Test , Composition Test , Trace Matrix Test

FU-ASCIIDOC-GENERATOR

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

Key	Value
Kind	Catalog Functional Unit
Name	asciidoc generator
Mentioned In	Architecture Intent View , Security View , Traceability View , Asciidoc Diagrams Core Test , Asciidoc Diagrams Runtime Test , Asciidoc Linking Units Test , Asciidoc Test , Asciidoc Views Test , Catalog Systems Test , Composition Test , E2e Test , Ears Preflight Test , Load Test , Project Test , Render Test , Validate Test

FU-CLI-ORCHESTRATION

Command entrypoints that orchestrate loading, validation, generation, and export flows.

Key	Value
Kind	Catalog Functional Unit
Name	cli orchestration

Key	Value
Mentioned In	Architecture Intent View , Security View , Traceability View , Asciidoc Diagrams Core Test , Asciidoc Diagrams Runtime Test , Asciidoc Linking Units Test , Asciidoc Test , Asciidoc Views Test , Catalog Systems Test , E2e Test , Ears Preflight Test , Gemara Export Test , Gemara Oscal Test , Load Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Project Test , Render Test , Structurizr Export Test , Threat Model Export Test , Trlc Export Test , Validate Test

FU-CODEMAP-INFERENCE

Infers runtime, code, and verification evidence from repository sources and test markers.

Key	Value
Kind	Catalog Functional Unit
Name	codemap inference
Mentioned In	Architecture Intent View , Security View , Traceability View , Composition Test , Inferred Code Test , Inferred Layers Test , Inferred Verification Test , Scan Test , Trace Matrix Test

FU-GEMARA-EXPORTER

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

Key	Value
Kind	Catalog Functional Unit
Name	gemara exporter
Mentioned In	Architecture Intent View , Gemara Export Test , Gemara Oscal Test

FU-LOBSTER-EXPORTER

Generates LOBSTER traceability inputs and reports from model and verification links.

Key	Value
Kind	Catalog Functional Unit
Name	lobster exporter
Mentioned In	Architecture Intent View , Security View , Traceability View , Lobster Export Test , Trlc Export Test

FU-MCP-SERVER

Serves model-backed MCP tools with validated inputs and structured responses.

Key	Value
Kind	Catalog Functional Unit
Name	mcp server
Mentioned In	Composition Test, Gemara Export Test, Gemara Oscal Test, Main Test, Server Test

FU-MODEL-LOADER

Loads and normalizes architecture, catalog, requirements, and design documents.

Key	Value
Kind	Catalog Functional Unit
Name	model loader
Mentioned In	Architecture Intent View, Security View, Traceability View, Catalog Systems Test, Composition Test, Composition Test, E2e Test, Ears Preflight Test, Load Test, Validate Test

FU-OSCAL-EXPORTER

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

Key	Value
Kind	Catalog Functional Unit
Name	oscal exporter
Mentioned In	Architecture Intent View, Catalog Systems Test, E2e Test, Ears Preflight Test, Load Test, Oscal Chain Test, Oscal Ssp Test, Validate Test

FU-STRUCTURIZR-EXPORTER

Exports Structurizr DSL and supports validation/static-site workflows.

Key	Value
Kind	Catalog Functional Unit
Name	structurizr exporter
Mentioned In	Architecture Intent View, Security View, Catalog Systems Test, E2e Test, Ears Preflight Test, Load Test, Structurizr Export Test, Threat Model Export Test, Validate Test

FU-SYSTEM-COMPOSITION

Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model.

Key	Value
Kind	Catalog Functional Unit
Name	system composition
Mentioned In	Architecture Intent View, Composition Test, Composition Test

FU-THREAT-EXPORTER

Exports threat model views to Threat Dragon and Open OTM representations.

Key	Value
Kind	Catalog Functional Unit
Name	threat exporter
Mentioned In	Architecture Intent View, Security View, Catalog Systems Test, E2e Test, Ears Preflight Test, Load Test, Threat Model Export Test, Validate Test

FU-TRLC-EXPORTER

Exports requirements and model structures to TRLC artifacts.

Key	Value
Kind	Catalog Functional Unit
Name	trlc exporter
Mentioned In	Architecture Intent View, Security View, Traceability View, Lobster Export Test, Trlc Export Test

FU-VALIDATION-ENGINE

Validates IDs, references, and model consistency across authored entities and mappings.

Key	Value
Kind	Catalog Functional Unit
Name	validation engine
Mentioned In	Architecture Intent View, Security View, Traceability View, Catalog Systems Test, Composition Test, E2e Test, Ears Preflight Test, Load Test, Threat Model Export Test, Trace Matrix Test, Validate Test

FU-VIEW-PROJECTION

Builds view-specific projections for architecture communication and traceability outputs.

Key	Value
Kind	Catalog Functional Unit
Name	view projection
Mentioned In	Architecture Intent View, Asciidoc Diagrams Core Test, Asciidoc Diagrams Runtime Test, Asciidoc Linking Units Test, Asciidoc Test, Asciidoc Views Test, Project Test, Render Test

MODE-CI-VALIDATION

Automated pipeline mode focused on deterministic checks and artifacts.

Key	Value
Kind	Catalog Mode
Name	ci validation mode is active
Mentioned In	REQ-EMG-009

MODE-LOCAL-DEVELOPMENT

Local developer mode for iterative model and code updates.

Key	Value
Kind	Catalog Mode
Name	local development mode is active
Mentioned In	REQ-EMG-001

REF-GO-TOOLCHAIN

Go compiler, test runner, and module tooling used by the repository.

Key	Value
Kind	Catalog Referenced Element
Name	go toolchain
Mentioned In	Architecture Intent View

REF-LOBSTER-TOOLCHAIN

LOBSTER core and TRLC integration tools used for traceability reporting.

Key	Value
Kind	Catalog Referenced Element

Key	Value
Name	lobster toolchain
Mentioned In	Architecture Intent View

REF-OPEN-OTM-SCHEMA

Open Threat Model schema used to validate OTM output artifacts.

Key	Value
Kind	Catalog Referenced Element
Name	open otm schema
Mentioned In	Architecture Intent View

REF-STRUCTURIZR-VALIDATOR

Structurizr DSL validation runtime used to verify generated DSL.

Key	Value
Kind	Catalog Referenced Element
Name	structurizr validator
Mentioned In	Architecture Intent View, REQ-EMG-005

REF-THREAT-DRAGON-SCHEMAS

External JSON schemas used to validate Threat Dragon exports.

Key	Value
Kind	Catalog Referenced Element
Name	threat dragon schemas
Mentioned In	Architecture Intent View

REF-TRLC-TOOLCHAIN

TRLC parser and runtime used for requirements language validation.

Key	Value
Kind	Catalog Referenced Element
Name	trlc toolchain

Key	Value
Mentioned In	Architecture Intent View

STATE-ARTIFACTS-FRESH

State where generated outputs reflect the latest authored model inputs.

Key	Value
Kind	Catalog State
Name	generated artifacts are fresh
Mentioned In	n/a

STATE-MODEL-INVALID

State where validation diagnostics include one or more errors.

Key	Value
Kind	Catalog State
Name	model is invalid
Mentioned In	n/a

STATE-MODEL-VALID

State where authored model passes validation checks.

Key	Value
Kind	Catalog State
Name	model is valid
Mentioned In	REQ-EMG-003

SYS-ENGINEERING-MODEL-GO

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog System
Name	engineering model go

Key	Value
Mentioned In	REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-009 , REQ-EMG-010 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030

TERM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	n/a

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc diagram
Mentioned In	n/a

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc document
Mentioned In	n/a

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc section
Mentioned In	n/a

TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term
Name	authored mapping
Mentioned In	Security View

TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

Key	Value
Kind	Engineering Model Term
Name	model bundle
Mentioned In	n/a

TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	catalog
Mentioned In	Architecture Intent View

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

Key	Value
Kind	Engineering Model Term
Name	catalog entry
Mentioned In	n/a

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

Key	Value
Kind	Engineering Model Term
Name	CLI command
Mentioned In	n/a

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

Key	Value
Kind	Engineering Model Term
Name	compliance mapping

Key	Value
Mentioned In	n/a

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term
Name	control
Mentioned In	Architecture Intent View, Security View

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

Key	Value
Kind	Engineering Model Term
Name	control verification
Mentioned In	n/a

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term
Name	data object
Mentioned In	n/a

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

Key	Value
Kind	Engineering Model Term
Name	architecture decision
Mentioned In	n/a

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target
Mentioned In	n/a

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

Key	Value
Kind	Engineering Model Term
Name	design document
Mentioned In	`Architecture Intent View`

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

Key	Value
Kind	Engineering Model Term
Name	design view
Mentioned In	n/a

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	`Security View`, `State Lifecycle View`

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

Key	Value
Kind	Engineering Model Term
Name	evidence
Mentioned In	Architecture Intent View, Security View, Traceability View, State Lifecycle View, Interaction Flow View, Audit Redaction Test

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	Architecture Intent View, Interaction Flow View, Pr Opened Review Flow

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step
Mentioned In	n/a

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	functional unit
Mentioned In	`State Lifecycle View`

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

Key	Value
Kind	Engineering Model Term
Name	generation workflow
Mentioned In	n/a

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	inference hint
Mentioned In	n/a

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface

Key	Value
Mentioned In	REQ-EMG-021 , REQ-EMG-024

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

Key	Value
Kind	Engineering Model Term
Name	lint run
Mentioned In	n/a

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

Key	Value
Kind	Engineering Model Term
Name	LOBSTER activity trace
Mentioned In	n/a

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

Key	Value
Kind	Engineering Model Term
Name	MCP server
Mentioned In	Architecture Intent View , Security View , Traceability View

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

Key	Value
Kind	Engineering Model Term
Name	MCP stdio transport
Mentioned In	n/a

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

Key	Value
Kind	Engineering Model Term
Name	MCP tool
Mentioned In	n/a

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

Key	Value
Kind	Engineering Model Term
Name	MCP tool response
Mentioned In	n/a

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

Key	Value
Kind	Engineering Model Term
Name	model
Mentioned In	`Architecture Intent View`, `Security View`, `REQ-EMG-004`, `REQ-EMG-014`, `REQ-EMG-015`, `REQ-EMG-020`, `REQ-EMG-027`, `REQ-EMG-028`, `REQ-EMG-030`, `Load Test`

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

Key	Value
Kind	Engineering Model Term
Name	Open OTM document
Mentioned In	n/a

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

Key	Value
Kind	Engineering Model Term
Name	OSCAL assessment results
Mentioned In	n/a

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL POA&M
Mentioned In	n/a

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL SSP
Mentioned In	`REQ-EMG-013`

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

Key	Value
Kind	Engineering Model Term
Name	POA&M item
Mentioned In	n/a

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	reference index
Mentioned In	n/a

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

Key	Value
Kind	Engineering Model Term
Name	repository index
Mentioned In	n/a

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	Architecture Intent View , Security View , Traceability View , Interaction Flow View , REQ-EMG-007 , REQ-EMG-010 , REQ-EMG-023 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	risk
Mentioned In	Architecture Intent View, Security View, Audit Record Persistence Test

TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

Key	Value
Kind	Engineering Model Term
Name	security context
Mentioned In	n/a

TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

Key	Value
Kind	Engineering Model Term
Name	source block
Mentioned In	n/a

TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state

Key	Value
Mentioned In	Security View , State Lifecycle View

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

Key	Value
Kind	Engineering Model Term
Name	Structurizr DSL
Mentioned In	n/a

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat assumption
Mentioned In	n/a

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

Key	Value
Kind	Engineering Model Term
Name	Threat Dragon document
Mentioned In	n/a

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat mitigation
Mentioned In	n/a

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

Key	Value
Kind	Engineering Model Term
Name	threat model export
Mentioned In	n/a

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat out-of-scope decision
Mentioned In	n/a

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	threat scenario
Mentioned In	n/a

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

Key	Value
Kind	Engineering Model Term
Name	trace marker
Mentioned In	`REQ-EMG-010`

TERM-TRLC-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

Key	Value
Kind	Engineering Model Term
Name	TRLC package
Mentioned In	REQ-EMG-006

TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term
Name	trust boundary
Mentioned In	Security View, REQ-EMG-008

TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

Key	Value
Kind	Engineering Model Term
Name	validation
Mentioned In	Architecture Intent View, REQ-EMG-001, REQ-EMG-009, REQ-EMG-011, Ota Campaign Test

TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	validation diagnostic
Mentioned In	n/a

TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

Key	Value
Kind	Engineering Model Term
Name	view
Mentioned In	`Architecture Intent View`, `Security View`, `Traceability View`, `State Lifecycle View`, `Interaction Flow View`, `REQ-EMG-024`, `REQ-EMG-026`, `REQ-EMG-027`, `Project Test`

13.3. Inferred Runtime References

13.4. Inferred Code References

aws-sdk-go-v2/secretsmanager (secrets_configuration.ts)

Inferred library_external dependency owned by FU-SECRETS-CONFIGURATION from src/secrets_configuration.ts.

Key	Value
Kind	Code library_external
Owner	FU-SECRETS-CONFIGURATION
Source	src/secrets_configuration.ts
Mentioned In	n/a

github-rest-sdk (pr_context_assembly.ts)

Inferred library_external dependency owned by FU-PR-CONTEXT-ASSEMBLY from src/pr_context_assembly.ts.

Key	Value
Kind	Code library_external
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	src/pr_context_assembly.ts
Mentioned In	n/a

github-rest-sdk/publication-client (review_publication.ts)

Inferred library_external dependency owned by FU-REVIEW-PUBLICATION from src/review_publication.ts.

Key	Value
Kind	Code library_external
Owner	FU-REVIEW-PUBLICATION
Source	src/review_publication.ts
Mentioned In	n/a

github.com/defenseunicorns/go-oscal/src/types/oscal-1-1-3 (gemara_oscal.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara.go)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/gemara.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_evaluation.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_evaluation.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_evaluation.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_export.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_export_test.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_extended.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_extended.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_extended.go

Key	Value
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_l1l3.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_l1l3.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_l1l3.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara/fetcher (gemara_export_test.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara/gemaraconv (gemara_oscal.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

github.com/goccy/go-yaml (gemara_export.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

github.com/hashicorp/hcl/v2 (inferred_layers.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

github.com/hashicorp/hcl/v2/hclparse (inferred_layers.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

github.com/labeth/ears-lint-go (ears_preflight.go)

Inferred library_external dependency owned by FU-VALIDATION-ENGINE from ears_preflight.go.

Key	Value
Kind	Code library_external
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (composition.go)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/composition.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (gemara.go)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/gemara.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engdoc/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engdoc/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-THREAT-EXPORTER from cmd/engdragon/main.go.

Key	Value
Kind	Code library_external
Owner	FU-THREAT-EXPORTER
Source	cmd/engdragon/main.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/enggemara/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/enggemara/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-LOBSTER-EXPORTER from cmd/englobster/main.go.

Key	Value
Kind	Code library_external
Owner	FU-LOBSTER-EXPORTER
Source	cmd/englobster/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-OSCAL-EXPORTER from cmd/engoscal/main.go.

Key	Value
Kind	Code library_external
Owner	FU-OSCAL-EXPORTER
Source	cmd/engoscal/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-STRUCTURIZR-EXPORTER from cmd/engstruct/main.go.

Key	Value
Kind	Code library_external
Owner	FU-STRUCTURIZR-EXPORTER
Source	cmd/engstruct/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-ALLOCATION-TRACE from cmd/engtrace/main.go.

Key	Value
Kind	Code library_external
Owner	FU-ALLOCATION-TRACE
Source	cmd/engtrace/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-TRLC-EXPORTER from cmd/engtrlc/main.go.

Key	Value
Kind	Code library_external
Owner	FU-TRLC-EXPORTER
Source	cmd/engtrlc/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go (main.go)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engview/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engview/main.go
Mentioned In	n/a

github.com/santhosh-tekuri/jsonschema/v5 (threat_model_export_test.go)

Inferred library_external dependency owned by FU-THREAT-EXPORTER from threat_model_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-THREAT-EXPORTER
Source	threat_model_export_test.go
Mentioned In	n/a

github.com/tree-sitter/go-tree-sitter (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

github.com/tree-sitter/tree-sitter-go/bindings/go (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

github.com/tree-sitter/tree-sitter-rust/bindings/go (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go

Key	Value
Mentioned In	n/a

github.com/tree-sitter/tree-sitter-typescript/bindings/go (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

gopkg.in/yaml.v3 (checkout_api.go)

Inferred library_external dependency owned by FU-CHECKOUT from src/checkout_api.go.

Key	Value
Kind	Code library_external
Owner	FU-CHECKOUT
Source	src/checkout_api.go
Mentioned In	n/a

gopkg.in/yaml.v3 (inferred_layers.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

gopkg.in/yaml.v3 (load.go)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/load.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/load.go
Mentioned In	n/a

serde_json::json (payment_engine.rs)

Inferred library_external dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment_engine.rs.

Key	Value
Kind	Code library_external
Owner	FU-PAYMENT-AUTHORIZATION
Source	src/payment_engine.rs
Mentioned In	n/a

serde_json::json (review_orchestrator.rs)

Inferred library_external dependency owned by FU-REVIEW-ORCHESTRATION from src/review_orchestrator.rs.

Key	Value
Kind	Code library_external
Owner	FU-REVIEW-ORCHESTRATION
Source	src/review_orchestrator.rs
Mentioned In	n/a

zod (risk_scorer.ts)

Inferred library_external dependency owned by FU-RISK-SCORING from src/risk_scorer.ts.

Key	Value
Kind	Code library_external
Owner	FU-RISK-SCORING
Source	src/risk_scorer.ts
Mentioned In	n/a

./support/audit_envelope (risk_scorer.ts)

Inferred library_first_party dependency owned by FU-RISK-SCORING from src/risk_scorer.ts.

Key	Value
Kind	Code library_first_party
Owner	FU-RISK-SCORING
Source	src/risk_scorer.ts
Mentioned In	n/a

crate::domain::events::PaymentEvent (payment_engine.rs)

Inferred library_first_party dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment_engine.rs.

Key	Value
Kind	Code library_first_party
Owner	FU-PAYMENT-AUTHORIZATION
Source	src/payment_engine.rs
Mentioned In	n/a

github.com/labeth/engineering-model-go/codemap (inferred_code.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_code.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_code.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/codemap (inferred_verification.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_verification.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_verification.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/codemap (server.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/server.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/server.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/mcp (main.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from cmd/engmcp/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	cmd/engmcp/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_composition.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_decisions.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_decisions.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_decisions.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_design_refs.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_design_refs.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_design_refs.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_diagrams_core.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_core.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_core.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_diagrams_core_test.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_core_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_core_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_diagrams_runtime.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_runtime.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_runtime.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_diagrams_runtime_test.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_runtime_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_runtime_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_gemara.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_gemara.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_gemara.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_linking_units.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_linking_units.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_linking_units.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_test.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_views.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_views.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_views.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (asciidoc_views_test.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_views_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_views_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (catalog_description_validation.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from catalog_description_validation.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	catalog_description_validation.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (catalog_systems_test.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from catalog_systems_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	catalog_systems_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (composition.go)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (composition_test.go)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (ears_preflight.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from ears_preflight.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (ears_preflight_test.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from ears_preflight_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/gemara.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_evaluation.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_evaluation.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_evaluation.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_export.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_export_test.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_extended.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_extended.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_extended.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_l1l3.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_l1l3.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_l1l3.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_oscal.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (gemara_oscal_test.go)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_oscal_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (generate.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	generate.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (inferred_code.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_code.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_code.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (inferred_layers.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (inferred_layers_test.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_layers_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers_test.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (inferred_verification.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_verification.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_verification.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (inferred_verification_test.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_verification_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_verification_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (oscal_ar.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_ar.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_ar.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (oscal_compliance.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_compliance.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_compliance.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (oscal_poam.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_poam.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_poam.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (oscal_ssp.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_ssp.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_ssp.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (oscal_ssp_test.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_ssp_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_ssp_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (project.go)

Inferred library_first_party dependency owned by FU-VIEW-PROJECTION from view/project.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VIEW-PROJECTION
Source	view/project.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (project_test.go)

Inferred library_first_party dependency owned by FU-VIEW-PROJECTION from view/project_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VIEW-PROJECTION
Source	view/project_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (server.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/server.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/server.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (structurizr_export.go)

Inferred library_first_party dependency owned by FU-STRUCTURIZR-EXPORTER from structurizr_export.go.

Key	Value
Kind	Code library_first_party
Owner	FU-STRUCTURIZR-EXPORTER
Source	structurizr_export.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (threat_model_export.go)

Inferred library_first_party dependency owned by FU-THREAT-EXPORTER from threat_model_export.go.

Key	Value
Kind	Code library_first_party
Owner	FU-THREAT-EXPORTER
Source	threat_model_export.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (trace_matrix.go)

Inferred library_first_party dependency owned by FU-ALLOCATION-TRACE from trace_matrix.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ALLOCATION-TRACE
Source	trace_matrix.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (trace_matrix_test.go)

Inferred library_first_party dependency owned by FU-ALLOCATION-TRACE from trace_matrix_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ALLOCATION-TRACE
Source	trace_matrix_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (trlc_export.go)

Inferred library_first_party dependency owned by FU-TRLC-EXPORTER from trlc_export.go.

Key	Value
Kind	Code library_first_party
Owner	FU-TRLC-EXPORTER
Source	trlc_export.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (validate.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from validate/validate.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	validate/validate.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/model (validate_test.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from validate/validate_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	validate/validate_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/render/diagramstyle (asciidoc.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/render/diagramstyle (asciidoc_diagrams_core.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_core.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_core.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/render/diagramstyle (render.go)

Inferred library_first_party dependency owned by FU-VIEW-PROJECTION from render/mermaid/render.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VIEW-PROJECTION
Source	render/mermaid/render.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/render/mermaid (generate.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	generate.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (asciidoc.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (asciidoc_composition.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (asciidoc_test.go)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (catalog_description_validation.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from catalog_description_validation.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	catalog_description_validation.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (checkout_api.go)

Inferred library_first_party dependency owned by FU-CHECKOUT from src/checkout_api.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CHECKOUT
Source	src/checkout_api.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (composition.go)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (composition.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/composition.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (composition_test.go)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (e2e_test.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from e2e_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	e2e_test.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (ears_preflight.go)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from ears_preflight.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (generate.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	generate.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (inferred_code.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_code.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_code.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (inferred_layers.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (inferred_verification.go)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_verification.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_verification.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from cmd/engdoc/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engdoc/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-THREAT-EXPORTER from cmd/engdragon/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-THREAT-EXPORTER
Source	cmd/engdragon/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from cmd/engoscal/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	cmd/engoscal/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-STRUCTURIZR-EXPORTER from cmd/engstruct/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-STRUCTURIZR-EXPORTER
Source	cmd/engstruct/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-ALLOCATION-TRACE from cmd/engtrace/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ALLOCATION-TRACE
Source	cmd/engtrace/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (main.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from cmd/engview/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engview/main.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (oscal_ar.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_ar.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_ar.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (oscal_compliance.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_compliance.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_compliance.go
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (oscal_poam.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_poam.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_poam.go

Key	Value
Mentioned In	n/a

github.com/labeth/engineering-model-go/validate (oscal_ssp.go)

Inferred library_first_party dependency owned by FU-OSCAL-EXPORTER from oscal_ssp.go.

Key	Value
Kind	Code library_first_party
Owner	FU-OSCAL-EXPORTER
Source	oscal_ssp.go
Mentioned In	n/a

13.5. Verification References

VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367

executes authorized checkout flow from session start through payment authorization success

Key	Value
Kind	e2e
Status	not-run
Source	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Name	Authorized Checkout Flow
Mentioned In	Interaction Flow View, Authorized Checkout Flow

VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

Key	Value
Kind	e2e
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Name	Ota Rollout Flow
Mentioned In	Interaction Flow View, Ota Rollout Flow

VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6

exercises end-to-end pull request opened flow from webhook ingest to published review

Key	Value
Kind	e2e
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Name	Pr Opened Review Flow
Mentioned In	Interaction Flow View , Pr Opened Review Flow

VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4

checks audit record persistence includes payment id and computed risk score

Key	Value
Kind	integration
Status	not-run
Source	examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts
Name	Audit Record Persistence Test
Mentioned In	Interaction Flow View , Audit Record Persistence Test

VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4

checks audit evidence redaction for secrets and sensitive payload fields

Key	Value
Kind	integration
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts
Name	Audit Redaction Test
Mentioned In	Interaction Flow View , Audit Redaction Test

VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163

validates fraud gate escalation and block decisions before authorization completion

Key	Value
Kind	integration
Status	not-run
Source	examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs
Name	Fraud Gate Test
Mentioned In	Interaction Flow View , Fraud Gate Test

VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8

validates telemetry logging pipeline persistence and operational reporting signals

Key	Value
Kind	integration
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts
Name	Logging Pipeline Test
Mentioned In	Interaction Flow View , Logging Pipeline Test

VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003

verifies policy-only fallback when Bedrock inference is unavailable

Key	Value
Kind	integration
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs
Name	Policy Only Fallback Test
Mentioned In	Interaction Flow View , Policy Only Fallback Test

VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84

Inferred from test source artifact.

Key	Value
Kind	test

Key	Value
Status	not-run
Source	asciidoc_diagrams_core_test.go
Name	Asciidoc Diagrams Core Test
Mentioned In	Interaction Flow View , Asciidoc Diagrams Core Test

VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_diagrams_runtime_test.go
Name	Asciidoc Diagrams Runtime Test
Mentioned In	Interaction Flow View , Asciidoc Diagrams Runtime Test

VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_linking_units_test.go
Name	Asciidoc Linking Units Test
Mentioned In	Interaction Flow View , Asciidoc Linking Units Test

VER-INF-TEST-ASCIIDOC_TEST-F60765

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run

Key	Value
Source	asciidoc_test.go
Name	Asciidoc Test
Mentioned In	Interaction Flow View , Asciidoc Test

VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_views_test.go
Name	Asciidoc Views Test
Mentioned In	Interaction Flow View , Asciidoc Views Test

VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	catalog_systems_test.go
Name	Catalog Systems Test
Mentioned In	Interaction Flow View , Catalog Systems Test

VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671

validates checkout decline response contract includes reason and retry guidance

Key	Value
Kind	test
Status	not-run
Source	examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go

Key	Value
Name	Checkout Decline Response Test
Mentioned In	Interaction Flow View , Checkout Decline Response Test

VER-INF-TEST-COMPOSITION_TEST-8FE8F1

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	composition_test.go
Name	Composition Test
Mentioned In	Interaction Flow View , Composition Test

VER-INF-TEST-COMPOSITION_TEST-AB66B8

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	mcp/composition_test.go
Name	Composition Test
Mentioned In	Interaction Flow View , Composition Test

VER-INF-TEST-E2E_TEST-BE56F8

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	e2e_test.go
Name	E2e Test

Key	Value
Mentioned In	Interaction Flow View , E2e Test

VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	ears_preflight_test.go
Name	Ears Preflight Test
Mentioned In	Interaction Flow View , Ears Preflight Test

VER-INF-TEST-GEMARA_EXPORT_TEST-45C982

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	gemara_export_test.go
Name	Gemara Export Test
Mentioned In	Interaction Flow View , Gemara Export Test

VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	gemara_oscal_test.go
Name	Gemara Oscal Test
Mentioned In	Interaction Flow View , Gemara Oscal Test

VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F

validates GitHub check-run contract shape and required review result fields

Key	Value
Kind	test
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go
Name	Github Check Run Contract Test
Mentioned In	Interaction Flow View , Github Check Run Contract Test

VER-INF-TEST-INFERRED_CODE_TEST-3CA82E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_code_test.go
Name	Inferred Code Test
Mentioned In	Interaction Flow View , Inferred Code Test

VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_layers_test.go
Name	Inferred Layers Test
Mentioned In	Interaction Flow View , Inferred Layers Test

VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_verification_test.go
Name	Inferred Verification Test
Mentioned In	Interaction Flow View , Inferred Verification Test

VER-INF-TEST-LOAD_TEST-A58EB3

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/load_test.go
Name	Load Test
Mentioned In	Interaction Flow View , Load Test

VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	lobster_export_test.go
Name	Lobster Export Test
Mentioned In	Interaction Flow View , Lobster Export Test

VER-INF-TEST-MAIN_TEST-6B61AE

Inferred from test source artifact.

Key	Value
Kind	test

Key	Value
Status	not-run
Source	cmd/engmcp/main_test.go
Name	Main Test
Mentioned In	Interaction Flow View , Main Test

VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	oscal_chain_test.go
Name	Oscal Chain Test
Mentioned In	Interaction Flow View , Oscal Chain Test

VER-INF-TEST-OSCAL_SSP_TEST-4B08EF

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	oscal_ssp_test.go
Name	Oscal Ssp Test
Mentioned In	Interaction Flow View , Oscal Ssp Test

VER-INF-TEST-PROJECT_TEST-2ECF72

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run

Key	Value
Source	view/project_test.go
Name	Project Test
Mentioned In	Interaction Flow View , Project Test

VER-INF-TEST-RENDER_TEST-EECC23

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	render/mermaid/render_test.go
Name	Render Test
Mentioned In	Interaction Flow View , Render Test

VER-INF-TEST-SCAN_TEST-B04F81

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	codemap/scan_test.go
Name	Scan Test
Mentioned In	Interaction Flow View , Scan Test

VER-INF-TEST-SERVER_TEST-389957

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	mcp/server_test.go

Key	Value
Name	Server Test
Mentioned In	Interaction Flow View , Server Test

VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	structurizr_export_test.go
Name	Structurizr Export Test
Mentioned In	Interaction Flow View , Structurizr Export Test

VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2

validates telemetry ingest contract fields and accepted request semantics

Key	Value
Kind	test
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go
Name	Telemetry Ingest Contract Test
Mentioned In	Interaction Flow View , Telemetry Ingest Contract Test

VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	threat_model_export_test.go
Name	Threat Model Export Test

Key	Value
Mentioned In	Interaction Flow View , Threat Model Export Test

VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	trace_matrix_test.go
Name	Trace Matrix Test
Mentioned In	Interaction Flow View , Trace Matrix Test

VER-INF-TEST-TRLC_EXPORT_TEST-DD1345

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	trlc_export_test.go
Name	Trlc Export Test
Mentioned In	Interaction Flow View , Trlc Export Test

VER-INF-TEST-VALIDATE_TEST-498C59

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	validate/validate_test.go
Name	Validate Test
Mentioned In	Interaction Flow View , Validate Test

VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD

validates checkout decline reason normalization and retry policy messaging

Key	Value
Kind	unit
Status	not-run
Source	examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go
Name	Checkout Decline Reason Test
Mentioned In	Interaction Flow View , Checkout Decline Reason Test

VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB

verifies OTA campaign signature validation and rollout trigger behavior

Key	Value
Kind	unit
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go
Name	Ota Campaign Test
Mentioned In	Interaction Flow View , Ota Campaign Test

VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE

verifies rate-limit deferral behavior for pull request review processing

Key	Value
Kind	unit
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go
Name	Rate Limit Deferral Test
Mentioned In	Interaction Flow View , Rate Limit Deferral Test